

claude-windows / 068560d3-1ca2-4387-981a-d2d6bca55414

Metadata

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Transcript

1. user (2026-06-16T06:26:59.417Z)

LENS = HONESTY / over-claim. Do the docstrings, the test assertion messages, doc \$6, and the tracker over-state what a green suite proves? Is the \$6 'behaviour-locking NOT correctness' caveat actually carried in the test PASS/FAIL MESSAGES (not only docs)? Does the 'cross-OS determinism' claim overreach given test_golden_real_fixtures.py is NOT in the cross-OS job? Is the 'owner-accepted (researched risk), not retained-counsel' framing honest and consistently stated? Could any synthetic-tier number (F1=1.0) be mistaken for field quality anywhere?

AUDIT the SHIPPED 'Golden Regression Fixtures' feature (4 merged PRs #186/#189/#191/#192) for any gap the per-PR verification missed.

Read files in the git worktree at C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gf-doc\ (master HEAD 8d3e185): backend/eval/golden_fixtures.py (the core); tests/test_golden_fixtures.py; tests/test_golden_real_fixtures.py; eval_baselines/fixtures/ (3 synthetic fixtures + manifest.json + README.md); .github/workflows/ci.yml (the golden-crossos job); docs/GOLDEN_REGRESSION_FIXTURES.md (\$4c/\$5/\$6 claims). For analysis also read backend/eval/{experiment,distill_local,metrics,gt_loader,regression}.py and backend/pipeline/detectors/{tracknet_runner,rally_gate}.py.

THE FEATURE: a video-free regression harness. replay_fixture = parse_trajectory_csv -> distill_local.build_candidates(PINNED preset) -> VideoCandidates(5 shuttle cols only) -> experiment.predict(CONTROL, pure-stdlib) -> metrics.score. Synthetic Tier-0 fixtures + characterization (parse-compare: exact on int/structural, approx 1e-6 on float).

freeze_real_fixture = de-attribution (random fx<hex> slug, metadata strip, metric-invariant time-origin reset, STRICT gt_loader, positive no-person/audio/MD5/source-path scan) + the M2 refusal gate (raise unless owned AND minors_free AND non-blank license). The golden-crossos CI job runs `--check` + tests/test_golden_fixtures.py on ubuntu/windows/macOS.

ALREADY VERIFIED ? do NOT re-report as new findings: full suite 759 pass/7 skip; coverage 80.49%; ruff+mypy clean; cross-OS green on all 3 OSe for the COMMITTED fixtures; CONTROL avoids numpy *execution*.

Find what that MISSED: real gaps / bugs / edge-cases / privacy holes / determinism holes / over-claims. Cite file:line. Do NOT invent issues ? if a lens is clean, say so honestly and set verdict accordingly.

2. assistant (2026-06-16T06:27:02.043Z)

I'll audit the shipped Golden Regression Fixtures feature through the HONESTY/over-claim lens. Let me start by pulling the worktree and reading the core files.

3. user (2026-06-16T06:27:03.456Z)

8d3e185 feat(eval): golden fixtures P1 ? de-attribution/minimization tool + M2 refusal gate (#192)

4. assistant (2026-06-16T06:27:10.031Z)

Let me read the core files in parallel.

5. user (2026-06-16T06:27:11.284Z)

```
1      """Golden Regression Fixtures (M1) ? video-free, CONTROL-only characterization harness.
2
3      `docs/GOLDEN_REGRESSION_FIXTURES.md` ($4, $6, $7). This module is the **single shared
4      replay code path** used by both the freeze tool and the characterization gate, and it
5      deliberately stays on the **pure-stdlib CONTROL path** (D5 in that doc): the learned
6      variants train a numpy/OpenBLAS logistic regression whose near-threshold decisions flip
7      across CPUs, so they can never anchor a cross-machine byte/parse snapshot. Here we
replay
8      only `experiment.CONTROL` (`is_learned() == False`), which keeps all candidate
intervals,
9      merges overlaps, and uses **no numpy**.
10
11      The replay reuses the already-shipped eval stack VERBATIM::
12
13          parse_trajectory_csv ? distill_local.build_candidates(proposal=pinned)
14              ? VideoCandidates(name, intervals, features={5 shuttle cols},
gts=load_gt_intervals)
15              ? experiment.predict(CONTROL, vc) ? metrics.score(preds, gts, iou)
16
17      Tier-0 / privacy invariants (D3, $4c, $6 of the doc), enforced here by construction:
18
19      * Fixtures are **synthetic** ? deterministic formulae, NO randomness ? so they carry
zero
20      provenance/licensing risk and are safe to commit publicly (owner Q7 "synthetic-only").
21      * Every fixture is tagged ``kind="synthetic"`, ``minors_free=true``,
``license="synthetic"`.
22      * ``expected.json`` carries **only the 5 shuttle feature columns** ? there is, by
design, no
23      person / pose / gait / face / audio field anywhere in the schema (positive-assertion
test).
24      * The windowing preset params ``(vt, mg, mwd, max_win)`` are **pinned inside each
fixture's
25      provenance** and passed explicitly to ``build_candidates`` ? we do NOT rely on the
named
26      ``windowing.SERVED`` constant, so a future SERVED retune cannot silently churn
fixtures.
27
28      $6 honest-limits caveat (carried in the CLI / test messages, not only the docs): a green
29      suite proves only that the deterministic post-trajectory transforms are unchanged for
the
30      frozen synthetic cases ? NOT that the detector/decoder/AI handoff/DB are correct, and
NOT
31      that rally detection is good. Synthetic-tier numbers measure plumbing correctness, never
32      field quality (never cite them in QUALITY_ITERATIONS.md / the paper).
33
34      Pure + offline + stdlib. The replay imports `tracknet_runner` (which is itself stdlib-
only
35      for the parse/window math ? no TF), but pulls in no numpy/cv2 on the CONTROL path.
36      """
37      from __future__ import annotations
38
39      import argparse
40      import json
41      import math
42      import os
43      import re
44      import sys
45      from dataclasses import dataclass
46      from pathlib import Path
47      from typing import Any, Dict, List, Optional, Sequence, Tuple
48
```

```

49      # --- shipped eval stack, reused verbatim (CONTROL / pure-stdlib path only)
-----
50      from backend.eval import distill_local
51      from backend.eval.experiment import CONTROL, VideoCandidates, predict
52      from backend.eval.gt_loader import load_gt_intervals
53      from backend.eval.metrics import score
54      from backend.eval.regression import gt_hash
55      from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
56
57      Interval = Tuple[float, float]
58
59      #: Bump when the fixture/expected schema changes shape. Loaders refuse a fixture whose
60      #: ``schema_version`` exceeds this (forward-incompatible), per the schema-drift guard.
61      CURRENT_SCHEMA = 1
62
63      #: Repo-root-relative fixtures dir, resolved from THIS file (never cwd) so the harness
64      #: works
65      #: from any working directory / a fresh clone. ``parents[2]`` = backend/eval ? backend ?
66      root.
67      FIXTURES_DIR: Path = Path(__file__).resolve().parents[2] / "eval_baselines" / "fixtures"
68
69      #: The pinned SERVED windowing the synthetic fixtures freeze at: (velocity_thresh,
70      #: merge_gap,
71      #: min_window_duration). Mirrors the historical SERVED values (0.02, 2.0, 2.0) but is a
72      #: LITERAL
73      #: here on purpose ? the spec forbids depending on the named ``windowing.SERVED``
74      #: constant so a
75      #: future retune of SERVED can never silently re-baseline a committed fixture. Each
76      #: fixture also
77      #: re-states these in its provenance; the replay reads them from there.
78      PINNED_VT = 0.02
79      PINNED_MERGE_GAP = 2.0
80      PINNED_MIN_WINDOW = 2.0
81      PINNED_MAX_WINDOW = 0.0 # 0 = bounded-windowing OFF (W1), matching the SERVED default
82
83      #: The 5 fps/resolution-invariant shuttle feature columns produced by distill_local. Re-
84      #: stated
85      #: here (not imported as a mutable alias) so a fixture's expected.json is pinned to
86      #: exactly these.
87      SHUTTLE_FEATURES: Tuple[str, ...] = (
88          "duration", "peak_velocity", "mean_velocity", "active_ratio", "net_crossings",
89      )
90      assert tuple(distill_local.FEATURE_NAMES) == SHUTTLE_FEATURES, (
91          "distill_local.FEATURE_NAMES drifted from the pinned shuttle feature set"
92      )
93
94      #: Forbidden key substrings ? biometric / identity / non-shuttle streams that MUST NOT
95      #: appear
96      #: anywhere in a Tier-0 fixture (D3 / §4c). The privacy assertion scans for these.
97      FORBIDDEN_KEYS: Tuple[str, ...] = ("person", "pose", "gait", "face", "audio", "player")
98
99      _FLOAT_ROUND_DP = 6 # all written floats rounded to this many decimals (determinism)
100     _FLOAT_TOL = 1e-6 # abs tolerance when parse-comparing recomputed vs committed floats
101
102     #: A 32-hex-char MD5 (the ``video_id`` shape, ``compute_video_id``) ? the leak-scan
103     #: refuses to
104     #: emit a fixture whose written text contains one (it would be an attribution back-
105     #: channel, §4c/§5).
106     _MD5_RE = re.compile(r"\b[0-9a-fA-F]{32}\b")
107
108     # =====
109     # Synthetic trajectory generation (deterministic ? NO random)
110     # =====

```

```

102
103     @dataclass(frozen=True)
104     class RallySpec:
105         """One in-flight rally segment: the shuttle oscillates across the net midline.
106
107         ``start_frame`` ? first frame index; ``n_frames`` ? length; ``period`` ? oscillation
108         period in frames (governs net-crossing count); ``amplitude`` ? px excursion from the
109         net midline along the play axis. Deterministic sine motion ? fixed
110         velocity/crossings.
111         """
112         start_frame: int
113         n_frames: int
114         period: float = 30.0
115         amplitude: float = 400.0
116
117
118     @dataclass(frozen=True)
119     class GapSpec:
120         """Dead-time between rallies ? what separates one window from the next.
121
122         ``kind="still"`` ? shuttle visible but resting (velocity ? 0 ? no active frames);
123         ``kind="occluded"`` ? shuttle not visible (Visibility=0 ? trajectory broken). Both
124         end an active run so the next rally forms a distinct window.
125         """
126         start_frame: int
127         n_frames: int
128         kind: str = "still" # "still" | "occluded"
129
130
131
132     @dataclass(frozen=True)
133     class FixtureSpec:
134         """A full synthetic scenario: a frame-indexed sequence of rallies and gaps + its GT.
135
136         ``gts`` are the hand-authored ground-truth rally intervals (seconds) ? the synthetic
137         "labels.csv". For a synthetic fixture they line up with the rallies by construction,
138         which is exactly what makes the quality-floor a *plumbing-correctness* check, not a
139         field-quality one (§6).
140         """
141         slug: str
142         description: str
143         fps: float
144         frame_width: float
145         frame_height: float
146         segments: Tuple[Any, ...] # ordered (RallySpec | GapSpec)
147         gts: Tuple[Interval, ...]
148         net_midline: float = 640.0
149
150
151
152     def _rally_points(spec: RallySpec, net_midline: float, frame_height: float
153                      ) -> List[Tuple[int, int, float, float]]:
154         """Deterministic in-flight shuttle: sine oscillation across the net midline (play
155         axis = x)."""
156         rows: List[Tuple[int, int, float, float]] = []
157         y_mid = frame_height / 2.0
158         for i in range(spec.n_frames):
159             x = net_midline + spec.amplitude * math.sin(2.0 * math.pi * i / spec.period)
160             # A small orthogonal wobble keeps y-spread < x-spread so the play axis is
161             unambiguously x.
162             y = y_mid + 30.0 * math.sin(2.0 * math.pi * i / (spec.period / 2.0))
163             rows.append((spec.start_frame + i, 1, x, y))
164         return rows

```

```

164
165 def _gap_points(spec: GapSpec, net_midline: float, frame_height: float
166                 ) -> List[Tuple[int, int, float, float]]:
167     """Dead-time rows: resting-visible (still) or absent (occluded)."""
168     rows: List[Tuple[int, int, float, float]] = []
169     y_mid = frame_height / 2.0
170     for i in range(spec.n_frames):
171         if spec.kind == "occluded":
172             rows.append((spec.start_frame + i, 0, 0.0, 0.0))
173         elif spec.kind == "still":
174             rows.append((spec.start_frame + i, 1, net_midline, y_mid))
175         else: # pragma: no cover - guarded by builder
176             raise ValueError(f"unknown gap kind {spec.kind!r} (use 'still'/'occluded')")
177     return rows
178
179
180 def make_synthetic_trajectory(spec: FixtureSpec) -> List[Tuple[int, int, float, float]]:
181     """Build the deterministic ``(Frame, Visibility, X, Y)`` rows for a synthetic
182     scenario."""
183     rows: List[Tuple[int, int, float, float]] = []
184     for seg in spec.segments:
185         if isinstance(seg, RallySpec):
186             rows.extend(_rally_points(seg, spec.net_midline, spec.frame_height))
187         elif isinstance(seg, GapSpec):
188             rows.extend(_gap_points(seg, spec.net_midline, spec.frame_height))
189         else: # pragma: no cover - guarded by dataclass shape
190             raise TypeError(f"unknown segment type {type(seg).__name__}")
191     rows.sort(key=lambda r: r[0])
192     return rows
193
194 def write_trajectory_csv(rows: Sequence[Tuple[int, int, float, float]], path: Path) ->
195 None:
196     """Write trajectory rows as the TrackNet CSV ``(Frame,Visibility,X,Y)``, LF + 6dp
197     floats."""
198     lines = ["Frame,Visibility,X,Y"]
199     for frame, vis, x, y in rows:
200         lines.append(f"{int(frame)},{int(vis)},{_fmt(x)},{_fmt(y)}")
201     path.write_text("\n".join(lines) + "\n", encoding="utf-8", newline="\n")
202
203 def write_labels_csv(gts: Sequence[Interval], path: Path) -> None:
204     """Write GT intervals as a ``start,end`` labels CSV (the strict gt_loader format),
205     LF."""
206     lines = ["start,end"]
207     for s, e in gts:
208         lines.append(f"{_fmt(s)},{_fmt(e)}")
209     path.write_text("\n".join(lines) + "\n", encoding="utf-8", newline="\n")
210
211 # =====
212 # Built-in synthetic scenarios (the 3 committed slugs)
213 # =====
214
215 def _fr(seconds: float, fps: float) -> int:
216     return int(round(seconds * fps))
217
218
219 def builtin_specs() -> List[FixtureSpec]:
220     """The committed synthetic scenarios. Deterministic; chosen so each exercises a
221     distinct
222     windowing behaviour (one window / dead-time split / occlusion split)."""
223     fps = 30.0
224     fw, fh = 1280.0, 720.0

```

```

224         net = 640.0
225
226         # 1) clean_single_rally: one ~4s in-flight rally ? exactly 1 candidate window,
crossings>0.
227         single = FixtureSpec(
228             slug="clean_single_rally",
229             description="One continuous ~4s rally; shuttle oscillates across the net ? 1
window.",
230             fps=fps, frame_width=fw, frame_height=fh, net_midline=net,
231             segments=(RallySpec(start_frame=0, n_frames=120),),
232             # Window is [0.033, 3.967]; GT is the rally span the synthetic motion fills.
233             gts=((0.0, 4.0),),
234         )
235
236         # 2) multi_rally_with_deadtime: rally, 3s STILL dead-time, rally ? 2 windows split
by the
237         #     low-velocity gap (> merge_gap=2.0s) ? proves dead-time separates windows.
238         r0 = RallySpec(start_frame=0, n_frames=120)
239         gap_still = GapSpec(start_frame=120, n_frames=_fr(3.0, fps), kind="still")
240         r1 = RallySpec(start_frame=120 + _fr(3.0, fps), n_frames=120)
241         multi = FixtureSpec(
242             slug="multi_rally_with_deadtime",
243             description="Two ~4s rallies separated by 3s of still (resting-shuttle) dead-
time ? 2 windows.",
244             fps=fps, frame_width=fw, frame_height=fh, net_midline=net,
245             segments=(r0, gap_still, r1),
246             gts=((0.0, 4.0), (7.0, 11.0)),
247         )
248
249         # 3) occlusion_break: rally, 3s OCCLUDED (Visibility=0), rally ? 2 windows split by
the
250         #     visibility break ? proves occlusion (not just low velocity) separates windows.
251         o0 = RallySpec(start_frame=0, n_frames=120)
252         gap_occ = GapSpec(start_frame=120, n_frames=_fr(3.0, fps), kind="occluded")
253         o1 = RallySpec(start_frame=120 + _fr(3.0, fps), n_frames=120)
254         occ = FixtureSpec(
255             slug="occlusion_break",
256             description="Two ~4s rallies separated by 3s of shuttle occlusion (Visibility=0)
? 2 windows.",
257             fps=fps, frame_width=fw, frame_height=fh, net_midline=net,
258             segments=(o0, gap_occ, o1),
259             gts=((0.0, 4.0), (7.0, 11.0)),
260         )
261
262         return [single, multi, occ]
263
264
265     def _spec_by_slug() -> Dict[str, FixtureSpec]:
266         return {s.slug: s for s in builtin_specs()}
267
268
269     # =====
270     # The shared replay (CONTROL / pure-stdlib path)
271     # =====
272
273
274     def _fmt(x: float) -> str:
275         """Render a float with up to 6dp, no scientific notation, ``-0.0`` normalised to
``0.0``."""
276         r = round(float(x) + 0.0, _FLOAT_ROUND_DP)
277         if r == 0.0:
278             r = 0.0
279         return f"{r:.6f}"
280
281

```

```

282     def _round(x: float) -> float:
283         r = round(float(x) + 0.0, _FLOAT_ROUND_DP)
284         return 0.0 if r == 0.0 else r
285
286
287     def _pinned_proposal(prov: Optional[Dict[str, Any]]) -> Tuple[float, float, float]:
288         """The windowing proposal tuple, read from the fixture provenance when present (the
pinned
289         params travel with the fixture), else the module-level pinned defaults. NEVER
windowing.SERVED."""
290         if prov and "windowing" in prov:
291             w = prov["windowing"]
292             return (float(w["vt"]), float(w["mg"]), float(w["mwd"]))
293         return (PINNED_VT, PINNED_MERGE_GAP, PINNED_MIN_WINDOW)
294
295
296     def replay_fixture(slug_or_dir: Any) -> Dict[str, Any]:
297         """Replay one fixture through the shipped CONTROL path; return its recomputed
snapshot.
298
299         Resolves ``slug_or_dir`` (a slug string or a fixture directory ``Path``) to its
committed
300         ``trajectory.csv`` + ``labels.csv`` + ``provenance.json``, then:
301
302         parse_trajectory_csv ? build_candidates(proposal=pinned) ? VideoCandidates(5
shuttle
303         cols only) ? predict(CONTROL, vc) ? score(preds, gts, iou).
304
305         Returns ``{slug, schema_version, iou, candidates:[{start,end,shuttle:{5 names}}],
306         control_segments:[[s,e]], score:{...}, gt_hash}``. CONTROL keeps every candidate
then
307         merges overlaps; NO numpy is touched.
308         """
309         fdir = _resolve_dir(slug_or_dir)
310         slug = fdir.name
311         traj_csv = fdir / "trajectory.csv"
312         labels_csv = fdir / "labels.csv"
313         prov = _read_json(fdir / "provenance.json") if (fdir / "provenance.json").is_file()
else None
314
315         fps, fw = _fps_fw(prov, slug)
316         iou = float(prov["iou"]) if (prov and "iou" in prov) else 0.5
317         proposal = _pinned_proposal(prov)
318
319         intervals, feature_rows = distill_local.build_candidates(
320             str(traj_csv), fps=fps, frame_width=fw, proposal=proposal)
321
322         # Build VideoCandidates DIRECTLY with ONLY the 5 shuttle feature columns. We never
call
323         # fusion_compare.load_videos / ablation.load_videos ? those read c['person']
unconditionally
324         # and would KeyError on these person-free fixtures (HARD CONSTRAINT 1).
325         features: Dict[str, List[float]] = {
326             name: [row[i] for row in feature_rows] for i, name in
enumerate(SHUTTLE_FEATURES)
327         }
328         gts = load_gt_intervals(str(labels_csv), strict=True)
329         vc = VideoCandidates(name=slug, intervals=intervals, features=features, gts=gts)
330
331         preds = predict(CONTROL, vc) # CONTROL: keep all ? merge_overlaps; pure-stdlib, no
numpy
332         sc = score(preds, gts, iou)
333
334         candidates: List[Dict[str, Any]] = []
335         for j, (s, e) in enumerate(intervals):

```

```

336         shuttle = {name: _round(feature_rows[j][i]) for i, name in
enumerate(SHUTTLE_FEATURES)}
337         # net_crossings is conceptually an integer count ? keep it exact (int), per
constraint 2.
338         shuttle["net_crossings"] =
int(round(feature_rows[j][SHUTTLE_FEATURES.index("net_crossings")]))
339         candidates.append({"start": _round(s), "end": _round(e), "shuttle": shuttle})
340
341     return {
342         "slug": slug,
343         "schema_version": CURRENT_SCHEMA,
344         "iou": _round(iou),
345         "candidates": candidates,
346         "control_segments": [[_round(s), _round(e)] for s, e in preds],
347         "score": {
348             "iou_threshold": _round(sc.iou_threshold),
349             "num_pred": int(sc.num_pred),
350             "num_gt": int(sc.num_gt),
351             "true_positives": int(sc.true_positives),
352             "false_positives": int(sc.false_positives),
353             "false_negatives": int(sc.false_negatives),
354             "precision": _round(sc.precision),
355             "recall": _round(sc.recall),
356             "f1": _round(sc.f1),
357             "mean_iou": _round(sc.mean_iou),
358             "mean_start_error": _round(sc.mean_start_error),
359             "mean_end_error": _round(sc.mean_end_error),
360         },
361         "gt_hash": gt_hash(gts),
362     }
363
364
365     # =====
366     # Freeze / refresh (expected.json + provenance.json are correct-by-construction)
367     # =====
368
369
370     def _provenance(spec_or_dir: Any, slug: str, fps: float, fw: float, iou: float = 0.5
371                     ) -> Dict[str, Any]:
372         """The committed provenance block ? pins the windowing + privacy/licensing tags
($4c, §5)."""
373         desc = ""
374         if isinstance(spec_or_dir, FixtureSpec):
375             desc = spec_or_dir.description
376         return {
377             "slug": slug,
378             "schema_version": CURRENT_SCHEMA,
379             "kind": "synthetic",
380             "license": "synthetic",
381             "minors_free": True,
382             "description": desc,
383             "fps": _round(fps),
384             "frame_width": _round(fw),
385             "iou": _round(iou),
386             "windowing": {
387                 "vt": PINNED_VT,
388                 "mg": PINNED_MERGE_GAP,
389                 "mwd": PINNED_MIN_WINDOW,
390                 "max_win": PINNED_MAX_WINDOW,
391             },
392             "paths": {
393                 "trajectory": "trajectory.csv",
394                 "labels": "labels.csv",
395                 "expected": "expected.json",
396             },

```



```

397         "note": (
398             "Synthetic, deterministic, owned (no real footage). Tier-0 / CONTROL-only. "
399             "Plumbing-correctness fixture ? NOT a measure of rally-detection quality
($6).",
400         ),
401     }
402
403
404     def freeze_fixture(slug: str, fixtures_dir: Path = FIXTURES_DIR,
405                       iou: float = 0.5) -> Dict[str, Any]:
406         """Compute ``expected.json`` + ``provenance.json`` for one fixture from its
COMMITTED
407         ``trajectory.csv`` + ``labels.csv`` via the replay (so expected is correct-by-
construction).
408
409         Idempotent: rewrites the same bytes given the same committed inputs. Requires the
trajectory
410         + labels CSVs to already exist (use ``--freeze-synthetic`` to generate them first).
411         """
412         fdir = fixtures_dir / slug
413         traj_csv, labels_csv = fdir / "trajectory.csv", fdir / "labels.csv"
414         if not traj_csv.is_file() or not labels_csv.is_file():
415             raise FileNotFoundError(
416                 f"{slug}: trajectory.csv/labels.csv missing ? run --freeze-synthetic to
generate them")
417
418         # Provenance first (it pins the windowing the replay reads), then replay ? expected.
419         spec = _spec_by_slug().get(slug)
420         fps, fw = (spec.fps, spec.frame_width) if spec is not None else
_fps_fw_from_prov(fdir, slug)
421         prov = _provenance(spec if spec is not None else fdir, slug, fps, fw, iou)
422         _write_json(fdir / "provenance.json", prov)
423
424         expected = replay_fixture(fdir)
425         _write_json(fdir / "expected.json", expected)
426         return expected
427
428
429     def freeze_synthetic(fixtures_dir: Path = FIXTURES_DIR) -> List[str]:
430         """Generate trajectory.csv + labels.csv for every built-in scenario, freeze
expected.json +
431         provenance.json, and (re)write the manifest. Returns the slugs frozen."""
432         specs = builtin_specs()
433         fixtures_dir.mkdir(parents=True, exist_ok=True)
434         slugs: List[str] = []
435         for spec in specs:
436             fdir = fixtures_dir / spec.slug
437             fdir.mkdir(parents=True, exist_ok=True)
438             rows = make_synthetic_trajectory(spec)
439             write_trajectory_csv(rows, fdir / "trajectory.csv")
440             write_labels_csv(spec.gts, fdir / "labels.csv")
441             freeze_fixture(spec.slug, fixtures_dir=fixtures_dir)
442             slugs.append(spec.slug)
443         _write_manifest(specs, fixtures_dir)
444         return slugs
445
446
447     def refresh_snapshot(slug: Optional[str] = None, all_slugs: bool = False,
448                        fixtures_dir: Path = FIXTURES_DIR) -> List[str]:
449         """Re-freeze expected.json (+ provenance.json) from the COMMITTED trajectory.csv +
labels.csv ?
450         NO regeneration of the trajectory/labels. ? §6: this can rubber-stamp a real
regression; the
451         mitigation is reviewing the visible diff + recording a reason (the CLI requires
one)."""

```

```

452     if all_slugs:
453         targets = [d.name for d in sorted(fixture_dir.iterdir())
454                     if d.is_dir() and (d / "trajectory.csv").is_file()]
455     elif slug:
456         targets = [slug]
457     else:
458         raise ValueError("refresh_snapshot needs --slug S or --all")
459     for s in targets:
460         freeze_fixture(s, fixture_dir=fixture_dir)
461     return targets
462
463
464     # =====
465     # P1 ? De-attribution + minimization: OWNED real trajectory+labels ? committable Tier-0
466     # fixture, reusing the SAME replay_fixture snapshot. (docs/GOLDEN_REGRESSION_FIXTURES.md
467     # §4c anti-attribution, §5 threat model, §7 row P1, §6 honest limits.)
468     # =====
469
470
471     class RefusalError(ValueError):
472         """The provenance/biometric refusal gate (M2) rejected an unsafe freeze.
473
474         A subclass of ``ValueError`` (so existing ``except ValueError`` paths still catch
475         it) that
476         names *why* the unsafe public-fixture path was refused. The refusal is the code-
477         enforced
478         provenance gate (§4c "provenance hard-fail gate", §7 ?): the owner-recorded /
479         minors-excluded
480         / license flags are human-asserted, but the gate makes the unsafe path impossible to
481         reach
482         *silently* ? a fixture cannot be written unless all three are affirmatively set.
483         """
484
485     def _new_surrogate_slug() -> str:
486         """An opaque RANDOM surrogate slug ? ``fx_<16 hex>`` from ``os.urandom`` (§4c).
487
488         Deliberately NOT the MD5 ``video_id`` and NOT ``HMAC(salt, video_id?name)`` ? a
489         random,
490         NON-reversible id with no algebraic link to the source, so it can't be brute-forced
491         over a
492         tiny known roster even if a salt leaked. The surrogate?source map (if kept) lives
493         off-git.
494         """
495         return f"fx_{os.urandom(8).hex()}"
496
497     #: The fixed schema filenames the writers ALWAYS emit (as ``paths`` literals). Source-
498     path tokens
499     #: that collide with these are NOT a leak ? exclude them so a short source stem like
500     ``traj`` can't
501     #: false-positive against the literal
502     ``trajectory.csv``/``labels.csv``/``expected.json``.
503     _SCHEMA_FILENAME_LITERALS: Tuple[str, ...] = (
504         "trajectory", "trajectory.csv", "labels", "labels.csv", "expected", "expected.json",
505         "provenance", "provenance.json",
506     )
507
508     def _scan_forbidden(text: str, *, where: str, source_paths: Sequence[str]) -> None:
509         """POSITIVE privacy assertion (§4c, red-team ?): raise if ``text`` leaks anything
510         attributable.
511
512         Scans the written JSON *text* for (a) any ``FORBIDDEN_KEYS`` biometric/identity
513         substring,

```

```

505         (b) a 32-hex MD5 (the ``video_id`` shape), or (c) any identifying token of a source
path
506         (full path / basename / stem) ? excluding the fixed schema filename literals the
writers
507         always emit. The trajectory is shuttle-only by nature; this guards against a
*regression*
508         re-introducing a person/audio/pose stream or an id/path back-channel ? never the
silent default.
509         """
510         low = text.lower()
511         for key in FORBIDDEN_KEYS:
512             if key in low:
513                 raise RefusalError(
514                     f"{where}: forbidden key substring {key!r} found in written output ? "
515                     f"Tier-0 fixtures are shuttle-only; person/pose/gait/face/audio/player
MUST NOT "
516                     f"appear (docs/GOLDEN_REGRESSION_FIXTURES.md §4c).")
517         m = _MD5_RE.search(text)
518         if m:
519             raise RefusalError(
520                 f"{where}: a 32-hex MD5 ({m.group(0)!r}) found in written output ? that is
the "
521                 f"video_id shape and an attribution back-channel; never write it (§4c/§5).")
522         for sp in source_paths:
523             if not sp:
524                 continue
525             p = Path(sp)
526             for token in (sp, p.name, p.stem):
527                 t = (token or "").strip().lower()
528                 # Skip a token that is itself a substring of a schema-literal the writers
always emit
529                 # (e.g. a source stem ``traj`` ? ``trajectory.csv``) ? that collision is not
a leak.
530                 if not t or any(t in lit for lit in _SCHEMA_FILENAME_LITERALS):
531                     continue
532                 if t in low:
533                     raise RefusalError(
534                         f"{where}: source-path token {token!r} found in written output ? the
source "
535                         f"filename/path must never be persisted (§4c metadata strip).")
536
537
538     def _reset_trajectory_rows(points: Sequence[Any], t0_frame: int
539                               ) -> List[Tuple[int, int, float, float]]:
540         """Shift every ``TrajectoryPoint`` frame by ``-t0_frame`` ?
``(Frame,Visibility,X,Y)`` rows.
541
542         Visibility/X/Y are passed through untouched (the shuttle geometry is metric-
invariant under a
543         pure time shift); only the absolute frame index moves toward 0, severing the match
timeline.
544         """
545         rows: List[Tuple[int, int, float, float]] = []
546         for p in points:
547             rows.append((int(p.frame) - t0_frame, 1 if p.visible else 0, float(p.x),
float(p.y)))
548         rows.sort(key=lambda r: r[0])
549         return rows
550
551
552     def _reset_labels(gts: Sequence[Interval], t0_sec: float) -> List[Interval]:
553         """Shift every label by ``-t0_sec`` (same offset as the trajectory ? metric-
invariant)."""
554         return [(s - t0_sec, e - t0_sec) for s, e in gts]
555

```

```

556
557     def freeze_real_fixture(
558         label: str,
559         trajectory_csv_path: Any,
560         labels_csv_path: Any,
561         *,
562         license: str,
563         minors_free: bool,
564         owned: bool,
565         fps: float,
566         frame_width: float,
567         fixtures_dir: Path = FIXTURES_DIR,
568         slug: Optional[str] = None,
569         time_origin_reset: bool = True,
570         source_note: str = "",
571     ) -> Dict[str, Any]:
572         """Turn an OWNED, minors-free real trajectory + golden labels into a committable,
de-attributed
573         Tier-0 fixture ? reusing M1's ``replay_fixture`` for the snapshot. ($4c / $5 / $7
row P1.)
574
575         The refusal gate (M2) makes the unsafe path impossible to reach silently: a fixture
is emitted
576         ONLY when ``minors_free is True`` AND ``owned is True`` AND ``license`` is a non-
blank string.
577         The committed bytes carry NO filename / video_id / match / player / capture-date ?
only an
578         opaque random surrogate slug, the shuttle-only trajectory (time-origin reset by
default), the
579         reset labels, and a ``kind="cleared_real"`` provenance block. A positive privacy
assertion
580         re-reads the written ``expected.json`` + ``provenance.json`` and refuses if anything
attributable
581         slipped in.
582
583         NOTE ($6 honest limits): the random surrogate + reset timeline make the fixture non-
attributable
584         only *relative to the unpublished source* ? it is NOT absolutely anonymous (EDPS v.
SRB). De-
585         attribution does not cure provenance/licensing; this sits BEHIND counsel sign-off
(P2) and the
586         owner-asserted Fork-A / minors / biometric-exclusion flags.
587
588         Returns the recomputed ``expected.json`` snapshot dict (identical to
``replay_fixture``).
589         """
590         # --- (a) REFUSAL GATE (M2) ? the code-enforced provenance hard-fail ($4c / $7 ?)
-----
591         reasons: List[str] = []
592         if minors_free is not True:
593             reasons.append("minors_free must be True (DPDP child = under 18; behavioural
monitoring "
594                             "of a minor is prohibited ? §3, guardrail 'exclude minors')")
595         if owned is not True:
596             reasons.append("owned must be True (only owner-recorded Fork-A source is safe to
commit; "
597                             "broadcast/scraped = Fork B, do NOT commit ? §3 D1)")
598         if not (isinstance(license, str) and license.strip()):
599             reasons.append("license must be a non-blank string (per-record provenance tag ?
$4c "
600                             "'provenance hard-fail gate')")
601         if reasons:
602             raise RefusalError(
603                 "REFUSED to freeze a real Tier-0 fixture ? unsafe provenance (de-attribution
does NOT "

```

```

604         "cure provenance/licensing; this gate sits behind counsel sign-off,
$6/P2):\n - "
605         + "\n - ".join(reasons))
606
607     traj_path = Path(str(trajjectory_csv_path))
608     labels_path = Path(str(labels_csv_path))
609
610     # --- (b) opaque RANDOM surrogate slug ? never the video_id MD5, never the source
name ---
611     out_slug = slug if slug is not None else _new_surrogate_slug()
612
613     # --- (c) read via the SHIPPED readers; labels via the STRICT loader (never lenient)
----
614     points = TrackNetRunner.parse_trajectory_csv(str(traj_path))
615     if not points:
616         raise ValueError(f"{traj_path}: no parseable trajectory rows
(Frame,Visibility,X,Y)")
617     gts = load_gt_intervals(str(labels_path), strict=True)
618
619     # --- (d) consistent time-origin reset (default ON; MUST be metric-invariant, $4c)
-----
620     # Subtract the SAME offset from BOTH the trajectory frames and the label seconds:
because
621     # metrics.score is built from pure differences (interval_iou, |start-start|, |end-
end|), the
622     # score is bit-identical with vs without reset ? only the absolute timeline is
severed. The
623     # quality-floor / characterization snapshot is therefore unchanged (asserted in the
tests).
624     if time_origin_reset:
625         t0_frame = min(int(p.frame) for p in points)
626         t0_sec = t0_frame / float(fps)
627     else:
628         t0_frame, t0_sec = 0, 0.0
629     rows = _reset_trajectory_rows(points, t0_frame)
630     reset_gts = _reset_labels(gts, t0_sec)
631
632     # --- (e) write the de-attributed fixture (REUSE M1 writers + replay_fixture)
-----
633     fdir = fixtures_dir / out_slug
634     fdir.mkdir(parents=True, exist_ok=True)
635     write_trajectory_csv(rows, fdir / "trajectory.csv")
636     write_labels_csv(reset_gts, fdir / "labels.csv")
637
638     iou = 0.5
639     prov = {
640         "slug": out_slug,
641         "schema_version": CURRENT_SCHEMA,
642         "kind": "cleared_real",
643         "label": str(label),
644         "license": license,
645         "minors_free": True,
646         "owned": True,
647         "fps": _round(fps),
648         "frame_width": _round(frame_width),
649         "iou": _round(iou),
650         "time_origin_reset": bool(time_origin_reset),
651         "windowing": {
652             "vt": PINNED_VT,
653             "mg": PINNED_MERGE_GAP,
654             "mwd": PINNED_MIN_WINDOW,
655             "max_win": PINNED_MAX_WINDOW,
656         },
657         "paths": {
658             "trajectory": "trajectory.csv",

```

```

659         "labels": "labels.csv",
660         "expected": "expected.json",
661     },
662     "gt_hash": gt_hash(reset_gts),
663     "source_note": str(source_note),
664     "note": (
665         "De-attributed REAL Fork-A fixture (owner-recorded, owned, minors-excluded,
biometric-"
666         "free, shuttle-only). Opaque RANDOM surrogate slug; absolute timeline reset.
NON-"
667         "attributable only relative to the UNPUBLISHED source (EDPS v. SRB) ? never
'anonymous'. "
668         "Tier-0 / CONTROL-only. $6: behaviour-locking, not correctness; sits behind
counsel (P2).",
669     ),
670 }
671 _write_json(fdir / "provenance.json", prov)
672
673 expected = replay_fixture(fdir)          # REUSE M1 ? correct-by-construction
snapshot
674 _write_json(fdir / "expected.json", expected)
675
676 # --- (f) POSITIVE privacy assertion AFTER writing (scan the written text, $4c / ?)
-----
677 source_paths = [str(traj_path), str(labels_path)]
678 _scan_forbidden((fdir / "expected.json").read_text(encoding="utf-8"),
679                 where=f"{out_slug}/expected.json", source_paths=source_paths)
680 _scan_forbidden((fdir / "provenance.json").read_text(encoding="utf-8"),
681                 where=f"{out_slug}/provenance.json", source_paths=source_paths)
682
683 # --- (g) append to the manifest with kind="cleared_real" (don't clobber synthetic)
-----
684 _append_manifest_entry(prov, fixtures_dir)
685 return expected
686
687
688 def _manifest_entry_for(prov: Dict[str, Any]) -> Dict[str, Any]:
689     """A manifest row for a single fixture, derived from its provenance block."""
690     slug = prov["slug"]
691     return {
692         "slug": slug,
693         "fps": _round(float(prov["fps"])),
694         "frame_width": _round(float(prov["frame_width"])),
695         "windowing": dict(prov.get("windowing", {})),
696         "kind": prov.get("kind", "synthetic"),
697         "schema_version": int(prov.get("schema_version", CURRENT_SCHEMA)),
698         "paths": {
699             "dir": slug,
700             "trajectory": f"{slug}/trajectory.csv",
701             "labels": f"{slug}/labels.csv",
702             "expected": f"{slug}/expected.json",
703             "provenance": f"{slug}/provenance.json",
704         },
705     }
706
707
708 def _append_manifest_entry(prov: Dict[str, Any], fixtures_dir: Path) -> None:
709     """Add/replace one fixture's manifest row WITHOUT clobbering the others (e.g.
synthetic)."""
710     entries = load_manifest(fixtures_dir)
711     entries = [e for e in entries if e.get("slug") != prov["slug"]]
712     entries.append(_manifest_entry_for(prov))
713     entries.sort(key=lambda e: str(e.get("slug")))
714     _write_json(fixtures_dir / "manifest.json", entries)
715

```

```

716
717 # =====
718 # Manifest + loaders
719 # =====
720
721
722 def _write_manifest(specs: Sequence[FixtureSpec], fixtures_dir: Path) -> None:
723     entries = [{
724         "slug": spec.slug,
725         "fps": _round(spec.fps),
726         "frame_width": _round(spec.frame_width),
727         "windowing": {
728             "vt": PINNED_VT, "mg": PINNED_MERGE_GAP,
729             "mwd": PINNED_MIN_WINDOW, "max_win": PINNED_MAX_WINDOW,
730         },
731         "kind": "synthetic",
732         "schema_version": CURRENT_SCHEMA,
733         "paths": {
734             "dir": spec.slug,
735             "trajectory": f"{spec.slug}/trajectory.csv",
736             "labels": f"{spec.slug}/labels.csv",
737             "expected": f"{spec.slug}/expected.json",
738             "provenance": f"{spec.slug}/provenance.json",
739         },
740     } for spec in specs]
741     _write_json(fixtures_dir / "manifest.json", entries)
742
743
744 def load_manifest(fixtures_dir: Path = FIXTURES_DIR) -> List[Dict[str, Any]]:
745     """Load the fixtures manifest (returns [] if absent)."""
746     path = fixtures_dir / "manifest.json"
747     if not path.is_file():
748         return []
749     data = _read_json(path)
750     if not isinstance(data, list): # pragma: no cover - corrupt manifest
751         raise ValueError(f"{path}: manifest must be a JSON array")
752     return data
753
754
755 def load_fixture(slug: str, fixtures_dir: Path = FIXTURES_DIR) -> Dict[str, Any]:
756     """Load one fixture's committed artifacts: `{provenance, expected, dir}`. Refuses
a fixture
757     whose ``schema_version`` exceeds ``CURRENT_SCHEMA`` (forward-incompatible)."""
758     fdir = fixtures_dir / slug
759     prov = _read_json(fdir / "provenance.json")
760     expected = _read_json(fdir / "expected.json")
761     sv = int(expected.get("schema_version", prov.get("schema_version", 0)))
762     if sv > CURRENT_SCHEMA:
763         raise ValueError(
764             f"{slug}: schema_version {sv} > CURRENT_SCHEMA {CURRENT_SCHEMA} ? upgrade
the harness")
765     return {"slug": slug, "dir": fdir, "provenance": prov, "expected": expected}
766
767
768 # =====
769 # Parse-compare (NEVER byte-compare): exact ints/structure, abs-tol floats
770 # =====
771
772
773 def compare_expected(recomputed: Dict[str, Any], committed: Dict[str, Any]) ->
List[str]:
774     """Return a list of human-readable mismatch messages (empty == match).
775
776     Parse-compare per HARD CONSTRAINT 2: structural/int fields (candidate count,
net_crossings,

```

```

777         TP/FP/FN, segment count, num_pred/num_gt, slug, schema_version, gt_hash) must be
EXACTLY
778         equal; float fields compared with abs tolerance 1e-6. Never a byte compare ?
CRLF/float-format
779         safe (the repo runs with core.autocrlf=true)."""
780         out: List[str] = []
781
782         def exact(path: str, a: Any, b: Any) -> None:
783             if a != b:
784                 out.append(f"{path}: {a!r} != {b!r} (exact)")
785
786         def approx(path: str, a: Any, b: Any) -> None:
787             try:
788                 if abs(float(a) - float(b)) > _FLOAT_TOL:
789                     out.append(f"{path}: {a!r} != {b!r} (|?| > {_FLOAT_TOL})")
790             except (TypeError, ValueError):
791                 out.append(f"{path}: non-numeric float field {a!r} vs {b!r}")
792
793         exact("slug", recomputed.get("slug"), committed.get("slug"))
794         exact("schema_version", recomputed.get("schema_version"),
committed.get("schema_version"))
795         exact("gt_hash", recomputed.get("gt_hash"), committed.get("gt_hash"))
796         approx("iou", recomputed.get("iou"), committed.get("iou"))
797
798         rc, cc = recomputed.get("candidates", []), committed.get("candidates", [])
799         if len(rc) != len(cc):
800             out.append(f"candidates: count {len(rc)} != {len(cc)} (exact)")
801         else:
802             for i, (r, c) in enumerate(zip(rc, cc)):
803                 approx(f"candidates[{i}].start", r.get("start"), c.get("start"))
804                 approx(f"candidates[{i}].end", r.get("end"), c.get("end"))
805                 rs, cs = r.get("shuttle", {}), c.get("shuttle", {})
806                 if set(rs) != set(cs):
807                     out.append(f"candidates[{i}].shuttle: keys {sorted(rs)} != {sorted(cs)}
(exact)")
808             else:
809                 for k in rs:
810                     if k == "net_crossings":
811                         exact(f"candidates[{i}].shuttle.net_crossings", rs[k], cs[k])
812                     else:
813                         approx(f"candidates[{i}].shuttle.{k}", rs[k], cs[k])
814
815         rseg, cseg = recomputed.get("control_segments", []),
committed.get("control_segments", [])
816         if len(rseg) != len(cseg):
817             out.append(f"control_segments: count {len(rseg)} != {len(cseg)} (exact)")
818         else:
819             for i, (r, c) in enumerate(zip(rseg, cseg)):
820                 approx(f"control_segments[{i}][0]", r[0], c[0])
821                 approx(f"control_segments[{i}][1]", r[1], c[1])
822
823         rsc, csc = recomputed.get("score", {}), committed.get("score", {})
824         int_fields = ("num_pred", "num_gt", "true_positives", "false_positives",
"false_negatives")
825         float_fields = ("iou_threshold", "precision", "recall", "f1", "mean_iou",
826                         "mean_start_error", "mean_end_error")
827         for k in int_fields:
828             exact(f"score.{k}", rsc.get(k), csc.get(k))
829         for k in float_fields:
830             approx(f"score.{k}", rsc.get(k), csc.get(k))
831
832         return out
833
834
835         # =====

```



```

836 # Small JSON / path helpers (deterministic writers: sort_keys, LF, trailing newline)
837 # =====
838
839
840 def _write_json(path: Path, obj: Any) -> None:
841     text = json.dumps(obj, indent=2, sort_keys=True, ensure_ascii=False)
842     path.write_text(text + "\n", encoding="utf-8", newline="\n")
843
844
845 def _read_json(path: Path) -> Any:
846     with open(path, encoding="utf-8") as f:
847         return json.load(f)
848
849
850 def _resolve_dir(slug_or_dir: Any) -> Path:
851     if isinstance(slug_or_dir, Path):
852         return slug_or_dir
853     p = Path(str(slug_or_dir))
854     if p.is_dir():
855         return p
856     return FIXTURES_DIR / str(slug_or_dir)
857
858
859 def _fps_fw(prov: Optional[Dict[str, Any]], slug: str) -> Tuple[float, float]:
860     if prov and "fps" in prov and "frame_width" in prov:
861         return float(prov["fps"]), float(prov["frame_width"])
862     spec = _spec_by_slug().get(slug)
863     if spec is not None:
864         return spec.fps, spec.frame_width
865     raise ValueError(f"{slug}: no fps/frame_width in provenance and not a built-in
spec")
866
867
868 def _fps_fw_from_prov(fdir: Path, slug: str) -> Tuple[float, float]:
869     prov_path = fdir / "provenance.json"
870     prov = _read_json(prov_path) if prov_path.is_file() else None
871     return _fps_fw(prov, slug)
872
873
874 # =====
875 # CLI
876 # =====
877
878
879 def _cmd_check(fixtures_dir: Path) -> int:
880     manifest = load_manifest(fixtures_dir)
881     if not manifest:
882         print("[golden-fixtures] no fixtures found ? run --freeze-synthetic first",
file=sys.stderr)
883         return 1
884     failures = 0
885     for entry in manifest:
886         slug = entry["slug"]
887         committed = _read_json(fixtures_dir / slug / "expected.json")
888         recomputed = replay_fixture(fixtures_dir / slug)
889         mismatches = compare_expected(recomputed, committed)
890         if mismatches:
891             failures += 1
892             print(f"[FAIL] {slug}: {len(mismatches)} mismatch(es) "
f"(characterization = behaviour-locking, NOT correctness ? $6):")
893             for m in mismatches:
894                 print(f"    - {m}")
895         else:
896             sc = recomputed["score"]
897             print(f"[ok] {slug}: {len(recomputed['candidates'])} candidates, "

```

```

899         f"{len(recomputed['control_segments'])} segments, F1={sc['f1']:.6f}")
900     print(f"[golden-fixtures] {len(manifest) - failures}/{len(manifest)} fixtures match
"
901         f"(synthetic-tier = plumbing correctness, not field quality ? $6)")
902     return 0 if failures == 0 else 1
903
904
905     def main(argv: Optional[Sequence[str]] = None) -> int:
906         ap = argparse.ArgumentParser(
907             description="Golden Regression Fixtures (M1) ? synthetic, CONTROL-only replay
harness.")
908         g = ap.add_mutually_exclusive_group(required=True)
909         g.add_argument("--freeze-synthetic", action="store_true",
910             help="generate synthetic trajectory+labels for all built-in
scenarios, then "
911                 "freeze expected.json+provenance.json + write manifest.json")
912         g.add_argument("--refresh-snapshot", action="store_true",
913             help="re-freeze expected.json from COMMITTED trajectory+labels (no
regeneration)")
914         g.add_argument("--check", action="store_true",
915             help="replay all fixtures + parse-compare vs committed expected.json
(exit 0/1)")
916         g.add_argument("--freeze-real", action="store_true",
917             help="(P1) de-attribute an OWNED real trajectory+labels into a
committable Tier-0 "
918                 "fixture; REFUSED without --owned --minors-free and a non-blank
--license")
919         ap.add_argument("--slug", help="single slug for --refresh-snapshot, or the surrogate
slug for "
920                 "--freeze-real (default: a random fx<hex> id)")
921         ap.add_argument("--all", action="store_true", help="all slugs for --refresh-
snapshot")
922         ap.add_argument("--reason", help="recorded reason for --refresh-snapshot (review the
diff!)")
923         # --freeze-real (P1 / M2 refusal gate) options:
924         ap.add_argument("--trajectory", help="--freeze-real) source trajectory CSV
(Frame,Visibility,X,Y)")
925         ap.add_argument("--labels", help="--freeze-real) source golden labels CSV
(start,end)")
926         ap.add_argument("--license", dest="license_", default=None,
927             help="--freeze-real) the source license tag (required, non-blank)")
928         ap.add_argument("--owned", action="store_true",
929             help="--freeze-real) assert owner-recorded Fork-A source
(required)")
930         ap.add_argument("--minors-free", dest="minors_free", action="store_true",
931             help="--freeze-real) assert the source contains NO minors
(required)")
932         ap.add_argument("--fps", type=float, help="--freeze-real) source fps")
933         ap.add_argument("--frame-width", dest="frame_width", type=float, help="--freeze-
real) source frame width (px)")
934         ap.add_argument("--label", default="cleared_real", help="--freeze-real) non-
identifying scenario label")
935         ap.add_argument("--source-note", dest="source_note", default="",
936             help="--freeze-real) non-identifying note (NEVER a path/name/id)")
937         ap.add_argument("--no-time-origin-reset", dest="time_origin_reset",
action="store_false",
938             help="--freeze-real) DISABLE the anti-attribution time-origin reset
(default ON)")
939         ap.set_defaults(time_origin_reset=True)
940         ap.add_argument("--fixtures-dir", type=Path, default=FIXTURES_DIR,
941             help="override the fixtures directory (default: repo
eval_baselines/fixtures)")
942         args = ap.parse_args(list(argv) if argv is not None else None)
943
944         fixtures_dir: Path = args.fixtures_dir

```

```

945
946     if args.freeze_synthetic:
947         slugs = freeze_synthetic(fixtures_dir)
948         print(f"[golden-fixtures] froze {len(slugs)} synthetic fixture(s): {'',
'.join(slugs)}")
949         print(f"  -> {fixtures_dir}")
950         return 0
951
952     if args.refresh_snapshot:
953         if not args.slug and not args.all:
954             ap.error("--refresh-snapshot requires --slug S or --all")
955         if not args.reason:
956             print("[golden-fixtures] WARNING: no --reason given. --refresh-snapshot can
"
957                 "rubber-stamp a real regression ? review the visible diff ($6).",
file=sys.stderr)
958         targets = refresh_snapshot(slug=args.slug, all_slugs=args.all,
fixtures_dir=fixtures_dir)
959         print(f"[golden-fixtures] refreshed snapshot for: {'', '.join(targets)}"
+ (f" (reason: {args.reason})" if args.reason else ""))
960         return 0
961
962     if args.freeze_real:
963         missing = [n for n, v in (("--trajectory", args.trajectory),
964                                 ("--labels", args.labels),
965                                 ("--fps", args.fps),
966                                 ("--frame-width", args.frame_width)) if v is None]
967         if missing:
968             ap.error("--freeze-real requires " + ", ".join(missing))
969         try:
970             expected = freeze_real_fixture(
971                 args.label, args.trajectory, args.labels,
972                 license=args.license_ if args.license_ is not None else "",
973                 minors_free=args.minors_free, owned=args.owned,
974                 fps=args.fps, frame_width=args.frame_width,
975                 fixtures_dir=fixtures_dir, slug=args.slug,
976                 time_origin_reset=args.time_origin_reset, source_note=args.source_note)
977         except RefusalError as e:
978             # The code-enforced provenance gate ($4c / $7 ?): refuse loudly, exit non-
979             zero.
980             print(f"[golden-fixtures] REFUSED (--freeze-real provenance gate):\n{e}",
file=sys.stderr)
981             return 2
982             print(f"[golden-fixtures] froze de-attributed REAL fixture '{expected['slug']}'
"
983                 f"(kind=cleared_real) -> {fixtures_dir / expected['slug']}")
984             print(" $6: behaviour-locking, NOT correctness; non-attributable only relative
to the "
985                 "UNPUBLISHED source ? never 'anonymous'. Sits behind counsel sign-off
(P2).")
986             return 0
987
988     # --check
989     return _cmd_check(fixtures_dir)
990
991
992 if __name__ == "__main__":
993     raise SystemExit(main())
994

```

6. user (2026-06-16T06:27:11.631Z)

```

1      # Golden Regression Fixtures ? video-free, non-attributable regression suite
2
3      > **Status:** `IN PROGRESS` ? owner-gated . **Owner:** Avi . **Created:** 2026-06-16 .

```

****Last updated:**** 2026-06-16 . ****Legal:**** owner ****accepted the researched risk**** 2026-06-16
(conditions §3 stand; this is the owner's informed acceptance, ****not**** a retained-counsel opinion)

4 > ****Lifecycle:**** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE, per archive policy)

5 > ****Tracking anchors:**** §7 progress tracker is the source of truth; indexed in `docs/README.md`; pointer in memory `current-state` (and `docs/NEXT\_STEPS.md` once work starts).

6 > ****Relation to existing docs:**** EXTENDS (does not supersede) `docs/GOLDEN\_DATA\_SHARING.md` (#175); reuses the eval stack documented in `docs/EVALUATION.md` / `docs/EXPERIMENT\_HARNESS.md`; quality floor ties to `docs/QUALITY\_ITERATIONS.md`.

7 > ****Honesty note:**** §3 (Legal) is ****researched general information, NOT legal advice**** ? it gates on retained counsel (§8). §2/§4/§7 mark which claims are ****[verified]**** against code vs ****[design]**** proposals.

8

9 ---

10

11 ## 0. TL;DR

12

13 Let a contributor on ****another machine with no video, GPU, WSL, or DB**** run the rally-segmentation regression suite from ****committed structured files****, so refactors and non-quality PRs get a red/green side-effect signal. Two layers, built once and grown by accretion:

14

15 1. ****Mechanics**** ? freeze the deterministic pipeline at the ****post-detector trajectory seam**** and replay it through the ***already-shipped*** eval stack (`experiment.py` / `metrics.py` / `windowing.py` / `regression.py`), with two auto-discovered gates: a ****characterization/snapshot**** gate (catches any behaviour change) and a ****quality-floor**** gate (catches quality drops).

16 2. ****Privacy/legal filter (SEALED-FIXTURES)**** ? only ****de-attributed, minimized, owner-source, biometric-free**** numbers are ever committed; richer/sensitive streams stay key-gated or out-of-repo. Non-attributability comes from ****deletion-at-source****, not encryption.

17

18 ****The convergence that makes this clean:**** the ***privacy*** answer (publish shuttle-only, drop person/pose streams) and the ***determinism*** answer (freeze only the pure-stdlib `CONTROL` path, keep the numpy/BLAS learned scorer off the cross-machine gate) point at the ****same minimal public tier****. That coincidence is the spine of the design.

19

20 ****Two findings already actionable, independent of the rest:****

21 - ? ****Live de-attribution bug**** ***(verified)***: real player/venue names are committed today in `eval\_baselines/heuristic\_lovo\_n6.json`, `eval\_baselines/gemini\_wrapper\_2026-06-10.json`, and `backend/eval/eval\_partial\_labels.py` (the last also has a personal `C:/Users/avidu/...` path) ? and in git history. ? ****Deliverable P0.****

22 - ? ****Determinism trap**** ***(verified, corroborated by two independent red-teams)***: the learned variants (`shuttle\_only`, `fusion`) train a numpy/OpenBLAS `DYNAMIC\_ARCH` logistic regression whose decisions flip near the 0.5 threshold across CPUs ? would false-RED on a contributor's machine. ? the cross-machine gate must be ****CONTROL/static-only****.

23

24 ---

25

26 ## 1. Problem & goal

27

28 The rally-quality measurement is split across two machines by capability (`docs/GOLDEN\_DATA\_SHARING.md`):

29 - `backend/eval/fusion\_golden.py` ****extracts**** `\*\_golden\_features.json` ? needs the ****video + trajectory CSV + labels + GPU/WSL**** (GPU box only).

30 - `backend/eval/fusion\_compare.py` / `backend/eval/ablation.py` ****re-score**** those JSONs on ****pure CPU, no video**** ? any machine.

31

32 The CPU re-scoring is ***already*** video-free, but the inputs live in gitignored `output/` / private OneDrive, so the regression tests that depend on them ****skip everywhere**** (`tests/test\_ablation.py::test\_litmus\_reproduces\_gate\_a`, `tests/test\_fusion\_compare.py`). The committed `eval\_baselines/` baselines exist but `regression.py` still scores DB-stored predictions (needs a pipeline run on the video).

33

34 ****Goal:**** commit a tiny, version-tagged, ****numeric-only, non-attributable**** per-fixture corpus + auto-discovered gates so a clean `git clone` / CI gets a real red/green regression

signal with ****zero video/GPU/DB****, while never leaking attributable or sensitive data.

35

36 ---

37

38 **## 2. Decisions locked**

39

	#	Decision	Source	Implication
40	#	Decision	Source	Implication
41	--- ----- ----- -----			
42	D1	**Corpus is owner-recorded / owned**	owner answer 2026-06-16	Legal analysis takes **Fork A** (clean): no third-party copyright/ToS infringement; concern is trade-secret/moat + privacy. Strongest non-attributability (no public reference to correlate against).
43	D2	**Repo private now, may open-source later**	owner answer 2026-06-16	Design for **public-grade hygiene from day one** ? git history is permanent; nothing scrubable-later.
44	D3	**Adults, consent not formalized**	owner answer 2026-06-16	**Person/pose/gait streams stay OUT of the public tier** (DPDP/GDPR derived-personal-data risk without consent). Public tier = **shuttle + abstract intervals + non-biometric features only** .
45	D4	**Freeze seam = post-detector trajectory**	*(design)*	workflow-1 synthesis Widest deterministic surface; reuse the shipped eval stack verbatim; the raw per-frame CSV is **never committed** (only ~5 aggregate, fps/res-invariant shuttle scalars/window).
46	D5	**Cross-machine gate is CONTROL/static-only**	*(design)*	both red-teams [verified cause] Learned variants (numpy/BLAS) are **owner-box/Tier-1 only** , asserted with tolerance, never as a cross-machine byte-snapshot.
47	D6	**Non-attributability = deletion-at-source, encryption = public-only barrier**	*(design)*	workflow-2 synthesis A contributor who runs the tests necessarily reads plaintext+key (DRM/analog-hole). So we **delete** attribution, not hide it. Encryption only stops the no-key public.
48	D7	**Two tiers**	*(design)*	workflow-2 synthesis **Tier-0** public/no-key (shuttle-only, de-attributed). **Tier-1** key-gated/out-of-repo (person/audio, full corpus), **skips cleanly when absent** .

49

50 ---

51

52 **## 3. Legal foundation (researched ? gates on counsel, §8)**

53

54 ****Core verdict (settled across IN/US/EU):**** the derived numbers ? per-frame shuttle (x,y), rally start/end intervals, audio/court/aggregate-person feature values ? ****do NOT inherit the source video's copyright.**** They are uncopyrightable ***facts/measurements***, not authored expression, and not a §101 "derivative work."

55

	Jurisdiction	Copyright of the data	Database right	Contract / ToS	Net
56	Jurisdiction	Copyright of the data	Database right	Contract / ToS	Net
57	--- --- --- ---				
58	**India**	(primary) Not copyrightable as facts (*EBC v. D.B. Modak*); but compilations get thin protection on a low "skill/labour" bar (*Burlington* , *OLX v. Padawan*) ? only bites if **lifted from a 3rd-party dataset** , not self-derived **None** (no sui-generis right) ? a real permissiveness edge Anti-scraping ToS (untested in court) + **IT Act s.43/s.66** civil/criminal exposure for 3rd-party ingestion **Owned-source + minors-excluded + no person/pose = clean**			
59	**US**	Not copyrightable (*Feist* ; §102(b); *NBA v. Motorola*) None (*Feist* killed sweat-of-the-brow) **Dominant risk.** ToS enforceable where copyright/CFAA fail (*ProCD* , *hiQ* ? injunction + **forced deletion**). Remedy for redistribution is takedown, not nominal damages Publishable as facts **iff** owned/permissioned source + no barring ToS + no person/biometric/minor stream			
60	**EU / UK**	Not copyrightable (*Football Dataco v. Yahoo*) **Genuinely contestable.** *Football Dataco v. Sportradar* : recording pre-existing on-court facts = **obtaining** , not "creating" ? our "we created it" premise is **overconfident** ; harm test = *CV-Online Latvia* Ryanair v. PR Aviation: ToS can ban extraction/redistribution even with no IP right Get EU+UK advice before redistributing numeric streams at scale			

61

62 ****Provenance is the decisive fork.**** ****Fork A** (owner-recorded, unpublished) ? the only path safe to commit publicly; collapses to a trade-secret ***choice*** (Indian breach-of-confidence / Contract Act 1872 / IT Act s.72 ? ****not**** US DTSA) + privacy. ****Fork B** (broadcast/YouTube/scraped) ? do ****not**** commit; the extraction act needs a license or now-****contested**** US fair use (***Thomson Reuters v. Ross*** 2025; USCO May-2025), and unlawful acquisition independently sinks it (***Bartz v. Anthropic***, ~\$1.5B settlement). Caveat: owner

footage **shot at a third party's event** (tournament/club) carries ticket/venue/organiser/broadcaster terms ? treat as Fork B until cleared.

63

64 **Publish-by-stream (privacy):**

65 - ? **Publish-OK:** shuttle (x,y)+Visibility (inanimate object), abstract rally/GT

66 **intervals** (identity-stripped), non-biometric audio/court/aggregate-person features detached from identity.

67 - ?? **Conditional (effectively never in v1):** person bounding boxes that *persistently single out a player* ? flips to GDPR Art.9 the moment it identifies. This architecture is prone to that flip ? treat per-player tracking as a tripwire.

68 - ? **Never publish:** pose/skeleton/**gait** (re-identifies at 80?95% even with faces blurred ? *Weiss et al. 2025*), face crops, any named-player?movement mapping. Matches the existing "gait/biometrics strictly future (GDPR Art.9)" guardrail.

69 - ? **Minors flip everything:** DPDP "child" = under 18; s.9(3) **prohibits behavioural monitoring** (the Rule-12 exemptions don't cover this project). A persistent per-player signature on a minor is, conservatively, prohibited. **Exclude minors** from any persisted stream beyond inanimate shuttle/interval data.

70 **Non-attributability is relative, not absolute.** A per-video trajectory is a fingerprint (de Montjoye: 4 points ? 95% re-ID). It's only non-attributable because the **Fork-A source stays unpublished** (no public reference to correlate against). Per *EDPS v. SRB* (CJEU C-413/23 P, 2025) it **remains personal data to the holder** who can re-run the detector. So: conditional, revocable (publish the source and the link snaps back), forward-looking (re-assess as the public-video environment grows). Never market it as "anonymous."

71

72 *Key authorities:* Feist 499 U.S. 340; NBA v. Motorola 105 F.3d 841; 17 U.S.C. §101/§102(b); Sega v. Accolade 977 F.2d 1510; Football Dataco v. Yahoo (C-604/10) & v. Sportradar (C-173/11); BHB v. William Hill (C-203/02); CV-Online Latvia (C-762/19); Ryanair v. PR Aviation (C-30/14); Thomson Reuters v. Ross (D. Del. 2025); EBC v. D.B. Modak; DPDP Act 2023 s.3/s.9 + Rules 2025; EDPS v. SRB (C-413/23 P). Full URLs in the research artifact (§10).

73

74 ---

75

76 ## 4. Architecture (design)

77

78 ### 4a. Freeze seam + reuse map

79 Freeze at the **trajectory CSV ? pure-Python replay**. One shared helper `backend/eval/golden\_fixtures.py::replay\_fixture()` is the *single* code path used by both the freeze tool and both gates:

80 `parse\_trajectory\_csv ? windowing.build\_windows(preset) ? distill\_local.build\_candidates ? rally\_gate net\_crossings ? experiment.predict(CONTROL) ? merge\_overlaps ? metrics.score`.

81 Reuses **verbatim**: `metrics.score/interval\_iou`, `segmentation\_metrics`, `experiment.predict/compare/Variant.spec\_hash`, `windowing.WindowingPreset` (default SERVED), `regression.gt\_hash/save\_baseline + incommensurability guards`, `eval\_stats` (seed=0), `eval\_baselines/` committed-baseline precedent.

82

83 ### 4b. Tiered model (SEALED-FIXTURES)

84 | Tier | Where | Key? | Contents | Test behaviour |

85 |---|---|---|---|---|

86 | **Tier-0** | `eval\_baselines/fixtures/` (tracked) | none | opaque `fixture\_id`, fps/frame_width (quantized), relative `start`/`end`, **5 shuttle scalars**, label, relative `gts`. **No person/audio/pose.** | **Unconditional** ? runs on every PR/fork; turns today's skipped litmus into a real public gate |

87 | **Tier-1** | OneDrive (default) **or** SOPS-encrypted in-repo | token/age key | full corpus **with** person/audio; authoritative ?F1/CI/sign-test + exact Gate-A litmus | **Skips cleanly** when key/data absent (generalize `\_golden\_present()` ? `\_tier1\_present()`) |

88

89 A fresh no-key clone is always **green**; coverage never depends on a secret.

90

91 ### 4c. Anti-attribution measures (Tier-0)

92 - **Opaque RANDOM surrogate id** (`os.urandom` + a private surrogate?source map kept off-git) ? *not* the MD5 `video\_id`, and *not* HMAC(salt, video_id?name) (red-team: brute-forceable over a tiny known roster if the salt leaks). Strip the `name` field (today = source filename).

93 - **Metadata strip** ? no filename, video_id, match/event/player identity, capture date.

```

94     - **Time-origin reset** ? subtract per-fixture `t0` from all start/end/gts. **Provably
metric-invariant** (`metrics.interval_iou` and start/end-error are pure differences) ? bit-
identical scores, severed absolute timeline.
95     - **Stream minimization** ? Tier-0 keeps shuttle block only; person/audio excluded;
pose/gait/face have **no field in the schema by design**.
96     - **Raw-trajectory exclusion** ? never commit the per-frame `(Frame,Visibility,X,Y)`
CSV.
97     - **Quantize fps/frame_width** ? red-team: at n=6 the raw values are a camera/venue
quasi-identifier.
98     - **Provenance hard-fail gate** ? the de-attribution pass refuses to emit a public
fixture for any record not tagged owner-recorded(Fork-A) + minors-excluded + biometric-excluded.
99
100    ### 4d. The two gates
101    - **Characterization/snapshot** ? replay fixture, **parse-compare** JSON (never byte-
compare ? CRLF/float-format safe; `core.autocrlf=true` here), exact on int/structural fields,
`approx(abs=1e-6)` on floats; stamp + check windowing-preset/label/gt fingerprints
(incommensurable ? loud fail). Catches any behaviour change even at constant F1. **CONTROL path
only** across machines.
102    - **Quality-floor** ? score vs committed labels; assert per-video + mean F1 ? floor in
`eval_baselines/regression_baseline.json` (created here; carries a **windowing fingerprint** so
a served-default change is flagged incommensurable, not silently compared); emit `eval_stats`
paired-delta-CI + sign test (NUMBERS INFORM); add a **metrics-vs-base-branch** paired delta.
Synthetic-tier floors measure **correctness, not field quality** (documented).
103
104    ### 4e. Serialization & optional protobuf
105    JSON with `sort_keys=True` + pre-quantized floats + LF is sufficient. **Protobuf is
optional (01)** ? adopt only if cross-OS byte-stable snapshots are wanted (proto3
`rally.fixture.v1`, codegen + `protoc && git diff --exit-code` guard, `*.pb binary -text`).
Protobuf is **encoding, not secrecy** (trivially `--decode_raw`).
106
107    ### 4f. Encryption & keys (Tier-1 only)
108    **Default = out-of-repo + token** (existing OneDrive `RALLY_GOLDEN_DIR` seam ? zero
sensitive bytes in git history, sidesteps the legal "redistribution" act). **Alternative = SOPS
+ age** (per-recipient, revocable, value-level diffs) for Fork-A-clean-but-private data wanting
in-repo versioning. Rejected: git-crypt (whole-repo re-key), git-lfs (storage ? secrecy). CI
secret exposed **only to the Tier-1 job on trusted branches, never to fork-PR runners**. The
**HMAC/surrogate salt** (if used) and the surrogate?source map live with the private corpus,
rotated per release. Document explicitly that Tier-1 crypto defends **only** the no-key public.
109
110    ---
111
112    ## 5. Threat model (who's protected vs not)
113
114    | Actor | Protected against | NOT protected (by design / residual) |
115    |---|---|---|
116    | **TM1 no-key public** | map fixture?source/match/player; decrypt Tier-1; confirm a
source even holding the candidate file | read Tier-0 numbers + run shuttle-only harness
(intended; useless-in-isolation) |
117    | **TM2 keyed contributor** | recover filename/video_id/match/face/pose/gait ? *never
written* (survives the analog hole) | read every value the suite reads (unpreventable; not
pretended otherwise) |
118    | **TM3 CI runner** | leaked log can't leak attribution it never had | runs the suite
(intended); Tier-1 secret never on fork runners |
119    | **TM4 malicious keyed insider** | correlation needs a public reference; Fork-A source
is unpublished ? "not reasonably likely" | **residual:** conditional + revocable ? publish/leak
the source later and the link can snap back (forward-looking re-test duty) |
120    | **TM5 re-ID adversary** | source unpublished + only ~5 invariant scalars/window +
reset timeline ? no back-match | becomes live if a Fork-A source is ever published |
121    | **TM6 accidental plaintext commit** | required CI leak-scan rejects MD5/known-
name/non-surrogate/plaintext-SOPS | if the required check is disabled, a leak lands |
122
123    ---
124
125    ## 6. Honest limits ? what a green suite does NOT prove
126

```

127 - Green proves only that the ****deterministic post-trajectory transforms**** are unchanged for the ****frozen cases****. It does ****not**** cover: the WASB/TrackNet detector, `\_probe\_fps\_width`/cv2 frame-seek, the AI/provider handoff, chunking/re-encode, the ffmpeg stitch, DB persistence/migrations, or the API filter. *A refactor that breaks decoding or the detector passes these gates green.* The owner-box `regression.py` DB run remains the only full-chain check.

128 - Characterization = ****behaviour-locking, not correctness**** ? it freezes current behaviour ***including current bugs***. "Passing" = "behaviour unchanged", never "behaviour correct". This caveat must ride in the ****test pass/fail message****, not only the docs.

129 - ****Synthetic-tier floors = plumbing correctness, NOT field quality.**** Never cite synthetic F1 in `QUALITY\_ITERATIONS.md` / the paper as rally-detection performance (tag `kind: synthetic`).

130 - Tier-0 public coverage is ****shuttle-only****; the person/audio fusion ?F1 and exact Gate-A values need Tier-1 (skipped on no-key clones).

131 - Data is ****not "anonymous"***** absolutely (EDPS v. SRB) ? relative, conditional, revocable, forward-looking.

132 - Encryption gives ****no secrecy from a running contributor****; protobuf gives ****no secrecy at all****.

133 - De-attribution does ****not**** cure provenance/licensing ? it sits ***behind*** the per-record provenance gate + counsel sign-off.

134 - `--refresh-snapshot` can rubber-stamp a real regression; mitigation is ****review of the visible diff**** + a separate, deliberate floor-baseline update. Require a recorded reason.

135

136 ---

137

138 ## 7. Deliverables & progress tracker ? ****source of truth****

139

140 Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. One ****small PR per row****, branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.

141

142 | ID | Deliverable | Depends on | Counsel-gated? | Status | PR |

143 |----|-----|-----|-----|-----|----|

144 | ****P0**** | ****Leak remediation + required CI leak-scan.**** Scrub real names/paths in `eval\_baselines/heuristic\_lovo\_n6.json`, `gemini\_wrapper\_2026-06-10.json`, `backend/eval/eval\_partial\_labels.py`, `tests/test\_cleanup\_caches.py` ? opaque surrogate keys + private map; add CI leak-scan (MD5 / known-name / non-surrogate / plaintext-SOPS) as a ****required status check****; decide on git history (PR #77, accept vs `filter-repo`). | ? | No (owned data hygiene) | ? | ? |

145 | ****M1**** | ****Fixture framework + synthetic Tier-0 + shared replay helper**** (also delivered M2-core + M3 ? see below). `backend/eval/golden\_fixtures.py` (`replay\_fixture`, schema dispatch, `Path(\_\_file\_\_)`-relative) + 3 ****tool-generated**** synthetic fixtures + manifest + README + `.gitattributes` LF pin + `tests/test\_golden\_fixtures.py`. CONTROL-only / pure-stdlib; cross-OS verified. | ? | No (synthetic) | ? | [#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) |

146 | ****M2**** | ****Freeze/refresh extractor.**** Core (`freeze\_synthetic` / `refresh\_snapshot` / `--check` CLI + idempotence test) shipped in M1 (#189); the `--license`/`--minors-free`/`--owned` ****refusal gate**** shipped with P1 (this PR). | M1 | No | ? | [#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) · **_this PR_** |

147 | ****M3**** | ****Characterization gate**** ? `tests/test\_golden\_fixtures.py`: parse-compared (exact-int / approx-float 1e-6), ****CONTROL-only****, positive no-person/audio assertion, perturbation + idempotence. ****Shipped in M1 (#189).**** (Windowing/label fingerprint guard + row-count cap ? fold into M4.) | M1 | No | ? | [#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) |

148 | ****MX**** | ****Cross-OS determinism guard (CI).**** Lightweight `golden-crossos` CI job (ubuntu/windows/macos) running `--check` + the fixtures test ? locks the cross-OS guarantee (CONTROL replay is numpy-only, no GPU stack; a ***separate fast job***, not full-suitex3). | M1 | No | ? | [#191](https://github.com/avidullu/badminton-highlight-indexer/pull/191) |

149 | ****M4**** | ****Quality-floor gate + first `regression\_baseline.json`.**** `tests/test\_golden\_quality\_floor.py`; create `eval\_baselines/regression\_baseline.json` with ****windowing fingerprint****; metrics-vs-base paired-delta; corpus-level (not per-slug) LOVO for runtime. | M1, M2 | No | ? | ? |

150 | ****P1**** | ****De-attribution + minimization**** ? `golden\_fixtures.freeze\_real\_fixture` (extends the M1 module, not a separate script): ****opaque random `fx\_hex` surrogate slug****, metadata strip, metric-invariant ****time-origin reset****, strict GT loader, ****positive no-person/audio/MD5/source-path assertion****, manifest non-clobber; `--freeze-real` CLI with the


```

**refusal gate** (exit 2 unless owned + minors-free + license). Tooling-only, 12 synthetic
tests; **no real data committed**. | M1 | No (tooling) | ? | _this PR_ |
151 | **P2** | **Owner-accepted (researched-risk), 2026-06-16.** Owner accepted the
researched IP/privacy analysis (§3) to proceed **under its conditions** (owned/Fork-A source ·
minors excluded · biometric/person streams excluded · de-attributed). Owner's informed risk
acceptance ? **NOT a retained-counsel opinion**;; optional future counsel confirmation per §8. |
P1 | Owner-accepted | ? | ? |
152 | **P3** | **Tier-0 REAL subset.** Commit de-attributed, minors/biometric-excluded,
owned Fork-A real fixtures (unblocked by P2). Hard-gated **in code** by P1's refusal gate (owned
+ minors-free + license required) ? needs the owner's per-video clearance list. | P1, P2 |
conditions §3 | ? | ? |
153 | **O1** | *(optional)* **Protobuf + deterministic serialization.** `rally.fixture.v1`
proto3, codegen + `protoc`/`git diff --exit-code` guard, canonical writer, `.gitattributes`
binary. Only if cross-OS byte-stable snapshots are wanted. | M1 | No | ? | ? |
154 | **O2** | *(optional)* **Tier-1 key-gated path.** `scripts/fetch_golden.py` (OneDrive
token, default) and/or SOPS+age; `_tier1_present()`; `RALLY_GOLDEN_DIR` wired; Tier-1 CI job
gated on secret, never on fork runners. | M1 | No | ? | ? |
155 | **DOC** | **Finalize this doc + cross-links.** Add `eval_baselines/fixtures/` row to
`CODE_MAP §0` + TEST MAP; cross-link `EVALUATION.md` / `EXPERIMENT_HARNESS.md` /
`GOLDEN_DATA_SHARING.md` / `QUALITY_ITERATIONS.md`; update `docs/README.md` index + memory
`current-state`/handoff. | M1..M4 | No | ? | ? |
156
157 **Must-fix-before-any-public-commit (red-team), folded into the rows above:** ? numeric-
tolerance not byte-equality + multi-OS matrix + CONTROL-only (M3/D5/MX ?); ? scrub names in
**all** files + history decision (P0); ? leak-scan a **required** check, hooks are convenience
only (P0); ? person-optional loader + **positive** no-person assertion, don't trust the silent
`0.0` default (P1 ?); ? random surrogate id not HMAC-over-roster (P1 ?); ? quantize/drop
fps/frame_width (P1 ? fps/fw retained as scale constants; quantize is a P3 option); ? document
Tier-0 omits the fusion path (DOC); ? counsel + Fork-A + minors/biometric exclusion as hard
preconditions (P2 owner-accepted; enforced in code by P1's refusal gate ?).
158
159 ### Progress log (quantitative ? coverage must hold/improve, CI incl. the 3-OS cross-OS
job green)
160 | PR | Deliverable(s) | Suite | Coverage | Notes |
161 |---|-----|-----|-----|-----|
162 | [#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) | M1 +
M2-core + M3 | 746 pass / 7 skip | 80.41% | synthetic Tier-0 + CONTROL replay +
characterization; cross-OS proven (CI Linux replayed Windows-frozen fixtures) |
163 | [#191](https://github.com/avidullu/badminton-highlight-indexer/pull/191) | MX +
tracker/legal reconcile | full suite green | 80.41% | `golden-crossos` CI job green on
ubuntu+windows+macos ? cross-OS **enforced**;; legal ? owner-accepted |
164 | _this PR_ | P1 + M2 refusal gate | 759 pass / 7 skip | **80.49%** (+0.08) |
`freeze_real_fixture` + refusal gate; **+13 tests**;; tooling-only, no real data committed |
165
166 ---
167
168 ## 8. Open owner / counsel questions
169
170 **Counsel questions ? documented basis (the owner accepted the researched risk on
2026-06-16; see P2). These remain the record, and a retained-counsel confirmation is
optional/recommended before scaling beyond a small owned, minors-free, de-attributed subset:**
171 1. Does the EU/UK **sui-generis database right** subsist in our self-derived
shuttle/rally streams on the *Sportradar* "obtained vs created" reasoning, and does
redistributing a de-attributed Tier-0 subset cause *CV-Online* harm?
172 2. Post *Thomson Reuters v. Ross* + USCO-2025, is **US fair use for the extraction
step** defensible if the dataset could compete in a "data/analytics" market ? does publishing
only numbers (never footage) matter?
173 3. India: are anti-scraping ToS enforceable; **IT Act s.43/s.66** exposure for any 3rd-
party ingestion; does pending *ANI v. OpenAI* (~Apr 2026) move it? Confirm **DPDP s.9/Rule 10**
and whether per-player tracking is prohibited s.9(3) monitoring of a minor.
174 4. Sign-off on the **final committed Tier-0 subset + provenance log**;; is non-
attributability-via-private-source a defensible posture to assert given its
conditional/revocable nature?
175
176 **Owner (strategic):**

```

```

177     5. Publishing the Fork-A subset **forfeits the trade-secret moat** (Indian breach-of-
confidence framing) ? weighed against community/benchmark/paper aims. Your call.
178     6. **Forward-looking commitment:** do not publish a highlight reel of the *same* source
whose fixtures are committed (without re-clearing), and accept a periodic re-assessment as the
public-video environment grows.
179     7. **Scope of first public commit:** synthetic-only (zero provenance risk, recommended)
vs a counsel-cleared real subset.
180     8. **Tier-1 mechanism:** out-of-repo+token (recommended) vs SOPS+age in-repo. And: is
protobuf (01) wanted, or is canonical-JSON enough?
181
182     ---
183
184     ## 9. Definition of done
185
186     The project is **DONE** (flip `Status ? DONE`, archive) when:
187     - [ ] **P0, M1?M4, P1, DOC** are ? and merged.
188     - [ ] A clean `git clone` on **win + ubuntu + mac** runs `python run_all_tests.py` and
the **Tier-0 gates pass green with no key / no video / no GPU / no DB**.
189     - [ ] The **CI leak-scan is a required status check** and passes (no MD5 / real-name /
non-surrogate / plaintext-SOPS in the tree).
190     - [ ] Characterization + quality-floor gates are **parse-compared, CONTROL-only,
windowing-fingerprinted**, and demonstrably catch a seeded behaviour change.
191     - [ ] `docs/CODE_MAP.md $0` + TEST MAP updated; this doc cross-linked; `NEXT_STEPS.md` +
memory `current-state` point here.
192     - [ ] **Either** the public commit is synthetic-only, **or** (**P2 owner-accepted ? +
P3**) for a real-data commit, under the S3 conditions enforced **in code** by P1's refusal gate.
193     - [ ] $6 honest-limits caveat rides in the test pass/fail messages, not only the docs.
194
195     *Optional 01/02 may remain ? without blocking DONE (they're earned, not assumed).*
196
197     ---
198
199     ## 10. References
200
201     **Internal code anchors (verified):** `backend/eval/{regression.py,fusion_golden.py,fusi
on_compare.py,ablation.py,experiment.py,metrics.py>windowing.py,distill_local.py,gt_loader.py,ev
al_stats.py}` · `backend/pipeline/detectors/{base.py,tracknet_runner.py,rally_gate.py}` ·
`backend/utils/hashing.py::compute_video_id` · `eval_baselines/{heuristic_lovo_n6.json,gemini_wr
apper_2026-06-10.json,experiment_leaderboard.json}` ·
`tests/test_ablation.py::test_litmus_reproduces_gate_a` · `.github/workflows/ci.yml`.
202     **Internal docs:** `GOLDEN_DATA_SHARING.md` (#175), `EVALUATION.md`,
`EXPERIMENT_HARNESS.md`, `QUALITY_ITERATIONS.md`, `MULTI_SIGNAL_FUSION_PLAN.md`, `CODE_MAP.md
$0`.
203     **Legal authorities:** see S3 (full URLs in the research workflow artifact). *This
section is researched general information, not legal advice.*
204
205     ---
206
207     ### Changelog
208     - **2026-06-16** ? Initial draft synthesized from two multi-agent workflows (mechanics
design + 2 critiques; legal memo IN/US/EU + SEALED-FIXTURES architecture + red-team). Verified
the live name-leak and the BLAS-determinism cause against code. Status: DRAFT, owner-gated.
209

```

7. assistant (2026-06-16T06:27:14.448Z)

Now let me read the test files and the supporting eval modules.

8. user (2026-06-16T06:27:15.571Z)

```

1     """Golden Regression Fixtures (M1) ? characterization tests over the committed synthetic
Tier-0.
2
3     Parametrized over ``load_manifest()`` so every committed fixture is exercised. The gate
is

```

```

4      **parse-compare** (never byte-compare), **CONTROL-only** (pure stdlib, no numpy on the
replay
5      path), and asserts the Tier-0 privacy invariant positively (no
person/audio/pose/gait/face keys).
6
7      $6 honest-limits caveat: green here proves only that the deterministic post-trajectory
transforms
8      are unchanged for the frozen synthetic cases ? characterization is behaviour-locking,
NOT
9      correctness, and synthetic-tier numbers measure plumbing, not field quality.
10     """
11     import json
12
13     import pytest
14
15     from backend.eval import golden_fixtures as gf
16
17     # Slugs discovered from the committed manifest ? the suite grows by accretion
automatically.
18     MANIFEST = gf.load_manifest()
19     SLUGS = [entry["slug"] for entry in MANIFEST]
20
21
22     def test_manifest_is_non_empty():
23         """A committed manifest with at least one fixture must exist (else nothing is
gated)."""
24         assert SLUGS, "no fixtures in eval_baselines/fixtures/manifest.json ? run --freeze-
synthetic"
25
26
27     @pytest.mark.parametrize("slug", SLUGS)
28     def test_replay_matches_committed_expected(slug):
29         """Replay the fixture and parse-compare against its committed expected.json.
30
31         Characterization = behaviour-locking, NOT correctness ($6): a non-empty mismatch
list means
32         the deterministic post-trajectory behaviour CHANGED for this frozen case, not that
it is wrong.
33         """
34         committed = gf.load_fixture(slug)["expected"]
35         recomputed = gf.replay_fixture(slug)
36         mismatches = gf.compare_expected(recomputed, committed)
37         assert mismatches == [], (
38             f"{slug}: behaviour changed vs frozen snapshot (characterization, not
correctness ? $6):\n"
39             + "\n".join(f"  - {m}" for m in mismatches)
40         )
41
42
43     @pytest.mark.parametrize("slug", SLUGS)
44     def test_schema_version_le_current(slug):
45         """Every committed fixture's schema_version must be <= CURRENT_SCHEMA (forward-
compatible)."""
46         fx = gf.load_fixture(slug)
47         assert int(fx["expected"]["schema_version"]) <= gf.CURRENT_SCHEMA
48         assert int(fx["provenance"]["schema_version"]) <= gf.CURRENT_SCHEMA
49
50
51     @pytest.mark.parametrize("slug", SLUGS)
52     def test_no_person_or_audio_keys(slug):
53         """Positive privacy assertion (D3 / $4c): NO biometric/identity/non-shuttle key
appears
54         anywhere in the committed expected.json ? not trusting a silent 0.0 default."""
55         fdir = gf.FIXTURES_DIR / slug
56         raw = (fdir / "expected.json").read_text(encoding="utf-8").lower()

```

```

57         for forbidden in gf.FORBIDDEN_KEYS:
58             assert forbidden not in raw, (
59                 f"{slug}: forbidden key substring {forbidden!r} present in expected.json
(Tier-0 leak)")
60
61         expected = gf.load_fixture(slug)["expected"]
62         for i, cand in enumerate(expected["candidates"]):
63             # The only feature block is the shuttle block, and it holds EXACTLY the 5
shuttle cols.
64             assert set(cand.keys()) == {"start", "end", "shuttle"}, f"{slug}: candidate[{i}]
has extra keys"
65             assert set(cand["shuttle"].keys()) == set(gf.SHUTTLE_FEATURES), (
66                 f"{slug}: candidate[{i}].shuttle is not exactly the 5 shuttle features")
67
68
69     @pytest.mark.parametrize("slug", SLUGS)
70     def test_synthetic_provenance_tags(slug):
71         """Tier-0 hard tags: kind=synthetic, license=synthetic, minors_free=true (constraint
5)."""
72         prov = gf.load_fixture(slug)["provenance"]
73         assert prov["kind"] == "synthetic"
74         assert prov["license"] == "synthetic"
75         assert prov["minors_free"] is True
76         # Windowing pinned in the fixture (not the named SERVED constant) ? constraint 3.
77         assert set(prov["windowing"]) == {"vt", "mg", "mwd", "max_win"}
78
79
80     def test_control_is_not_learned():
81         """The replay must use the pure-stdlib CONTROL variant (no numpy/LogisticRegression
? D5)."""
82         from backend.eval.experiment import CONTROL
83         assert CONTROL.is_learned() is False
84
85
86     def test_perturbing_trajectory_changes_segments(tmp_path):
87         """Mutating a copy of a trajectory and replaying yields DIFFERENT segments ? proving
the
88         windows are RE-DERIVED from the CSV each run, not merely echoed from
expected.json."""
89         slug = SLUGS[0]
90         src = gf.FIXTURES_DIR / slug
91         work = tmp_path / slug
92         work.mkdir()
93         # Copy the fixture, then TRUNCATE the trajectory to its first half so the derived
window's
94         # `end` shrinks ? a guaranteed structural change in the re-derived segments.
(Deleting an
95         # interior hole would not move start/end, since merge_gap coalesces across it and
the frame
96         # indices of the surviving rows are unchanged.)
97         lines = (src / "trajectory.csv").read_text(encoding="utf-8").splitlines()
98         header, body = lines[0], lines[1:]
99         mutated = [header] + body[: len(body) // 2]
100         (work / "trajectory.csv").write_text("\n".join(mutated) + "\n", encoding="utf-8",
newline="\n")
101         (work / "labels.csv").write_text((src / "labels.csv").read_text(encoding="utf-8"),
encoding="utf-8", newline="\n")
102         (work / "provenance.json").write_text((src /
"provenance.json").read_text(encoding="utf-8"),
103         encoding="utf-8", newline="\n")
104
105
106         original = gf.load_fixture(slug)["expected"]
107         perturbed = gf.replay_fixture(work)
108         assert perturbed["control_segments"] != original["control_segments"], (
109             "perturbed trajectory produced identical segments ? replay is echoing, not re-

```

```

deriving")
110
111
112     def test_freeze_is_idempotent(tmp_path):
113         """Re-freezing every built-in fixture into a temp dir produces expected.json
structurally
114         equal to the committed one (correct-by-construction + deterministic)."""
115         slugs = gf.freeze_synthetic(fixtures_dir=tmp_path)
116         assert slugs, "freeze_synthetic produced no fixtures"
117         for slug in slugs:
118             committed = gf.load_fixture(slug)["expected"]
119             refrozen = json.loads((tmp_path / slug /
"expected.json").read_text(encoding="utf-8"))
120             # Structural (parsed) equality ? no float churn, exact same shape/values.
121             assert gf.compare_expected(refrozen, committed) == [], (
122                 f"{slug}: re-frozen expected.json differs from committed (non-idempotent
freeze)")
123             assert refrozen == committed, f"{slug}: re-frozen expected.json not structurally
identical"
124
125
126     def test_check_cli_passes(capsys):
127         """The --check CLI replays all fixtures and exits 0 when they match the committed
snapshots."""
128         rc = gf.main(["--check"])
129         assert rc == 0
130

```

9. user (2026-06-16T06:27:16.056Z)

```

1     """Golden Regression Fixtures (P1 + M2) ? de-attribution / minimization + the refusal
gate.
2
3     These tests exercise ``golden_fixtures.freeze_real_fixture`` (the P1 tooling that turns
an OWNED,
4     minors-free real trajectory + golden labels into a committable, de-attributed Tier-0
fixture) and
5     its M2 ``--license`` / ``--minors-free`` / ``--owned`` refusal gate.
6
7     Everything runs on SYNTHETIC input written into ``tmp_path`` ? NO real data, NO network,
NO GPU.
8     We reuse M1's synthetic generators (``builtin_specs`` / ``make_synthetic_trajectory``
and the
9     ``write_trajectory_csv`` / ``write_labels_csv`` writers) to manufacture the "source"
CSVs, then
10    prove:
11        * the refusal gate blocks every unsafe combination (M2);
12        * a successful freeze is de-attributed (random ``fx_`` slug, no id/name/path, no
FORBIDDEN_KEYS);
13        * the time-origin reset is metric-INVARIANT (score identical with/without; only
absolute
14        ``control_segments`` shift toward 0);
15        * the freeze round-trips (``replay_fixture`` + ``compare_expected`` == []).
16
17    $6 honest-limits caveat: a green suite proves only that the deterministic post-
trajectory transforms
18    are unchanged for the frozen cases ? characterization is behaviour-locking, NOT
correctness.
19    """
20    import pytest
21
22    from backend.eval import golden_fixtures as gf
23
24
25    # -----

```

```

26 # Helpers ? manufacture a SYNTHETIC "real" source (trajectory + labels CSV) in tmp_path.
27 # -----
28
29
30 def _write_source(tmp_path, *, frame_offset: int = 0):
31     """Write a synthetic source trajectory+labels CSV pair into ``tmp_path``.
32
33     Reuses M1's first built-in scenario (a clean single rally), optionally shifted by
34     ``frame_offset`` frames (and the matching label seconds) to model an arbitrary
absolute
35     capture timeline. Returns ``(traj_path, labels_path, fps, frame_width, gts)``.
36     """
37     spec = gf.builtin_specs()[0] # clean_single_rally
38     fps, fw = spec.fps, spec.frame_width
39     rows = gf.make_synthetic_trajectory(spec)
40     if frame_offset:
41         rows = [(f + frame_offset, v, x, y) for (f, v, x, y) in rows]
42     gts = list(spec.gts)
43     if frame_offset:
44         off_sec = frame_offset / fps
45         gts = [(s + off_sec, e + off_sec) for (s, e) in gts]
46
47     traj = tmp_path / "src_trajectory.csv"
48     labels = tmp_path / "src_labels.csv"
49     gf.write_trajectory_csv(rows, traj)
50     gf.write_labels_csv(gts, labels)
51     return traj, labels, fps, fw, gts
52
53
54 # -----
55 # M2 ? refusal gate
56 # -----
57
58
59 @pytest.mark.parametrize(
60     "kwargs, why",
61     [
62         ({"minors_free": False, "owned": True, "license": "owned"},
"minors_free=False"),
63         ({"minors_free": True, "owned": False, "license": "owned", "owned=False"},
64         ({"minors_free": True, "owned": True, "license": ""}, "license=''),
65         ({"minors_free": True, "owned": True, "license": " "}, "license=blank"),
66         ({"minors_free": True, "owned": True, "license": None}, "license=None"),
67     ],
68 )
69 def test_refusal_gate_blocks_unsafe(tmp_path, kwargs, why):
70     """freeze_real_fixture refuses (ValueError) unless minors_free AND owned AND a non-
blank license.
71
72     The refusal is the code-enforced provenance gate ($4c / $7 ?): the unsafe public-
fixture path
73     must be impossible to reach silently.
74     """
75     traj, labels, fps, fw, _ = _write_source(tmp_path)
76     with pytest.raises(ValueError) as ei:
77         gf.freeze_real_fixture(
78             "scenario", traj, labels,
79             fps=fps, frame_width=fw, fixtures_dir=tmp_path / "fixtures", **kwargs)
80     # RefusalError is a ValueError subclass; the message must name a reason.
81     assert isinstance(ei.value, gf.RefusalError), why
82     assert "REFUSED" in str(ei.value), why
83     # Nothing was written for a refused freeze.
84     assert not (tmp_path / "fixtures").exists() or not any((tmp_path /
"fixtures").iterdir()), (
85         f"{why}: a refused freeze must not write any fixture")

```

```

86
87
88 # -----
89 # P1 ? de-attribution
90 # -----
91
92
93 def test_real_fixture_is_deattributed(tmp_path):
94     """A successful safe freeze is de-attributed: kind=cleared_real, fx_-prefixed
surrogate slug,
95     no id/name/filename/source-path key, and NO FORBIDDEN_KEYS in expected.json."""
96     fixtures = tmp_path / "fixtures"
97     traj, labels, fps, fw, _ = _write_source(tmp_path)
98     expected = gf.freeze_real_fixture(
99         "clean_rally", traj, labels,
100         license="owned", minors_free=True, owned=True,
101         fps=fps, frame_width=fw, fixtures_dir=fixtures)
102
103     slug = expected["slug"]
104     assert slug.startswith("fx_"), f"surrogate slug must be opaque fx_<hex>, got
{slug!r}"
105
106     prov = gf._read_json(fixtures / slug / "provenance.json")
107     assert prov["kind"] == "cleared_real"
108     assert prov["minors_free"] is True and prov["owned"] is True
109     assert prov["license"] == "owned"
110     assert prov["slug"] == slug
111     # No attribution keys / no source filename anywhere in the provenance.
112     for banned in ("video_id", "name", "filename", "path", "match", "capture_date"):
113         assert banned not in prov, f"provenance must not carry an attribution key
{banned!r}"
114     prov_text = (fixtures / slug /
"provenance.json").read_text(encoding="utf-8").lower()
115     for tok in ("src_trajectory", "src_labels", str(traj).lower(), str(labels).lower()):
116         assert tok not in prov_text, f"source token {tok!r} leaked into provenance"
117
118     # expected.json carries no biometric/identity/non-shuttle stream.
119     exp_text = (fixtures / slug / "expected.json").read_text(encoding="utf-8").lower()
120     for forbidden in gf.FORBIDDEN_KEYS:
121         assert forbidden not in exp_text, f"FORBIDDEN_KEYS leak {forbidden!r} in
expected.json"
122
123     # The fixture is registered in the manifest with kind=cleared_real.
124     manifest = gf.load_manifest(fixtures)
125     rows = [e for e in manifest if e["slug"] == slug]
126     assert len(rows) == 1 and rows[0]["kind"] == "cleared_real"
127
128
129 def test_supplied_slug_is_honored(tmp_path):
130     """An explicitly supplied slug is used verbatim (the surrogate map is the caller's
concern)."""
131     fixtures = tmp_path / "fixtures"
132     traj, labels, fps, fw, _ = _write_source(tmp_path)
133     expected = gf.freeze_real_fixture(
134         "scenario", traj, labels, slug="fx_custom00",
135         license="owned", minors_free=True, owned=True,
136         fps=fps, frame_width=fw, fixtures_dir=fixtures)
137     assert expected["slug"] == "fx_custom00"
138     assert (fixtures / "fx_custom00" / "expected.json").is_file()
139
140
141 def test_freeze_real_does_not_clobber_existing_manifest(tmp_path):
142     """Freezing a second real fixture must APPEND to the manifest, not wipe the
first."""
143     fixtures = tmp_path / "fixtures"

```

```

144     traj, labels, fps, fw, _ = _write_source(tmp_path)
145     e1 = gf.freeze_real_fixture("a", traj, labels, slug="fx_aaaa0001",
146                               license="owned", minors_free=True, owned=True,
147                               fps=fps, frame_width=fw, fixtures_dir=fixtures)
148     e2 = gf.freeze_real_fixture("b", traj, labels, slug="fx_bbbb0002",
149                               license="owned", minors_free=True, owned=True,
150                               fps=fps, frame_width=fw, fixtures_dir=fixtures)
151     slugs = {e["slug"] for e in gf.load_manifest(fixtures)}
152     assert {e1["slug"], e2["slug"]} <= slugs
153
154
155     def test_positive_privacy_assertion_catches_leak_in_source_note(tmp_path):
156         """source_note is non-identifying by contract ? a path/name in it trips the positive
157         scan."""
158         fixtures = tmp_path / "fixtures"
159         traj, labels, fps, fw, _ = _write_source(tmp_path)
160         with pytest.raises(gf.RefusalError):
161             gf.freeze_real_fixture(
162                 "scenario", traj, labels, slug="fx_leak0001",
163                 license="owned", minors_free=True, owned=True,
164                 fps=fps, frame_width=fw, fixtures_dir=fixtures,
165                 source_note=str(traj)) # leaking the source path ? must be refused
166
167         # -----
168         # P1 (d) ? time-origin reset is metric-invariant
169         # -----
170
171
172     def test_time_origin_reset_is_metric_invariant(tmp_path):
173         """A large-offset source frozen WITH reset has IDENTICAL score to the SAME source
174         frozen
175         WITHOUT reset ? but its absolute control_segments shift toward 0 (severed
176         timeline)."""
177         offset_frames = 9000 # +5 minutes at 30fps ? a large absolute capture-timeline
178         offset
179         traj, labels, fps, fw, _ = _write_source(tmp_path, frame_offset=offset_frames)
180
181         with_reset = gf.freeze_real_fixture(
182             "scenario", traj, labels, slug="fx_reset_on",
183             license="owned", minors_free=True, owned=True,
184             fps=fps, frame_width=fw, fixtures_dir=tmp_path / "with",
185             time_origin_reset=True)
186         without_reset = gf.freeze_real_fixture(
187             "scenario", traj, labels, slug="fx_reset_off",
188             license="owned", minors_free=True, owned=True,
189             fps=fps, frame_width=fw, fixtures_dir=tmp_path / "without",
190             time_origin_reset=False)
191
192         # (1) Score is bit-identical ? metrics.score is built from pure differences.
193         assert with_reset["score"] == without_reset["score"], (
194             "time-origin reset must be metric-invariant (score built from pure
195             differences)")
196
197         # (2) Absolute timeline differs ? the reset pulls control_segments toward 0.
198         seg_on = with_reset["control_segments"]
199         seg_off = without_reset["control_segments"]
200         assert len(seg_on) == len(seg_off) and len(seg_on) >= 1
201         assert seg_on[0][0] < seg_off[0][0], "reset must shift the first segment start
202         toward 0"
203
204         # The shift equals the offset in seconds (within rounding).
205         shift = seg_off[0][0] - seg_on[0][0]
206         assert abs(shift - offset_frames / fps) < 1e-3, (
207             f"segment shift {shift} should equal the frame offset in seconds {offset_frames
208             / fps}")

```



```

202     # Reset trajectory starts at frame 0.
203     first_frame = int((tmp_path / "with" / "fx_reset_on" / "trajectory.csv")
204                       .read_text(encoding="utf-8").splitlines()[1].split(",")[0])
205     assert first_frame == 0, "with reset, the trajectory must start at Frame 0"
206
207
208     # -----
209     # P1 (e/f) ? round-trip / correct-by-construction
210     # -----
211
212
213     def test_freeze_real_round_trips_check(tmp_path):
214         """After freeze_real_fixture, replay_fixture + compare_expected vs the written
215         expected.json
216         returns [] ? idempotent / correct-by-construction (the same path M1's --check
217         uses)."""
218         fixtures = tmp_path / "fixtures"
219         traj, labels, fps, fw, _ = _write_source(tmp_path)
220         expected = gf.freeze_real_fixture(
221             "scenario", traj, labels, slug="fx_rt000001",
222             license="owned", minors_free=True, owned=True,
223             fps=fps, frame_width=fw, fixtures_dir=fixtures)
224
225         fdir = fixtures / expected["slug"]
226         committed = gf._read_json(fdir / "expected.json")
227         recomputed = gf.replay_fixture(fdir)
228         assert gf.compare_expected(recomputed, committed) == [], (
229             "a just-frozen real fixture must replay-compare clean (correct-by-
230             construction)")
231
232
233     def test_freeze_real_uses_strict_label_loader(tmp_path):
234         """The labels are read through the STRICT gt_loader ? a malformed label row fails
235         loudly,
236         never silently shrinks the GT set."""
237         fixtures = tmp_path / "fixtures"
238         traj, labels, fps, fw, _ = _write_source(tmp_path)
239         # Corrupt one label row (end <= start) ? strict loader must raise.
240         labels.write_text("start,end\n5.0,1.0\n", encoding="utf-8", newline="\n")
241         with pytest.raises(ValueError):
242             gf.freeze_real_fixture(
243                 "scenario", traj, labels, slug="fx_strict01",
244                 license="owned", minors_free=True, owned=True,
245                 fps=fps, frame_width=fw, fixtures_dir=fixtures)

```

10. assistant (2026-06-16T06:27:18.499Z)

Now the supporting eval modules. Let me read them in parallel.

11. user (2026-06-16T06:27:20.200Z)

```

1     """Distill Gemini (teacher) into a LEARNED local rally classifier (student).
2
3     Threshold-tuning the local knobs hit a low ceiling (see calibrate_local). This goes
4     further: learn "is this WASB candidate a real rally?" from Gemini silver labels,
5     using a few resolution/fps-INVARIANT local features, so the model transfers across
6     videos and runs 100% local at inference (no Gemini, privacy lever intact).
7
8     Flow (all local at runtime):
9     WASB trajectory -> recall-oriented candidate windows (propose many; classifier
10    filters) -> per-candidate features [duration, peak_velocity, mean_velocity,
11    active_ratio, net_crossings] -> LogisticRegression(prob >= threshold) -> merge
12    -> rally windows.
13

```

```

14     Training labels come from Gemini full-context labels via the SAME IoU matching the
15     evaluator uses (training.label_candidates). Honest leave-one-video-out: train on the
16     other videos' candidates, predict the held-out one, score vs its Gemini labels, and
17     compare against the threshold baseline. Reuses classifier.py, training.label_candidates,
18     rally_gate, metrics, calibrate_wasb.load_golden_gts, gemini_refine.merge_overlaps.
19
20     Run:
21     python -m backend.eval.distill_local --manifest videos.json --model-out
output/local_rally_model.json
22     """
23     from __future__ import annotations
24
25     import json
26     import argparse
27     from typing import Dict, List, Optional, Tuple
28
29
30     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
31     from backend.pipeline.detectors import rally_gate
32     from backend.eval.calibrate_wasb import load_golden_gts
33     from backend.eval.training import label_candidates
34     from backend.eval.classifier import LogisticRegression
35     from backend.eval.metrics import score
36     from backend.eval.gemini_refine import merge_overlaps
37     from backend.eval import windowing as _windowing
38
39     Interval = Tuple[float, float]
40
41     FEATURE_NAMES = ["duration", "peak_velocity", "mean_velocity", "active_ratio",
"net_crossings"]
42     # Recall-oriented proposal: deliberately permissive so real rallies are among the
43     # candidates (the classifier removes the junk). (velocity_thresh, merge_gap, min_dur).
44     # Single-sourced from backend.eval.windowing so eval and serve can never drift ? see
that
45     # module's docstring for the eval?serve bug these named presets close. The bare tuples
are
46     # preserved (byte-identical: PROPOSAL == (0.005, 1.0, 0.5), BASELINE_LOCAL == SERVED ==
47     # (0.02, 2.0, 2.0)) so every existing call site / golden JSON stays valid.
48     PROPOSAL = _windowing.PROPOSAL.as_tuple()
49     BASELINE_LOCAL = _windowing.SERVED.as_tuple() # the served (production) windowing, no
learning
50
51
52     def build_candidates(trajecory_csv: str, fps: float, frame_width: float,
53                         proposal: Tuple[float, float, float] = PROPOSAL) ->
Tuple[List[Interval], List[List[float]]]:
54     """WASB trajectory -> (candidate intervals, fps/resolution-invariant feature
rows)."""
55     points = TrackNetRunner.parse_trajectory_csv(trajecory_csv)
56     vt, mg, mwd = proposal
57     wins = TrackNetRunner.trajectory_to_action_windows(
58         points, fps=fps, frame_width=frame_width,
59         velocity_thresh=vt, merge_gap=mg, min_window_duration=mwd)
60     intervals = [(w["start"], w["end"]) for w in wins]
61     gated = rally_gate.gate_windows(points, intervals, fps, min_crossings=0) # for net-
crossings
62     X: List[List[float]] = []
63     for w, g in zip(wins, gated):
64         dur = max(1e-6, w["end"] - w["start"])
65         win_frames = max(1.0, dur * fps)
66         X.append([
67             dur, # rally length (sec) ?
68             float(w["peak_velocity"]), # already normalized by
69             frame_width

```

```

69         float(w["mean_velocity"]),
70         float(w["active_frame_count"]) / win_frames,      # active-shuttle ratio ?
fps-invariant
71         float(g.crossings),                                # net crossings (count) ?
the rally signal
72     ])
73     return intervals, X
74
75
76     def baseline_windows(trajecory_csv: str, fps: float, frame_width: float) ->
List[Interval]:
77         points = TrackNetRunner.parse_trajectory_csv(trajecory_csv)
78         vt, mg, mwd = BASELINE_LOCAL
79         wins = TrackNetRunner.trajectory_to_action_windows(
80             points, fps=fps, frame_width=frame_width,
81             velocity_thresh=vt, merge_gap=mg, min_window_duration=mwd)
82         return [(w["start"], w["end"]) for w in wins]
83
84
85     def load_video(spec: Dict, iou: float = 0.5) -> Dict:
86         fps, fw = float(spec["fps"]), float(spec["frame_width"])
87         intervals, X = build_candidates(spec["trajectory"], fps, fw)
88         gts = load_golden_gts(spec["labels"])
89         y = label_candidates(intervals, gts, iou)
90         return {"name": spec.get("name", spec["trajectory"]), "intervals": intervals, "X":
X,
91                 "y": y, "gts": gts, "baseline": baseline_windows(spec["trajectory"], fps,
fw)}
92
93
94     def predict_rallies(model: LogisticRegression, X: List[List[float]], intervals:
List[Interval],
95         threshold: float = 0.5) -> List[Interval]:
96         if not intervals:
97             return []
98         proba = model.predict_proba(X)
99         pos = [iv for iv, p in zip(intervals, proba) if p >= threshold]
100         return merge_overlaps(pos, gap_tol=1.0)
101
102
103     def run(specs: List[Dict], iou: float = 0.5, threshold: float = 0.5,
104         model_out: Optional[str] = None) -> dict:
105         videos = [load_video(s, iou) for s in specs]
106         print(f"=== Distilled local rally classifier vs Gemini silver-GT "
107             f"({len(videos)} videos, IoU>={iou}, prob>={threshold}) ===")
108         for v in videos:
109             ceil = score(v["intervals"], v["gts"], iou).recall
110             print(f"[{v['name']}] {len(v['intervals'])} candidates ({sum(v['y'])} pos) | "
111                 f"proposal recall ceiling {ceil:.3f} | {len(v['gts'])} Gemini rallies")
112
113         result: dict = {"videos": [v["name"] for v in videos]}
114         if len(videos) >= 2:
115             print("\n--- HONEST leave-one-video-out: learned vs threshold baseline (held-
out) ---")
116             clf, base = [], []
117             for i, held in enumerate(videos):
118                 others = [v for j, v in enumerate(videos) if j != i]
119                 X = [row for v in others for row in v["X"]]
120                 y = [t for v in others for t in v["y"]]
121                 model = LogisticRegression().fit(X, y, FEATURE_NAMES)
122                 preds = predict_rallies(model, held["X"], held["intervals"], threshold)
123                 sc = score(preds, held["gts"], iou)
124                 bsc = score(held["baseline"], held["gts"], iou)
125                 clf.append(sc)
126                 base.append(bsc)

```

```

127         print(f" held-out {held['name']}: LEARNED P={sc.precision:.3f}
R={sc.recall:.3f} F1={sc.f1:.3f}"
128         f" | baseline P={bsc.precision:.3f} R={bsc.recall:.3f}
F1={bsc.f1:.3f}")
129     def mean(xs, a):
130         return sum(getattr(s, a) for s in xs) / len(xs)
131     print(f" MEAN held-out: LEARNED P={mean(clf, 'precision'):.3f}
R={mean(clf, 'recall'):.3f} "
132         f"F1={mean(clf, 'f1'):.3f} | baseline P={mean(base, 'precision'):.3f} "
133         f"R={mean(base, 'recall'):.3f} F1={mean(base, 'f1'):.3f}")
134     result["mean_learned_f1"] = mean(clf, "f1")
135     result["mean_baseline_f1"] = mean(base, "f1")
136
137     # Final model on ALL videos (for production / running on a fresh held-out video).
138     Xall = [row for v in videos for row in v["X"]]
139     yall = [t for v in videos for t in v["y"]]
140     final = LogisticRegression().fit(Xall, yall, FEATURE_NAMES)
141     assert final.w is not None # set by fit()
142     weights = dict(zip(FEATURE_NAMES, [round(float(w), 3) for w in final.w]))
143     print(f"\nfinal model standardized weights (sign/size = influence): {weights}")
144     if model_out:
145         final.save(model_out)
146         print(f"final model (all videos) -> {model_out}")
147         result["model_out"] = model_out
148     return result
149
150
151 def main() -> None:
152     ap = argparse.ArgumentParser(description="Distill Gemini labels into a local rally
classifier.")
153     ap.add_argument("--manifest", required=True,
154                     help="JSON: [{name, trajectory, fps, frame_width, labels}, ...]
(labels = Gemini CSV)")
155     ap.add_argument("--iou", type=float, default=0.5)
156     ap.add_argument("--threshold", type=float, default=0.5, help="classifier probability
cutoff")
157     ap.add_argument("--model-out", default="output/local_rally_model.json")
158     args = ap.parse_args()
159     with open(args.manifest, encoding="utf-8") as f:
160         specs = json.load(f)
161     run(specs, iou=args.iou, threshold=args.threshold, model_out=args.model_out)
162
163
164 if __name__ == "__main__":
165     main()
166

```

12. user (2026-06-16T06:27:20.366Z)

```

1     """Experiment harness ? A/B + shadow + side-by-side for rally detection.
2
3     This is the thin **variant/experiment layer** that `docs/EXPERIMENT_HARNESS.md`
4     identifies as the one missing piece: ~80% of the machinery (scoring, LOVO, the
5     no-regression gate, the pinnable classifier artifact, the persisted-candidate
6     substrate) already exists ? this module composes it. **No new scoring math lives
7     here**; everything defers to:
8
9     - `backend.eval.metrics`      ? `score`, `match_intervals`, `interval_iou`
10    - `backend.eval.training`      ? `label_candidates` (y by the same IoU matcher)
11    - `backend.eval.classifier`    ? `LogisticRegression` (the pinnable JSON model)
12    - `backend.eval.gemini_refine` ? `merge_overlaps` (the blessed window merge)
13    - `backend.eval.regression`    ? `gt_hash` (label-set fingerprint)
14    - `backend.eval.calibration`   ? `pct_improvement`
15
16    The thesis (`EXPERIMENT_HARNESS.md` §2): separate the **expensive, risky DETECTION**

```

```

17 (per-frame signals ? run once on the GPU/WSL box, persisted into
18 `candidate_segments.stats`) from the **cheap, swappable DECISION** (given those
19 signals, where are the rallies). A `Variant` is a declarative re-scoring recipe;
20 given persisted candidate features it produces `List[Interval]` deterministically,
21 with no GPU and zero production risk. So most experiments are pure offline
22 re-scoring of stored data and never touch the served pipeline.
23
24 Three modes (`EXPERIMENT_HARNESS.md` §5):
25 - `compare()` ? labeled A/B vs golden labels: per-video + LOVO-honest
26   aggregate deltas. Mirrors `distill_local`'s honest train-on-others/predict-
27   held-out loop, but variant-parameterised.
28 - `compare_unlabeled()` ? SxS on NEW footage with no ground truth: where do two
29   variants agree / diverge (A-only, B-only, agreed). The divergences are what a
30   human spot-checks (and what two highlight reels would show side by side).
31 - the promotion gate stays `run_regression.py` + `regression_baseline.json`
32   (exit 0/1/3); **rollback = flip the variant/config, never `git revert`.**
33
34 Pure + offline + unit-tested. The only DB-touching helper is
35 `load_video_candidates`, which reads persisted candidates via
36 `Database.query_candidate_segments`.
37 """
38 from __future__ import annotations
39
40 import json
41 import logging
42 import os
43 from dataclasses import dataclass, field
44 from datetime import datetime, timezone
45 from hashlib import sha1
46 from typing import Any, Callable, Dict, List, Optional, Sequence
47
48 from backend.eval.calibration import pct_improvement
49 from backend.eval.classifier import LogisticRegression
50 from backend.eval.gemini_refine import merge_overlaps
51 from backend.eval.metrics import Interval, match_intervals, score
52 from backend.eval.regression import gt_hash
53 from backend.eval.training import label_candidates
54 from backend.eval import eval_stats as _stats # C1: CIs + paired sign test
55 from backend.eval import segmentation_metrics as _segm # C4: F1@k + segment-count
56 ratio
57 scores
58 from backend.eval.score_calibration import fit_isotonic, fit_platt # C2: calibrate cue
59
60 logger = logging.getLogger(__name__)
61
62 # Where a comparison run appends one reproducible record. Tracked location (NOT under
63 # the gitignored output/) so the leaderboard survives a clean checkout, mirroring
64 # regression.DEFAULT_BASELINE_PATH.
65 DEFAULT_LEADERBOARD_PATH = os.path.join("eval_baselines", "experiment_leaderboard.json")
66 _METRICS = ("precision", "recall", "f1", "mean_iou")
67
68 class ExperimentError(ValueError):
69     """A variant cannot be evaluated as configured (e.g. a learned variant asked to
70     score an unlabeled video with no pinned model, or a gate referencing an absent
71     feature)."""
72
73 # ----- Variant
74
75 @dataclass
76 class Variant:
77     """A declarative re-scoring recipe ? NOT a code branch (`EXPERIMENT_HARNESS.md` §4).
78

```

```

80 A variant turns one video's persisted candidate windows + features into rally
81 intervals. Every knob defaults to the **control** behaviour (no gate, no learned
82 filter), so a variant that merely *carries* a new, disabled cue is byte-identical
83 to control ? the core no-regression guarantee.
84
85 Fields:
86 - ``segmenter`` / ``config_overrides`` / ``cue_flags`` ? informational provenance
87   for the leaderboard and for selecting the served path (the existing
88   `segmenter_registry` + `IndexingConfig` do the actual selection in production).
89   New cues are recorded here default-OFF.
90 - ``min_crossings`` ? decision-gate **Gate A**: keep only windows whose persisted
91   ``net_crossings`` feature is  $\geq$  this. 0 = off. This is the eval-only gate from
92   `rally_gate.py` expressed as a re-scoring knob (kills service-feed / held-
shuttle
93   windows ? see `MULTI_SIGNAL_FUSION_PLAN.md` §3.1).
94 - ``min_players`` ? decision-gate **Gate B** (the future person cue): keep windows
95   whose persisted ``players_in_court`` is  $\geq$  this. 0 = off (no-op until the person
96   cue persists that feature).
97 - ``classifier_model`` ? path to a frozen `LogisticRegression` JSON. When set,
98   `compare()` scores videos with the SHIPPED model as-is (no per-fold training);
99   required for `compare_unlabeled` on a learned variant.
100 - ``feature_names`` ? the persisted features the learned filter consumes. When set
101   WITHOUT ``classifier_model``, `compare()` trains a fresh model per LOVO fold
102   (honest) ? the recipe, not a pinned artifact.
103 - ``threshold`` ? classifier probability cutoff. ``merge_gap`` ? post-filter
104   overlap-merge gap (seconds), the same `gemini_refine.merge_overlaps` default.
105 """"
106
107 name: str
108 segmenter: str = "wasb_hybrid"
109 config_overrides: Dict[str, Any] = field(default_factory=dict)
110 cue_flags: Dict[str, bool] = field(default_factory=dict)
111 min_crossings: int = 0
112 min_players: int = 0
113 classifier_model: Optional[str] = None
114 feature_names: Optional[List[str]] = None
115 threshold: float = 0.5
116 merge_gap: float = 1.0
117 # C2 / threshold-in-LOVO (default OFF ? unchanged behaviour). When set, the
operating
118 # point is chosen on the TRAINING fold, never the held-out one ? fixing the
first-A/B
119 # failure where a fixed 0.5 zeroed out a small-candidate video.
120 tune_threshold: bool = False # pick the F1-best threshold on the train fold
121 calibrate: Optional[str] = None # None | "platt" | "isotonic" ? calibrate
proba first
122
123 def is_learned(self) -> bool:
124     """True if this variant applies a learned classifier (pinned model or
recipe)."""
125     return self.classifier_model is not None or bool(self.feature_names)
126
127 def to_dict(self) -> dict:
128     return {
129         "name": self.name,
130         "segmenter": self.segmenter,
131         "config_overrides": self.config_overrides,
132         "cue_flags": self.cue_flags,
133         "min_crossings": self.min_crossings,
134         "min_players": self.min_players,
135         "classifier_model": self.classifier_model,
136         "feature_names": self.feature_names,
137         "threshold": self.threshold,
138         "merge_gap": self.merge_gap,
139         "tune_threshold": self.tune_threshold,

```

```

140         "calibrate": self.calibrate,
141     }
142
143     @classmethod
144     def from_dict(cls, d: dict) -> "Variant":
145         known = {f for f in cls.__dataclass_fields__} # type: ignore[attr-defined]
146         return cls(**{k: v for k, v in d.items() if k in known})
147
148     def spec_hash(self) -> str:
149         """Stable 12-char hash of the recipe ? identifies the variant in the leaderboard
150         and makes a result reproducible. The pinned **control** variant always hashes
151         the
152         same, so a regression is always traceable to a recipe change, never to drift."""
153         blob = json.dumps(self.to_dict(), sort_keys=True, default=str)
154         return sha1(blob.encode()).hexdigest()[:12]
155
156 #: The pinned, frozen baseline: candidates as detected, merged, no gate, no learned
157 #: filter. Always the comparison reference; "rollback" = make this the served variant.
158 CONTROL = Variant(name="control")
159
160
161 # ----- persisted candidates
162
163
164 @dataclass
165 class VideoCandidates:
166     """One video's persisted detection, ready for offline re-scoring.
167
168     ``intervals`` are the candidate windows; ``features`` is column-major
169     (``feature_name -> per-candidate value``, each list aligned to ``intervals``);
170     ``gts`` are golden labels (None for new/unlabeled footage). Build one from the DB
171     with ``load_video_candidates``, or directly in tests.
172     """
173
174     name: str
175     intervals: List[Interval]
176     features: Dict[str, List[float]] = field(default_factory=dict)
177     gts: Optional[List[Interval]] = None
178
179
180     def _feature_column(vc: VideoCandidates, name: str) -> List[float]:
181         """Persisted feature column, defaulting missing/short columns to 0.0 (matches
182         ``training.extract_yolo_features``, which lets a sparse/old row still vectorise)."""
183         col = vc.features.get(name)
184         n = len(vc.intervals)
185         if col is None:
186             logger.warning("variant references feature %r absent from %s ? treating as 0.0",
187                             name, vc.name)
188             return [0.0] * n
189         if len(col) != n:
190             raise ExperimentError(
191                 f"feature {name!r} has {len(col)} values but {vc.name} has {n} candidates")
192         return [float(x) for x in col]
193
194
195     def _feature_matrix(vc: VideoCandidates, feature_names: Sequence[str]) ->
196     List[List[float]]:
197         cols = [_feature_column(vc, name) for name in feature_names]
198         return [list(row) for row in zip(*cols)] if cols else [[] for _ in vc.intervals]
199
200     def _gate_mask(variant: Variant, vc: VideoCandidates) -> List[bool]:
201         """Boolean keep-mask from the decision gates (Gate A net-crossings, Gate B
202         players)."""

```

```

202     n = len(vc.intervals)
203     mask = [True] * n
204     if variant.min_crossings > 0:
205         col = _feature_column(vc, "net_crossings")
206         mask = [m and (col[i] >= variant.min_crossings) for i, m in enumerate(mask)]
207     if variant.min_players > 0:
208         col = _feature_column(vc, "players_in_court")
209         mask = [m and (col[i] >= variant.min_players) for i, m in enumerate(mask)]
210     return mask
211
212
213 def predict(variant: Variant, vc: VideoCandidates,
214             model: Optional[LogisticRegression] = None,
215             threshold: Optional[float] = None,
216             calibrator: Optional[Callable[[float], float]] = None) -> List[Interval]:
217     """Variant prediction: gate by persisted features, optionally filter by a learned
218     model, then merge. Pure and deterministic.
219
220     ``model`` (an already-fitted `LogisticRegression`) is supplied by `compare` per LOVO
221     fold; if omitted and the variant pins ``classifier_model``, that frozen model is
222     loaded. A learned variant with neither a supplied nor a pinned model raises ? there
223     is nothing to score with. ``threshold``/``calibrator`` (both fitted on the TRAINING
224     fold) override the variant defaults when the C2 / threshold-in-LOVO path supplies
225     them.
226     """
227     mask = _gate_mask(variant, vc)
228     if variant.feature_names:
229         if model is None:
230             if variant.classifier_model is None:
231                 raise ExperimentError(
232                     f"variant {variant.name!r} is learned but has no model to predict "
233                     f"with (pass a fitted model or set classifier_model)")
234             model = LogisticRegression.load(variant.classifier_model)
235             proba = [float(p) for p in model.predict_proba(_feature_matrix(vc,
236 variant.feature_names))]
237             if calibrator is not None:
238                 proba = [calibrator(p) for p in proba]
239             thr = variant.threshold if threshold is None else threshold
240             mask = [m and (proba[i] >= thr) for i, m in enumerate(mask)]
241             kept = [iv for iv, keep in zip(vc.intervals, mask) if keep]
242             return merge_overlaps(kept, gap_tol=variant.merge_gap)
243
244
245 def _calibrate_and_tune(variant: Variant, model: LogisticRegression,
246                        train_videos: Sequence[VideoCandidates], iou: float
247                        ) -> "tuple[Optional[Callable[[float], float]],
248 Optional[float]]":
249     """Fit a calibrator and/or pick the operating threshold on the TRAINING fold only
250     (C2 + threshold-in-LOVO). Returns ``(calibrator, threshold)`` ? either may be None
251     (use the variant default). Honest: never looks at the held-out video.
252     """
253     calibrator: Optional[Callable[[float], float]] = None
254     if variant.calibrate:
255         raw: List[float] = []
256         y: List[int] = []
257         for vc in train_videos:
258             raw.extend(float(p) for p in
259 model.predict_proba(_feature_matrix(vc, variant.feature_names or
260 [])))
261             y.extend(label_candidates(vc.intervals, vc.gts or [], iou))
262         if variant.calibrate == "platt":
263             calibrator = fit_platt(raw, y)
264         elif variant.calibrate == "isotonic":
265             calibrator = fit_isotonic(raw, y)
266         else:

```



```

263         raise ExperimentError(f"unknown calibrate={variant.calibrate!r} (use
'platt'/'isotonic')")
264     threshold: Optional[float] = None
265     if variant.tune_threshold:
266         best_t, best_f1 = variant.threshold, -1.0
267         for k in range(1, 20):                                     # grid 0.05..0.95 on the train
fold's rally F1
268             t = round(0.05 * k, 2)
269             f1s = [score(predict(variant, vc, model=model, threshold=t,
calibrator=calibrator),
270                        vc.gts or [], iou).f1 for vc in train_videos]
271             mean_f1 = sum(f1s) / len(f1s) if f1s else 0.0
272             if mean_f1 > best_f1:
273                 best_f1, best_t = mean_f1, t
274             threshold = best_t
275     return calibrator, threshold
276
277
278     # ----- variant scoring
279
280
281     def _train(variant: Variant, videos: Sequence[VideoCandidates], iou: float) ->
LogisticRegression:
282         """Fit the variant's classifier on the pooled candidates of ``videos`` (labels via
the
283         evaluator's own IoU matcher). The honest-LOVO training step from `distill_local`. """
284         X: List[List[float]] = []
285         y: List[int] = []
286         for vc in videos:
287             if vc.gts is None:
288                 raise ExperimentError(f"cannot train: {vc.name} has no golden labels")
289             X.extend(_feature_matrix(vc, variant.feature_names or []))
290             y.extend(label_candidates(vc.intervals, vc.gts, iou))
291         if not X:
292             raise ExperimentError("cannot train a learned variant on zero candidates")
293         return LogisticRegression().fit(X, y, variant.feature_names)
294
295
296     def evaluate_variant(videos: Sequence[VideoCandidates], variant: Variant,
297                        iou: float = 0.5, honest: bool = True) -> Dict[str, Any]:
298         """Score a variant on a labeled corpus. Returns per-video Scores + predictions + a
299         mean aggregate.
300
301         Prediction policy:
302         - **static variant** (no learned filter): predict each video directly ? no
training,
303         so no leakage; per-video scoring is already honest.
304         - **learned, pinned model** (`classifier_model` set): score every video with the
SHIPPED model. ? optimistic if those videos were in its training set ? use this
305         to
306         report "how the shipped artifact does here", not for the promotion decision.
307         - **learned recipe** (`feature_names`, no pinned model) + `honest=True`: LOVO
?
308         train on the OTHER videos, predict the held-out one. The number to trust.
309         - learned recipe + `honest=False`: train on all, predict each (overfit; sanity
only).
310         """
311         for vc in videos:
312             if vc.gts is None:
313                 raise ExperimentError(f"{vc.name} has no golden labels; use
compare_unlabeled")
314
315         per_video: List[Dict[str, Any]] = []
316         predictions: Dict[str, List[Interval]] = {}
317

```

```

318 recipe_lovo = bool(variant.feature_names) and variant.classifier_model is None
319 pinned_model = (LogisticRegression.load(variant.classifier_model)
320                 if variant.classifier_model is not None else None)
321 all_model = None
322 if recipe_lovo and not honest:
323     all_model = _train(variant, videos, iou)
324
325 for i, vc in enumerate(videos):
326     cal: Optional[Callable[[float], float]] = None
327     thr: Optional[float] = None
328     if recipe_lovo and honest:
329         others = [v for j, v in enumerate(videos) if j != i]
330         if not others:
331             raise ExperimentError("LOVO needs >=2 videos; pass honest=False or a
pinned model")
332         model: Optional[LogisticRegression] = _train(variant, others, iou)
333         if variant.tune_threshold or variant.calibrate: # operating point from
the train fold
334             assert model is not None # _train never returns
None
335             cal, thr = _calibrate_and_tune(variant, model, others, iou)
336         elif recipe_lovo: # honest=False ? one model over all
videos
337             model = all_model
338         else: # static variant or pinned model
339             model = pinned_model
340         preds = predict(variant, vc, model=model, threshold=thr, calibrator=cal)
341         assert vc.gts is not None # guarded at function top
342         sc = score(preds, vc.gts, iou)
343         per_video.append({"video": vc.name, "num_pred": len(preds), **{m: getattr(sc, m)
for m in _METRICS}})
344         predictions[vc.name] = preds
345
346         aggregate = {m: (sum(p[m] for p in per_video) / len(per_video) if per_video else
0.0)
347                     for m in _METRICS}
348     return {
349         "variant": variant.name,
350         "spec_hash": variant.spec_hash(),
351         "is_learned": variant.is_learned(),
352         "honest_lovo": recipe_lovo and honest,
353         "per_video": per_video,
354         "aggregate": aggregate,
355         "predictions": predictions,
356     }
357
358
359 def _ci_block(eval_v: Dict[str, Any]) -> Dict[str, Any]:
360     """Per-metric mean + bootstrap CI over the per-video scores (C1 ? never a bare
mean)."""
361     return {m: _stats.mean_with_ci([p[m] for p in eval_v["per_video"]]) for m in
_METRICS}
362
363
364 def _segmentation_block(videos: Sequence[VideoCandidates], eval_v: Dict[str, Any]) ->
Dict[str, Any]:
365     """Mean F1@k + segment-count ratio across videos (C4 ? the over-segmentation tIoU
hides)."""
366     gts_by_name = {v.name: (v.gts or []) for v in videos}
367     overlaps = _segm.DEFAULT_OVERLAPS
368     f1k: Dict[float, List[float]] = {ov: [] for ov in overlaps}
369     ratios: List[float] = []
370     for name, preds in eval_v.get("predictions", {}).items():
371         gts = gts_by_name.get(name, [])
372         got = _segm.f1_at_overlaps(preds, gts, overlaps)

```

```

373         for ov in overlaps:
374             f1k[ov].append(got[ov])
375         ratios.append(_segm.segment_count_ratio(preds, gts))
376
377     def _mean(xs: List[float]) -> float:
378         return sum(xs) / len(xs) if xs else 0.0
379     out: Dict[str, Any] = {f"f1@{int(ov * 100)}": _mean(vals) for ov, vals in
f1k.items()}
380     out["segment_count_ratio"] = _mean(ratios)
381     return out
382
383
384     def honest_summary(videos: Sequence[VideoCandidates], eval_a: Dict[str, Any],
385                       eval_b: Dict[str, Any]) -> Dict[str, Any]:
386         """C1 + C4 honest reporting layered over a compare() pair (additive,
EXPERIMENT_HARNESS §6).
387
388         ``per_video`` lists are video-aligned (``evaluate_variant`` iterates ``videos`` in
order),
389         so ``paired_delta_f1`` pairs B?A by video ? the promotion-grade view: a lift is real
when
390         the ?F1 CI clears 0 *and* the sign test agrees, not when a bare mean ticked up.
391         """
392         a_f1 = [p["f1"] for p in eval_a["per_video"]]
393         b_f1 = [p["f1"] for p in eval_b["per_video"]]
394         return {
395             "variant_a_ci": _ci_block(eval_a),
396             "variant_b_ci": _ci_block(eval_b),
397             "paired_delta_f1": _stats.paired_delta_ci(a_f1, b_f1),
398             "segmentation": {"variant_a": _segmentation_block(videos, eval_a),
399                             "variant_b": _segmentation_block(videos, eval_b)},
400         }
401
402
403     def compare(videos: Sequence[VideoCandidates], variant_a: Variant, variant_b: Variant,
404                iou: float = 0.5, honest: bool = True, honest_stats: bool = True) ->
Dict[str, Any]:
405         """Offline labeled A/B (EXPERIMENT_HARNESS.md §5.1). Scores both variants on the
same
406         golden corpus and reports the **B ? A** deltas, per video and in aggregate.
407
408         The deltas use the existing `metrics.score` (per video) +
`calibration.pct_improvement`;
409         learned variants are scored LOVO-honestly. The result is a self-contained record fit
for
410         `record_experiment` ? variant specs + hashes + the label-set fingerprint travel with
it.
411
412         ``honest_stats`` (default True) **adds** a ``"honest"`` block ? per-metric bootstrap
CIs, a
413         paired ?F1 CI + exact sign test (C1), and F1@k + segment-count ratio (C4) ?
*alongside* the
414         existing bare-mean fields (additive: nothing existing changes). Set False for the
legacy
415         shape. This is the measurement-honesty wiring (ANALYZERS_AND_RECONCILERS.md
§13/C1+C4): a
416         bare LOVO mean at n?6 is not trustworthy on its own.
417         """
418         if len(videos) < 1:
419             raise ExperimentError("compare needs at least one labeled video")
420         eval_a = evaluate_variant(videos, variant_a, iou, honest)
421         eval_b = evaluate_variant(videos, variant_b, iou, honest)
422
423         delta = {m: eval_b["aggregate"][m] - eval_a["aggregate"][m] for m in _METRICS}
424         delta_pct = {m: pct_improvement(eval_a["aggregate"][m], eval_b["aggregate"][m]) for

```

```

m in _METRICS}
425
426     by_video_a = {p["video"]: p for p in eval_a["per_video"]}
427     per_video_delta = [
428         {"video": p["video"], **{m: p[m] - by_video_a[p["video"]][m] for m in _METRICS}}
429         for p in eval_b["per_video"]]
430     ]
431
432     # One fingerprint over the whole label set ? detects "the deltas were measured
against
433     # different labels" the same way regression.gt_hash guards the no-regression
baseline.
434     all_gts: List[Interval] = [iv for vc in videos for iv in (vc.gts or [])]
435
436     result: Dict[str, Any] = {
437         "iou": iou,
438         "label_set_hash": gt_hash(all_gts),
439         "num_videos": len(videos),
440         "variant_a": eval_a,
441         "variant_b": eval_b,
442         "delta": delta,
443         "delta_pct": delta_pct,
444         "per_video_delta": per_video_delta,
445     }
446     if honest_stats:
447         result["honest"] = honest_summary(videos, eval_a, eval_b)
448     return result
449
450
451     # ----- SxS on unlabeled video
452
453
454     def compare_unlabeled(video: VideoCandidates, variant_a: Variant, variant_b: Variant,
455         agree_iou: float = 0.5) -> Dict[str, Any]:
456         """Side-by-side on NEW footage with no ground truth (`EXPERIMENT_HARNESS.md` §5.2).
457
458         With no labels there is no lift to measure ? instead this surfaces where the two
459         variants **agree vs diverge**, which is exactly what a human reviews (and what two
460         highlight reels would show side by side):
461
462         - ``agreed`` ? windows both variants found (matched at ``agree_iou``);
463         - ``a_only`` ? A fired, B suppressed (B's added precision, if A was over-firing);
464         - ``b_only`` ? B fired, A did not (B's new detections ? real recall or new FPS:
465           the human / a later golden label adjudicates).
466
467         Both variants must be predictable without labels: a learned variant therefore needs
a
468         pinned ``classifier_model`` (you cannot train on an unlabeled video). Reuses
469         ``metrics.match_intervals`` ? no new matching logic.
470         """
471         preds_a = predict(variant_a, video)
472         preds_b = predict(variant_b, video)
473         mr = match_intervals(preds_a, preds_b, agree_iou)
474
475         agreed = [{"a": preds_a[m.pred_index], "b": preds_b[m.gt_index], "iou": m.iou}
476             for m in mr.matches]
477         agree_iou = [m.iou for m in mr.matches]
478         a_only = [preds_a[i] for i in mr.unmatched_pred_indices]
479         b_only = [preds_b[i] for i in mr.unmatched_gt_indices]
480
481         def _dur(ivs: List[Interval]) -> float:
482             return sum(e - s for s, e in ivs)
483
484         return {
485             "video": video.name,

```

```

486         "agree_iou": agree_iou,
487         "variant_a": variant_a.name,
488         "variant_b": variant_b.name,
489         "num_a": len(preds_a),
490         "num_b": len(preds_b),
491         "num_agreed": len(agreed),
492         "mean_agree_iou": (sum(agree_ious) / len(agree_ious)) if agree_ious else 0.0,
493         "duration_a": _dur(preds_a),
494         "duration_b": _dur(preds_b),
495         "agreed": agreed,
496         "a_only": a_only,
497         "b_only": b_only,
498     }
499
500
501 # ----- leaderboard / DB
502
503
504 def record_experiment(result: Dict[str, Any], path: str = DEFAULT_LEADERBOARD_PATH,
505                      timestamp: Optional[str] = None) -> Dict[str, Any]:
506     """Append a compact, reproducible record of a `compare()` result to the JSON
507     leaderboard (`EXPERIMENT_HARNESS.md` §6: experiments are records). Generalises
508     `regression_baseline.json`; deliberately the size of a baseline file, not a service
509     (§9). Returns the row written. ``timestamp`` is injectable for deterministic tests.
510     """
511     row: Dict[str, Any] = {
512         "timestamp": timestamp or datetime.now(timezone.utc).isoformat(),
513         "iou": result.get("iou"),
514         "label_set_hash": result.get("label_set_hash"),
515         "num_videos": result.get("num_videos"),
516         "variant_a": {"name": result["variant_a"]["variant"],
517                      "spec_hash": result["variant_a"]["spec_hash"],
518                      "aggregate": result["variant_a"]["aggregate"]},
519         "variant_b": {"name": result["variant_b"]["variant"],
520                      "spec_hash": result["variant_b"]["spec_hash"],
521                      "aggregate": result["variant_b"]["aggregate"]},
522         "delta": result.get("delta"),
523     }
524     # Record the honest ?F1 (CI + sign test) when present, so the leaderboard carries
525     # the
526     # promotion-grade evidence, not just the bare-mean delta (additive ? C1).
527     honest = result.get("honest") or {}
528     if honest.get("paired_delta_f1"):
529         row["honest_delta_f1"] = honest["paired_delta_f1"]
530     entries: List[Dict[str, Any]] = []
531     if os.path.isfile(path):
532         with open(path, encoding="utf-8") as f:
533             entries = json.load(f)
534     entries.append(row)
535     os.makedirs(os.path.dirname(path) or ".", exist_ok=True)
536     with open(path, "w", encoding="utf-8") as f:
537         json.dump(entries, f, indent=2)
538     logger.info("recorded experiment %s vs %s (?f1=%3f) -> %s",
539               row["variant_a"]["name"], row["variant_b"]["name"],
540               (result.get("delta") or {}).get("f1", 0.0), path)
541     return row
542
543 def load_video_candidates(db: Any, video_id: str, *, detector: Optional[str] = None,
544                          gts: Optional[List[Interval]] = None,
545                          feature_names: Sequence[str] = ()) -> VideoCandidates:
546     """Bridge persisted detection ? `VideoCandidates` via
547     `Database.query_candidate_segments`.
548     ``feature_names`` selects which persisted features to lift out of each row's

```

```

``stats``
549     JSON (``"duration"`` is derived from the window bounds). This is the only DB-
touching
550     function; the comparison itself is pure and offline. Exercised owner-side on a real
DB,
551     but works against any `Database` (including an in-memory one in tests).
552     """
553     rows = db.query_candidate_segments(video_id, detector=detector)
554     intervals: List[Interval] = [(float(r["start_time"]), float(r["end_time"])) for r in
rows]
555     features: Dict[str, List[float]] = {}
556     for name in feature_names:
557         if name == "duration":
558             features[name] = [float(r["end_time"]) - float(r["start_time"]) for r in
rows]
559         else:
560             features[name] = [float((r.get("stats") or {}).get(name, 0.0)) for r in
rows]
561     return VideoCandidates(name=video_id, intervals=intervals, features=features,
gts=gts)
562

```

13. assistant (2026-06-16T06:27:23.363Z)

Now the remaining eval modules and detectors, plus the CI job and fixtures.

14. user (2026-06-16T06:27:24.908Z)

```

1     """Pure temporal interval-matching metrics for rally evaluation.
2
3     No I/O, no DB, no global state ? every function here is deterministic and unit
4     tested. An "interval" is a `(start, end)` tuple of floats in seconds with
5     start < end. We score a list of *predicted* rally intervals against a list of
6     *ground-truth* intervals via temporal IoU and one-to-one greedy matching.
7     """
8
9     from dataclasses import dataclass, field
10    from typing import List, Tuple, Optional
11
12    Interval = Tuple[float, float]
13
14
15    def interval_iou(a: Interval, b: Interval) -> float:
16        """Temporal intersection-over-union of two intervals.
17
18        Returns 0.0 when they don't overlap (or either is degenerate). IoU is
19        overlap / union, both measured as durations on the timeline.
20        """
21        a_start, a_end = a
22        b_start, b_end = b
23        if a_end <= a_start or b_end <= b_start:
24            return 0.0
25        inter = max(0.0, min(a_end, b_end) - max(a_start, b_start))
26        if inter <= 0.0:
27            return 0.0
28        union = (a_end - a_start) + (b_end - b_start) - inter
29        return inter / union if union > 0 else 0.0
30
31
32    @dataclass
33    class Match:
34        """One matched predicted?ground-truth pair."""
35        pred_index: int
36        gt_index: int
37        iou: float

```

```

38
39
40 @dataclass
41 class MatchResult:
42     """Outcome of matching predictions to ground truth at a given IoU threshold."""
43     matches: List[Match] = field(default_factory=list)
44     unmatched_pred_indices: List[int] = field(default_factory=list) # false positives
45     unmatched_gt_indices: List[int] = field(default_factory=list)    # false negatives
(misses)
46
47
48 def match_intervals(preds: List[Interval], gts: List[Interval],
49                     iou_threshold: float = 0.5) -> MatchResult:
50     """Greedy one-to-one matching of predictions to ground truth by descending IoU.
51
52     Every candidate (pred, gt) pair with IoU >= threshold is considered; pairs
53     are claimed highest-IoU first, and each prediction/GT can be used once. This
54     keeps a single prediction from being credited for two ground-truth rallies
55     (and vice versa).
56     """
57     candidates = []
58     for pi, p in enumerate(preds):
59         for gi, g in enumerate(gts):
60             iou = interval_iou(p, g)
61             # Require real overlap: at iou_threshold=0.0 a plain `iou >= threshold`
would
62             # match completely disjoint intervals (IoU 0.0), fabricating perfect scores.
63             if iou > 0.0 and iou >= iou_threshold:
64                 candidates.append((iou, pi, gi))
65             # Highest IoU first; ties broken by indices for determinism.
66             candidates.sort(key=lambda c: (-c[0], c[1], c[2]))
67
68     used_pred, used_gt = set(), set()
69     matches: List[Match] = []
70     for iou, pi, gi in candidates:
71         if pi in used_pred or gi in used_gt:
72             continue
73         used_pred.add(pi)
74         used_gt.add(gi)
75         matches.append(Match(pred_index=pi, gt_index=gi, iou=iou))
76
77     unmatched_pred = [i for i in range(len(preds)) if i not in used_pred]
78     unmatched_gt = [i for i in range(len(gts)) if i not in used_gt]
79     return MatchResult(matches=matches,
80                        unmatched_pred_indices=unmatched_pred,
81                        unmatched_gt_indices=unmatched_gt)
82
83
84 @dataclass
85 class Score:
86     """Aggregate metrics for one (predictions vs ground-truth) evaluation."""
87     iou_threshold: float
88     num_pred: int
89     num_gt: int
90     true_positives: int
91     false_positives: int
92     false_negatives: int
93     precision: float
94     recall: float
95     f1: float
96     mean_iou: float # over matched pairs (0.0 if none)
97     mean_start_error: float # mean |pred_start - gt_start| over matches, seconds
98     mean_end_error: float # mean |pred_end - gt_end| over matches, seconds
99
100     def as_dict(self) -> dict:

```

```

101         return self.__dict__.copy()
102
103
104     def score(preds: List[Interval], gts: List[Interval],
105              iou_threshold: float = 0.5,
106              match_result: Optional[MatchResult] = None) -> Score:
107         """Compute precision/recall/F1, mean matched IoU, and boundary errors.
108
109         Pass a precomputed `match_result` to avoid re-matching (e.g. when also
110         extracting the failure lists); otherwise matching is done here.
111         """
112         mr = match_result if match_result is not None else match_intervals(preds, gts,
113         iou_threshold)
114
115         tp = len(mr.matches)
116         fp = len(mr.unmatched_pred_indices)
117         fn = len(mr.unmatched_gt_indices)
118
119         precision = tp / (tp + fp) if (tp + fp) > 0 else 0.0
120         recall = tp / (tp + fn) if (tp + fn) > 0 else 0.0
121         f1 = (2 * precision * recall / (precision + recall)) if (precision + recall) > 0
122     else 0.0
123
124         if mr.matches:
125             mean_iou = sum(m.iou for m in mr.matches) / tp
126             mean_start_err = sum(abs(preds[m.pred_index][0] - gts[m.gt_index][0]) for m in
127         mr.matches) / tp
128             mean_end_err = sum(abs(preds[m.pred_index][1] - gts[m.gt_index][1]) for m in
129         mr.matches) / tp
130         else:
131             mean_iou = mean_start_err = mean_end_err = 0.0
132
133         return Score(
134             iou_threshold=iou_threshold,
135             num_pred=len(preds), num_gt=len(gts),
136             true_positives=tp, false_positives=fp, false_negatives=fn,
137             precision=precision, recall=recall, f1=f1,
138             mean_iou=mean_iou, mean_start_error=mean_start_err, mean_end_error=mean_end_err,
139         )
140
141
142     def pr_curve(preds: List[Interval], gts: List[Interval],
143                 thresholds: Optional[List[float]] = None) -> List[Score]:
144         """Score across a sweep of IoU thresholds (default 0.1..0.9 step 0.1).
145
146         Lets you see strict-vs-lenient behaviour rather than a single operating point.
147         """
148         if thresholds is None:
149             thresholds = [round(0.1 * i, 1) for i in range(1, 10)]
150         return [score(preds, gts, iou_threshold=t) for t in thresholds]
151

```

15. user (2026-06-16T06:27:25.277Z)

```

1     """Canonical ground-truth interval loader ? THE one GT/golden-CSV reader (ADR-008
2     prereq).
3
4     The repo historically carried two incompatible loaders:
5     - ``annotations.load_annotations``: positional ``start,end`` (the collector's
6     ``curate export`` format), raises on bad rows;
7     - ``calibrate_wasb.load_golden_gts``: DictReader requiring ``start_time/end_time``
8     (the rally-annotator golden format), silently skipping bad rows ? fed 7 modules.
9
10    A file in one convention fed to the other loader yielded zero/garbage intervals with no
11    error. This module accepts BOTH conventions (header-mapped, else positional), so every

```



```

11     quantity the ship-gate depends on (gt_hash, LOVO scores, the future consent fingerprint)
12     flows through one reader. Both legacy entry points now delegate here.
13
14     Accepted shapes (one rally per row; blank lines and ``#`` comments ignored):
15     - header with ``start``/``end`` OR ``start_time``/``end_time`` columns (any other
16       columns ? rally_number, ending_reason, sport, ? ? are ignored);
17     - no header: columns 0,1 positionally.
18     Timestamps: decimal seconds or ``MM:SS`` / ``HH:MM:SS``
19     (``annotations.parse_timestamp``).
20
21     """
22     import csv
23     import logging
24     from typing import List, Optional, Tuple
25
26     from backend.eval.annotations import parse_timestamp
27
28     logger = logging.getLogger(__name__)
29
30     Interval = Tuple[float, float]
31
32     _START_NAMES = ("start", "start_time")
33     _END_NAMES = ("end", "end_time")
34
35     def _header_columns(row: List[str]) -> Optional[Tuple[int, int]]:
36         """If `row` is a header naming start/end columns, return their indices, else
37         None."""
38         cells = [(c or "").strip().lower() for c in row]
39         start_idx = next((i for i, c in enumerate(cells) if c in _START_NAMES), None)
40         end_idx = next((i for i, c in enumerate(cells) if c in _END_NAMES), None)
41         if start_idx is not None and end_idx is not None:
42             return start_idx, end_idx
43         return None
44
45     def load_gt_intervals(csv_path: str, *, strict: bool = True) -> List[Interval]:
46         """Load GT rally intervals from either golden-CSV convention.
47
48         strict=True (corpus/gate use): raise ValueError with file:line on any malformed row
49         (bad timestamp, end <= start, negative start) so labeling mistakes fail loudly.
50         strict=False (legacy tuning-driver behavior): skip bad rows, but LOG how many were
51         skipped ? a silently shrinking GT set flatters precision.
52
53         Returns intervals sorted by start; overlapping GT intervals are logged as a warning
54         (they break one-to-one matching and usually indicate a labeling error).
55
56         """
57         intervals: List[Interval] = []
58         skipped = 0
59         cols: Optional[Tuple[int, int]] = None # (start_idx, end_idx); None until detected
60         saw_header = False
61
62         with open(csv_path, newline="", encoding="utf-8-sig") as f:
63             for lineno, row in enumerate(csv.reader(f), start=1):
64                 if not row or not (row[0] or "").strip() or row[0].strip().startswith("#"):
65                     continue
66                 if cols is None and not saw_header:
67                     header = _header_columns(row)
68                     if header is not None:
69                         cols = header
70                         saw_header = True
71                     continue # consume the header row
72                 cols = (0, 1) # headerless: positional start,end
73                 assert cols is not None # every branch above assigns it
74                 si, ei = cols
75                 try:

```

```

74         if len(row) <= max(si, ei):
75             raise ValueError(f"expected at least {max(si, ei) + 1} columns, got
{row!r}")
76         start = parse_timestamp(row[si])
77         end = parse_timestamp(row[ei])
78         if start < 0:
79             raise ValueError(f"negative start ({start})")
80         if start >= end:
81             raise ValueError(f"start ({start}) >= end ({end})")
82     except ValueError as e:
83         if strict:
84             raise ValueError(f"{csv_path}:{lineno}: {e}") from e
85         skipped += 1
86         continue
87     intervals.append((start, end))
88
89     if skipped:
90         logger.warning("%s: skipped %d malformed GT row(s) (strict=False)", csv_path,
skipped)
91
92     intervals.sort(key=lambda iv: iv[0])
93     for (s1, e1), (s2, e2) in zip(intervals, intervals[1:]):
94         if s2 < e1:
95             logger.warning("%s: overlapping GT intervals (%.3f-%.3f vs %.3f-%.3f) ? "
96                             "labeling error? Overlaps break one-to-one matching.",
97                             csv_path, s1, e1, s2, e2)
98     return intervals
99

```

16. user (2026-06-16T06:27:25.938Z)

```

1     """Regression runner ? verify the production tool against golden-set ground truth.
2
3     Implements the explicit ask: *"given a video, get a golden set created and verify
4     its precision and recall against the production tool."* This is deliberately a thin
5     wrapper over machinery that already exists:
6
7     ground truth ? AnnotationProvider (file / vendor / oracle) [backend.annotations]
8     predictions  ? the production pipeline's stored DB rows    [backend.eval.harness]
9     comparison   ? interval matching + P/R/F1                  [backend.eval.metrics]
10    baseline     ? a per-video snapshot of "known good" numbers [this module]
11
12    A "regression" is a precision/recall drop beyond `tolerance` versus the snapshotted
13    baseline. Tiers map to annotation providers: e.g. tier "file" ? FileProvider (CI),
14    later tier "vendor" ? HumanVendorProvider (authoritative), tier "auto" ? an
15    independent-lineage oracle (fast smoke alarm). See docs/GOLDEN_SET_IMPLEMENTATION.md.
16    """
17    import os
18    import json
19    import hashlib
20    import logging
21    from dataclasses import dataclass, asdict
22    from typing import Dict, List, Optional
23
24    from backend.infrastructure.database import Database
25    from backend.eval import metrics
26    from backend.eval.harness import get_predictions
27    from backend.annotations import annotation_registry
28
29    logger = logging.getLogger(__name__)
30
31    # Tracked location (NOT under the gitignored output/), so a fresh checkout / CI run can
32    # actually load a committed baseline to compare against.
33    DEFAULT_BASELINE_PATH = os.path.join("eval_baselines", "regression_baseline.json")
34    DEFAULT_TOLERANCE = 0.05

```

```

35
36
37     def gt_hash(ground_truth: List[metrics.Interval]) -> str:
38         """Stable short hash of the GT intervals ? detects a baseline taken against
different labels."""
39         norm = sorted((round(float(s), 3), round(float(e), 3)) for s, e in ground_truth)
40         return hashlib.sha1(json.dumps(norm).encode()).hexdigest()[:12]
41
42
43     @dataclass
44     class RegressionResult:
45         video_id: str
46         stage: str
47         iou_threshold: float
48         precision: float
49         recall: float
50         f1: float
51         # Baseline values compared against (None when no baseline existed yet).
52         baseline_precision: Optional[float]
53         baseline_recall: Optional[float]
54         tolerance: float
55         regressed: bool
56         reasons: List[str]
57         # True only when a baseline existed to compare against. Distinguishes a genuine PASS
58         # from "nothing to compare" so the CLI can refuse to silently green-light the
latter.
59         has_baseline: bool = False
60         gt_hash: Optional[str] = None
61
62         def as_dict(self) -> dict:
63             return asdict(self)
64
65
66     # ----- baseline store
67
68     def load_baselines(path: str = DEFAULT_BASELINE_PATH) -> Dict[str, dict]:
69         """Load the per-video baseline snapshot file (returns {} if absent)."""
70         if not os.path.isfile(path):
71             return {}
72         with open(path) as f:
73             return json.load(f)
74
75
76     def _baseline_key(video_id: str, stage: str) -> str:
77         return f"{video_id}::{stage}"
78
79
80     def save_baseline(video_id: str, stage: str, precision: float, recall: float,
81                     f1: float, path: str = DEFAULT_BASELINE_PATH,
82                     iou_threshold: Optional[float] = None, detector: Optional[str] = None,
83                     gt_fingerprint: Optional[str] = None) -> None:
84         """Snapshot a video/stage's current numbers as the new 'known good' baseline.
85
86         Records the iou_threshold, detector, and a GT fingerprint alongside the metrics so a
87         later run computed under different settings (or against different labels) is
detected
88         as incommensurable rather than silently compared.
89         """
90         baselines = load_baselines(path)
91         baselines[_baseline_key(video_id, stage)] = {
92             "video_id": video_id, "stage": stage,
93             "precision": precision, "recall": recall, "f1": f1,
94             "iou_threshold": iou_threshold, "detector": detector, "gt_hash": gt_fingerprint,
95         }
96         os.makedirs(os.path.dirname(path) or ".", exist_ok=True)

```

```

97         with open(path, "w") as f:
98             json.dump(baselines, f, indent=2, sort_keys=True)
99             logger.info("Saved baseline for %s (precision=%.3f recall=%.3f)",
100 _baseline_key(video_id, stage), precision, recall)
101
102 # ----- the runner
103
104 def regress(db: Database, video_id: str, ground_truth: List[metrics.Interval],
105             stage: str = "final", iou_threshold: float = 0.5,
106             detector: Optional[str] = None,
107             tolerance: float = DEFAULT_TOLERANCE,
108             baseline_path: str = DEFAULT_BASELINE_PATH) -> RegressionResult:
109     """Score stored predictions for one video against ground truth and compare to
baseline.
110
111     `ground_truth` is a list of (start, end) intervals ? typically obtained from an
112     AnnotationProvider (see `regress_with_provider`). A regression is flagged when
113     precision OR recall falls more than `tolerance` below the snapshotted baseline.
114     If no baseline exists yet, `regressed` is False (nothing to compare to) and the
115     caller can choose to snapshot the current numbers via `save_baseline`.
116     """
117     preds = get_predictions(db, video_id, stage, detector=detector)
118     s = metrics.score(preds, ground_truth, iou_threshold)
119     fp = gt_hash(ground_truth)
120
121     baselines = load_baselines(baseline_path)
122     b = baselines.get(_baseline_key(video_id, stage))
123
124     reasons: List[str] = []
125     regressed = False
126     has_baseline = b is not None
127     base_p = base_r = None
128     if b is not None:
129         base_p, base_r = b["precision"], b["recall"]
130         # Incommensurable comparison guard: a baseline taken at a different IoU/detector
or
131         # against different labels is not a valid reference ? flag it instead of
trusting it.
132         b_iou, b_det, b_hash = b.get("iou_threshold"), b.get("detector"),
b.get("gt_hash")
133         if b_iou is not None and abs(float(b_iou) - iou_threshold) > 1e-9:
134             regressed = True
135             reasons.append(f"baseline IoU {b_iou} != run IoU {iou_threshold}
(incommensurable)")
136         if b_det is not None and b_det != detector:
137             regressed = True
138             reasons.append(f"baseline detector {b_det!r} != run detector {detector!r}
(incommensurable)")
139         if b_hash is not None and b_hash != fp:
140             regressed = True
141             reasons.append(f"baseline GT hash {b_hash} != run GT hash {fp} (labels
changed)")
142         if s.precision < base_p - tolerance:
143             regressed = True
144             reasons.append(f"precision {s.precision:.3f} < baseline {base_p:.3f} -
{tolerance}")
145         if s.recall < base_r - tolerance:
146             regressed = True
147             reasons.append(f"recall {s.recall:.3f} < baseline {base_r:.3f} -
{tolerance}")
148         else:
149             reasons.append("no baseline recorded for this video/stage ? nothing to compare")
150
151     return RegressionResult(

```

```

152         video_id=video_id, stage=stage, iou_threshold=iou_threshold,
153         precision=s.precision, recall=s.recall, f1=s.f1,
154         baseline_precision=base_p, baseline_recall=base_r,
155         tolerance=tolerance, regressed=regressed, reasons=reasons,
156         has_baseline=has_baseline, gt_hash=fp,
157     )
158
159
160     def regress_with_provider(db: Database, video_id: str, video_ref: str,
161                             tier: str = "file", stage: str = "final",
162                             iou_threshold: float = 0.5,
163                             detector: Optional[str] = None,
164                             tolerance: float = DEFAULT_TOLERANCE,
165                             baseline_path: str = DEFAULT_BASELINE_PATH,
166                             guideline: Optional[str] = None,
167                             sport: str = "badminton",
168                             provider_kwargs: Optional[dict] = None) -> RegressionResult:
169         """End-to-end: obtain ground truth from the `tier` annotation provider, then
regress.
170
171         `tier` is an annotation-provider registry key ("file" today; "vendor"/"auto"
172         once those providers land). `video_ref` is whatever that provider expects (a CSV
173         path for FileProvider, a video_id/URL for a vendor). `video_id` keys the stored
174         predictions in the DB (the file MD5).
175         """
176         provider = annotation_registry.get(tier, **(provider_kwargs or {}))
177         gt = provider.annotate(video_ref, guideline=guideline, sport=sport)
178         return regress(db, video_id, gt, stage=stage, iou_threshold=iou_threshold,
179                       detector=detector, tolerance=tolerance, baseline_path=baseline_path)
180
181
182     def format_result(r: RegressionResult) -> str:
183         verdict = "REGRESSED" if r.regressed else "ok"
184         head = (f"[{verdict}] {r.video_id} ({r.stage}) "
185               f"precision={r.precision:.3f} recall={r.recall:.3f} f1={r.f1:.3f}")
186         if r.baseline_precision is not None:
187             head += f" | baseline p={r.baseline_precision:.3f} r={r.baseline_recall:.3f}
tol={r.tolerance}"
188         return head + ".join(f'\n    - {why}' for why in r.reasons)
189

```

17. assistant (2026-06-16T06:27:29.577Z)

Now the detectors and the CI job, plus the fixtures directory.

18. user (2026-06-16T06:27:30.612Z)

```

1     """TrackNet shuttle-trajectory runner (WSL2 subprocess adapter).
2
3     This bridges the Windows-side indexer to a TensorFlow TrackNetV4 install living
4     inside WSL2 (ADR-001 in docs/DECISIONS.md). The flow is:
5
6         1. Translate the Windows video path to its WSL-visible form (and optionally
7           stage it into the WSL filesystem to avoid slow /mnt/c per-frame reads).
8         2. Invoke ``src/predict.py`` inside the TrackNetV4 conda env via ``wsl``.
9         3. Read back the per-frame ``Frame,Visibility,X,Y`` CSV it writes.
10        4. Convert that trajectory into high-motion "action windows" ? the same
11           candidate-rally shape HybridYoloSegmenter produces ? so the rest of the
12           pipeline (AI handoff, DB population) is unchanged.
13
14     Nothing here imports TensorFlow; all model code stays in WSL.
15     """
16
17     import os
18     import csv

```

```

19     import shlex
20     import logging
21     import subprocess
22     from dataclasses import dataclass
23     from typing import List, Dict, Any, Optional, Tuple
24
25     from backend.pipeline.detectors.base import DetectorRunner, WslCommandMixin,
wsl_tilde_quote
26
27     logger = logging.getLogger(__name__)
28
29
30     @dataclass
31     class TrackNetConfig:
32         """Resolved settings for a TrackNet WSL run (built from config.json ->
indexing.tracknet)."""
33         distro: str = "Ubuntu"
34         repo_dir: str = "~/models/TrackNetV4"           # WSL path to the cloned repo
35         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
36         conda_env: str = "TrackNetV4"
37         weights_path: str = ""                          # WSL path to the .h5/.keras weights
(REQUIRED)
38         queue_length: int = 5
39         stage_in_wsl: bool = True                       # copy video into WSL fs first
(perf)
40         wsl_stage_dir: str = "~/clips"                  # where staged videos/outputs live
in WSL
41         timeout_sec: int = 1800                         # 30 min; kills a hung call (0 = no
timeout)
42         keep_frames: bool = False                      # retain staged video after success
(debug only)
43         min_free_gb: float = 0.0                       # warn if WSL free space below this
(0 = off)
44
45     @classmethod
46     def from_indexing_cfg(cls, idx_cfg: Dict[str, Any]) -> "TrackNetConfig":
47         tn = (idx_cfg or {}).get("tracknet", {}) or {}
48         return cls(
49             distro=tn.get("wsl_distro", "Ubuntu"),
50             repo_dir=tn.get("repo_dir", "~/models/TrackNetV4"),
51             conda_sh=tn.get("conda_sh", "~/miniconda3/etc/profile.d/conda.sh"),
52             conda_env=tn.get("conda_env", "TrackNetV4"),
53             weights_path=tn.get("weights_path", ""),
54             queue_length=int(tn.get("queue_length", 5)),
55             stage_in_wsl=bool(tn.get("stage_in_wsl", True)),
56             wsl_stage_dir=tn.get("wsl_stage_dir", "~/clips"),
57             timeout_sec=int(tn.get("timeout_sec", 1800)),
58             keep_frames=bool(tn.get("keep_frames", False)),
59             min_free_gb=float(tn.get("min_free_gb", 0.0)),
60         )
61
62
63     def to_wsl_mnt_path(win_path: str) -> str:
64         """Translate a Windows path (C:\\Users\\x) to its WSL /mnt form (/mnt/c/Users/x).
65
66         Already-POSIX paths (starting with / or ~) are returned unchanged so the
67         same helper is safe to call on either kind of path.
68         """
69         p = str(win_path)
70         if p.startswith("/") or p.startswith("~"):
71             return p
72         p = p.replace("\\", "/")
73         if len(p) >= 2 and p[1] == ":":
74             drive = p[0].lower()
75             return f"/mnt/{drive}{p[2:]}"

```

```

76         return p
77
78
79     @dataclass
80     class TrajectoryPoint:
81         frame: int
82         visible: bool
83         x: float
84         y: float
85
86
87     class TrackNetRunner(WslCommandMixin, DetectorRunner):
88         """Runs TrackNet predict.py in WSL and turns the output CSV into action windows.
89
90         Implements the common DetectorRunner contract so a future Linux-native or
91         alternative trajectory runner can be substituted without touching the hybrids.
92         """
93
94         def __init__(self, cfg: TrackNetConfig):
95             self.cfg = cfg
96
97         def healthcheck(self) -> Tuple[bool, str]:
98             """Verify the WSL env is usable: conda env exists, TF imports, GPU visible.
99
100             Returns (ok, message). Cheap to call before a long run.
101             """
102             cmd = (
103                 f"conda activate {self.cfg.conda_env} && "
104                 f"python -c \"import tensorflow as tf; "
105                 f"print('TF', tf.__version__, 'GPUs', "
106                 f"len(tf.config.list_physical_devices('GPU')))\\"
107             )
108             try:
109                 res = self._wsl_bash(cmd)
110             except FileNotFoundError:
111                 return False, "wsl` not found ? is this running on Windows with WSL2
112                 installed?"
113             except subprocess.TimeoutExpired:
114                 return False, "WSL healthcheck timed out."
115             out = (res.stdout or "").strip()
116             if res.returncode != 0:
117                 return False, f"WSL env check failed: {(res.stderr or out).strip()}"
118             return True, out
119
120         # ----- inference
121
122         def run_predict(self, video_win_path: str, output_win_dir: str,
123                        video_id: Optional[str] = None) -> Optional[str]:
124             """Run predict.py on a video. Returns the Windows path to the produced CSV, or
125             None.
126
127             ``output_win_dir`` is a Windows directory (under the repo's output/) that
128             WSL writes into via its /mnt view, so the Windows process can read the CSV
129             back with no copy. ``video_id`` (file MD5) namespaces the staged video ? and
130             therefore predict.py's ``<stem>_predict.csv`` output ? so two videos that share
131             a filename cannot collide on the output CSV.
132             """
133             if not self.cfg.weights_path:
134                 logger.error(
135                     "TrackNet weights_path is not set. Set indexing.tracknet.weights_path "
136                     "in config.json to the WSL path of your trained TrackNetV4 weights."
137                 )
138             return None
139
140         # Resolve relative dirs BEFORE the /mnt translation ? to_wsl_mnt_path passes

```

```

138         # relative paths through unchanged, so WSL would resolve them against the
139         # predict.py cwd instead of the repo (same bug class as wasb_runner).
140         output_win_dir = os.path.abspath(output_win_dir)
141         os.makedirs(output_win_dir, exist_ok=True)
142         # Normalize separators so basename/splitext work when tests (or collector)
143         # pass Windows-style paths on a Linux host.
144         video_win_path = str(video_win_path).replace("\\", "/")
145         base = os.path.basename(video_win_path)
146         stem, ext = os.path.splitext(base)
147
148         # predict.py only accepts .avi/.mp4 and names the CSV "<stem>_predict.csv".
149         if ext.lower() not in (".mp4", ".avi"):
150             logger.error("TrackNet predict.py only supports .mp4/.avi (got %s)", ext)
151             return None
152
153         wsl_out = to_wsl_mnt_path(output_win_dir)
154         # Namespace by video_id so two same-filename videos don't collide on the CSV.
155         # predict.py derives its output name from the (staged) video stem, so staging
156         # under the namespaced name propagates into "<key>_predict.csv".
157         key = f"{stem}_{video_id[:12]}" if video_id else stem
158         expected_csv_win = os.path.join(output_win_dir, f"{key}_predict.csv")
159
160         # Resolve the video path WSL will read.
161         if self.cfg.stage_in_wsl:
162             staged = self._stage_video(video_win_path, f"{key}{ext}")
163             if staged is None:
164                 return None
165             wsl_video = staged
166         else:
167             # No staging: predict.py reads /mnt directly and writes
168             # "<stem>_predict.csv".
169             # We cannot rename its output, so fall back to the un-namespaced name.
170             wsl_video = to_wsl_mnt_path(video_win_path)
171             expected_csv_win = os.path.join(output_win_dir, f"{stem}_predict.csv")
172
173         if self.cfg.min_free_gb > 0:
174             free = self._wsl_free_gb(self.cfg.wsl_stage_dir)
175             if free is not None and free < self.cfg.min_free_gb:
176                 logger.warning("Low WSL disk: %.1f GB free at %s (< min_free_gb=%.1f); "
177                               "consider running tools/cleanup_caches.py.",
178                               free, self.cfg.wsl_stage_dir, self.cfg.min_free_gb)
179
180         cmd = (
181             f"conda activate {self.cfg.conda_env} && "
182             f"cd {self.cfg.repo_dir}/src && "
183             f"python predict.py "
184             f"--video_path {wsl_tilde_quote(wsl_video)} "
185             f"--model_weights {wsl_tilde_quote(self.cfg.weights_path)} "
186             f"--output_dir {wsl_tilde_quote(wsl_out)} "
187             f"--queue_length {self.cfg.queue_length}"
188         )
189         logger.info("Running TrackNet inference on %s ...", base)
190         try:
191             res = self._wsl_bash(cmd)
192         except subprocess.TimeoutExpired:
193             logger.error("TrackNet inference timed out after %ss.",
194                         self.cfg.timeout_sec)
195             return None
196
197         if res.returncode != 0:
198             logger.error("TrackNet predict.py failed:\n%s", (res.stderr or
199                     res.stdout)[-2000:])
200             return None
201
202         if not os.path.exists(expected_csv_win):

```



```

200         logger.error("TrackNet finished but expected CSV not found at %s",
expected_csv_win)
201         return None
202         logger.info("TrackNet trajectory CSV: %s", expected_csv_win)
203
204         # Success ? the CSV is the durable output; drop the staged video copy (a pure,
205         # regenerable intermediate). On earlier failure we return above without
cleaning.
206         if self.cfg.stage_in_wsl and not self.cfg.keep_frames:
207             logger.info("Cache hygiene: removing staged video for %s "
208                         "(set indexing.tracknet.keep_frames=true to retain).", key)
209             self._wsl_rm_rf([wsl_video])
210             return expected_csv_win
211
212     def _stage_video(self, video_win_path: str, base: str) -> Optional[str]:
213         """Copy the input video into the WSL filesystem (avoids slow /mnt/c reads).
214
215         Returns the WSL path to the staged copy, or None on failure.
216         """
217         src_mnt = to_wsl_mnt_path(video_win_path)
218         reason = self.wsl_source_unreadable_reason(src_mnt)
219         if reason:
220             logger.error("Cannot stage video into WSL: %s", reason)
221             return None
222         dest = f"{self.cfg.wsl_stage_dir}/{base}"
223         cmd = f"mkdir -p {self.cfg.wsl_stage_dir} && cp {shlex.quote(src_mnt)}
{wsl_tilde_quote(dest)} && echo OK"
224         try:
225             res = self._wsl_bash(cmd)
226         except subprocess.TimeoutExpired:
227             logger.error("Staging video into WSL timed out.")
228             return None
229         if res.returncode != 0 or "OK" not in (res.stdout or ""):
230             logger.error("Failed to stage video into WSL: %s", (res.stderr or
res.stdout).strip())
231             return None
232         return dest
233
234     # ----- CSV -> windows
235
236     @staticmethod
237     def parse_trajectory_csv(csv_win_path: str) -> List[TrajectoryPoint]:
238         """Parse a TrackNet predict CSV (columns: Frame,Visibility,X,Y)."""
239         points: List[TrajectoryPoint] = []
240         with open(csv_win_path, newline="") as f:
241             reader = csv.DictReader(f)
242             for row in reader:
243                 try:
244                     points.append(TrajectoryPoint(
245                         frame=int(row["Frame"]),
246                         visible=int(row["Visibility"]) == 1,
247                         x=float(row["X"]),
248                         y=float(row["Y"]),
249                     ))
250                 except (KeyError, ValueError):
251                     continue
252             points.sort(key=lambda p: p.frame)
253         return points
254
255     @staticmethod
256     def trajectory_to_action_windows(
257         points: List[TrajectoryPoint],
258         fps: float,
259         frame_width: float,
260         chunk_start: float = 0.0,

```

```

261     velocity_thresh: float = 0.02,
262     merge_gap: float = 2.0,
263     min_window_duration: float = 2.0,
264     chunk_index: Optional[int] = None,
265     max_window_duration: float = 0.0,
266     min_split_gap: float = 0.5,
267 ) -> List[Dict[str, Any]]:
268     """Convert a shuttle trajectory into candidate rally windows.
269
270     A frame is "active" when the shuttle is visible AND its inter-frame
271     displacement (normalised by frame width, per second) exceeds
272     ``velocity_thresh`` ? i.e. the shuttle is in flight. Active frames within
273     ``merge_gap`` seconds of each other are grouped into one window; short
274     windows are dropped. Mirrors HybridYoloSegmenter's windowing so the dict
275     shape feeding the AI-handoff phase is identical.
276
277     ``max_window_duration`` (0 = OFF, the default ? behaviour unchanged): a guard
278     against the *over-merge* failure where "shuttle moving" is conflated with "rally in
279     progress" and a single window swallows several rallies + the dead time between them. When
280     set, any window longer than this is recursively SPLIT at its largest internal inactivity
281     gap (the longest stretch with no in-flight shuttle ? the most likely rally boundary),
282     provided that gap is at least ``min_split_gap`` seconds. This is the bounded-windowing
283     fix from ``docs/RALLY_QUALITY_RESEARCH.md`` §6 (W1); it never merges, only cuts, so
284     recall cannot drop, and it is config-gated default-OFF until measured on the golden set.
285     """
286     if fps <= 0:
287         fps = 30.0
288     if frame_width <= 0:
289         frame_width = 1280.0
290
291     # Compute per-frame normalised velocity from the last *visible* point.
292     active = [] # (frame_idx, velocity)
293     prev: Optional[TrajectoryPoint] = None
294     for p in points:
295         if not p.visible:
296             prev = None # break the trajectory; absent shuttle = not in flight
297             continue
298         if prev is not None:
299             dframes = p.frame - prev.frame
300             if dframes > 0:
301                 dist = ((p.x - prev.x) ** 2 + (p.y - prev.y) ** 2) ** 0.5
302                 velocity = (dist / frame_width) * (fps / dframes)
303                 if velocity > velocity_thresh:
304                     active.append((p.frame, velocity))
305             prev = p
306
307     if not active:
308         return []
309
310     max_gap_frames = int(merge_gap * fps)
311
312     def _finalize(group: List[Tuple[int, float]]) -> Dict[str, Any]:
313         vels = [v for _, v in group]
314         return {
315             "start": group[0][0] / fps + chunk_start,
316             "end": group[-1][0] / fps + chunk_start,
317             "active_frame_count": len(group),
318             "peak_velocity": max(vels),

```

```

319         "mean_velocity": sum(vels) / len(vels),
320         "track_id_count": 1, # single shuttle; kept for window-shape parity
321         "chunk_index": chunk_index,
322     }
323
324     groups: List[List[Tuple[int, float]]] = []
325     group = [active[0]]
326     for af in active[1:]:
327         if af[0] - group[-1][0] <= max_gap_frames:
328             group.append(af)
329         else:
330             groups.append(group)
331             group = [af]
332     groups.append(group)
333
334     if max_window_duration > 0:
335         groups = TrackNetRunner._split_overlong_groups(
336             groups, int(max_window_duration * fps), int(min_split_gap * fps))
337
338     windows = [_finalize(g) for g in groups]
339     return [w for w in windows if (w["end"] - w["start"]) >= min_window_duration]
340
341     @staticmethod
342     def _split_overlong_groups(groups: List[List[Tuple[int, float]]], max_win_frames:
343         int, min_split_frames: int) -> List[List[Tuple[int, float]]]:
344         """Recursively split any active-frame group longer than ``max_win_frames`` at
345         its largest internal gap (? ``min_split_frames``) ? an inactivity-aware cut at the most
346         likely rally boundary. Splitting only subdivides existing groups (never merges/extends), so
347         it cannot invent or drop coverage; a group with no gap ? the floor is left intact (a
348         genuinely long rally). Iterative worklist to bound stack depth."""
349         if max_win_frames <= 0:
350             return groups
351         out: List[List[Tuple[int, float]]] = []
352         work = list(groups)
353         while work:
354             g = work.pop()
355             if len(g) < 2 or (g[-1][0] - g[0][0]) <= max_win_frames:
356                 out.append(g)
357                 continue
358             best_i, best_gap = -1, -1
359             for i in range(1, len(g)):
360                 gap = g[i][0] - g[i - 1][0]
361                 if gap > best_gap:
362                     best_gap, best_i = gap, i
363             if best_i < 0 or best_gap < min_split_frames:
364                 out.append(g) # too long but no real dead-time gap ? keep as one rally
365                 continue
366             work.append(g[:best_i])
367             work.append(g[best_i:])
368         out.sort(key=lambda grp: grp[0][0])
369         return out
370

```

19. user (2026-06-16T06:27:31.567Z)

```

1     """Rally-start gate: suppress between-point false fires using shuttle net-crossings.
2
3     WASB (shuttle-only) over-detects: when a player flicks/tosses the shuttle back to
4     the opponent for service after losing a point, that single trajectory looks like a
5     rally. The discriminator is structural, not duration-based: a real rally has the

```

```

6 shuttle crossing the net MULTIPLE times (each shot lands on the other side); a
7 single service feed crosses -once.
8
9 This module is camera-agnostic and needs NO new model ? it works on the trajectory
10 we already produce (Frame,Visibility,X,Y). It estimates the play axis (the image
11 axis with the larger shuttle spread) and the net position (median shuttle coord on
12 that axis ? the shuttle clusters at the two ends and crosses the net region often,
13 so the median sits near the net), then counts sign changes of (coord - net) across
14 the visible points inside a candidate window, with a deadband to ignore jitter.
15
16 Gate A in the over-fire fix spectrum (B = player-stance state machine, future).
17 Pure functions only ? no I/O, unit-tested; intended as a post-filter on the
18 windows from trajectory_to_action_windows, or to fold into it later.
19 """
20 from __future__ import annotations
21
22 from dataclasses import dataclass
23 from typing import List, Optional, Sequence, Tuple
24
25 Interval = Tuple[float, float]
26
27
28 @dataclass
29 class GatedWindow:
30     start: float
31     end: float
32     crossings: int
33     visible_points: int
34     kept: bool
35
36
37 def _spread(vals: Sequence[float]) -> float:
38     """Robust spread = p95 - p5 (ignores a few outlier detections)."""
39     if len(vals) < 2:
40         return 0.0
41     s = sorted(vals)
42     lo = s[max(0, int(0.05 * (len(s) - 1)))]
43     hi = s[min(len(s) - 1, int(0.95 * (len(s) - 1)))]
44     return hi - lo
45
46
47 def _median(vals: Sequence[float]) -> float:
48     if not vals:
49         return 0.0
50     s = sorted(vals)
51     n = len(s)
52     return s[n // 2] if n % 2 else 0.5 * (s[n // 2 - 1] + s[n // 2])
53
54
55 def estimate_play_axis(points) -> Tuple[str, float, float]:
56     """Return (axis 'x'/'y', net_value, span) from visible points.
57
58     Play axis = the image axis along which the shuttle travels most (larger spread).
59     For a baseline camera that's Y (players top/bottom); for a side camera, X.
60     """
61     xs = [p.x for p in points if p.visible]
62     ys = [p.y for p in points if p.visible]
63     sx, sy = _spread(xs), _spread(ys)
64     if sx >= sy:
65         return "x", _median(xs), sx
66     return "y", _median(ys), sy
67
68
69 def count_net_crossings(window_points, axis: str, net: float, deadband: float) -> int:
70     """Count sign changes of (coord - net) across visible points, ignoring the

```

```

71     deadband zone near the net (so jitter at the net doesn't inflate the count)."""
72     crossings = 0
73     last_side = None
74     for p in window_points:
75         if not p.visible:
76             continue
77         v = (p.x if axis == "x" else p.y) - net
78         if abs(v) < deadband:
79             continue
80         side = v > 0
81         if last_side is not None and side != last_side:
82             crossings += 1
83         last_side = side
84     return crossings
85
86
87 def _points_in(points, fps: float, start: float, end: float):
88     return [p for p in points if start * fps <= p.frame <= end * fps]
89
90
91 def gate_windows(points, windows: List[Interval], fps: float,
92                 min_crossings: int = 2, deadband_frac: float = 0.06,
93                 estimate: Optional[Tuple[str, float, float]] = None) ->
List[GatedWindow]:
94     """Annotate each (start,end) window with its shuttle net-crossing count and a
95     keep decision (crossings >= min_crossings). Deadband is a fraction of the play
96     axis span. Net/axis are estimated once over the whole trajectory (stable).
97
98     `estimate` lets a caller scoring MANY windows off one trajectory pass a precomputed
99     `(axis, net, span)` (from `estimate_play_axis`) so the O(P log P) estimate isn't
redone
100     per window ? same result, avoids the per-window re-estimation."""
101     axis, net, span = estimate if estimate is not None else estimate_play_axis(points)
102     deadband = deadband_frac * span
103     out: List[GatedWindow] = []
104     for (s, e) in windows:
105         wp = _points_in(points, fps, s, e)
106         c = count_net_crossings(wp, axis, net, deadband)
107         vis = sum(1 for p in wp if p.visible)
108         out.append(GatedWindow(start=s, end=e, crossings=c, visible_points=vis,
109                               kept=c >= min_crossings))
110     return out
111
112
113 def apply_gate(points, windows: List[Interval], fps: float,
114               min_crossings: int = 2, deadband_frac: float = 0.06) -> List[Interval]:
115     """Convenience: return only the windows that pass the crossing gate."""
116     return [(g.start, g.end) for g in
117             gate_windows(points, windows, fps, min_crossings, deadband_frac) if g.kept]
118

```

20. user (2026-06-16T06:27:31.906Z)

```

1     name: CI
2
3     on:
4         push:
5             branches: [master]
6         pull_request:
7
8     permissions:
9         contents: read
10
11     concurrency:
12         group: ci-`${{ github.ref }}`

```

```

13     cancel-in-progress: true
14
15 jobs:
16     # Lint gate (ruff.toml at repo root): pyflakes/pycodestyle + bugbear.
17     # Also the repo-wide syntax guard ? ruff reports invalid-syntax on every
18     # tracked .py, including scripts/ and training/, which the backend-only
19     # py_compile smoke sweep does not reach.
20     lint:
21         name: Ruff lint
22         runs-on: ubuntu-latest
23         steps:
24             - uses: actions/checkout@v6
25             - uses: actions/setup-python@v6
26               with:
27                 python-version: "3.13"
28             - name: Install ruff
29               run: python -m pip install ruff
30             - name: Ruff check
31               run: ruff check . --output-format github
32
33     # Type gate (mypy.ini at repo root): lenient green baseline with an explicit
34     # per-module quarantine list ? ratchet by removing entries as modules clean
35     # up. Typed core deps are installed so mypy sees their real signatures
36     # (without them ignore_missing_imports would Any-stub half the checks).
37     typecheck:
38         name: Mypy (lenient baseline)
39         runs-on: ubuntu-latest
40         steps:
41             - uses: actions/checkout@v6
42             - uses: actions/setup-python@v6
43               with:
44                 python-version: "3.13"
45             - name: Install mypy + typed core deps
46               run: python -m pip install mypy pydantic fastapi uvicorn python-dotenv google-
genai numpy
47             - name: Mypy
48               run: mypy backend training
49
50     # Fast guard against the 7fbb0 class of breakage: a SyntaxError or
51     # module-level import error shipping while the suite stays green.
52     # Deliberately does NOT install cv2/torch/numpy ? the smoke tests must hold
53     # on machines without the GPU stack (test_import_smoke stubs what's absent).
54     import-smoke:
55         name: Syntax sweep + import smoke (no GPU stack)
56         runs-on: ubuntu-latest
57         steps:
58             - uses: actions/checkout@v6
59             - uses: actions/setup-python@v6
60               with:
61                 python-version: "3.13"
62             - name: Install minimal deps
63               run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-
genai
64             - name: Run import smoke tests
65               run: python -m pytest tests/test_import_smoke.py -v
66
67     full-suite:
68         name: Full test suite (offline, mocked)
69         runs-on: ubuntu-latest
70         steps:
71             - uses: actions/checkout@v6
72             - uses: actions/setup-python@v6
73               with:
74                 python-version: "3.13"
75             - cache: pip

```

```

76         - name: System libs for opencv-python
77         run: |
78             sudo apt-get update
79             sudo apt-get install -y libgl1 libglu1-mesa-dev || sudo apt-get install -y
libgl1 libglu1-mesa-dev
80         - name: Install CPU-only torch (no CUDA wheels on a CPU runner)
81         # torch/torchvision are NOT in pyproject deps ? their CPU/CUDA wheels
82         # need an out-of-band index that PEP 621 can't express. Keep installing
83         # them separately here, before the editable install pulls the rest.
84         run: python -m pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
85         - name: Install package (editable) + dev tooling
86         # Editable PEP 621 install: runtime deps + the `dev` group
87         # (pytest, pytest-cov, httpx for the fastapi TestClient, ruff, mypy).
88         run: python -m pip install -e .[dev]
89         # obsreport (private, soft dep) is absent here: reporting tests skip via
90         # pytest.importorskip ? run_all_tests.py prints which. The coverage floor
91         # is therefore calibrated against THIS lane's number (local runs with
92         # obsreport installed measure a little higher).
93         # Floor calibrated 2026-06-11 against this lane's measured 72% (local
94         # with obsreport: 73%). Ratchet it UP as coverage grows; never down
95         # without an owner decision.
96         - name: Run full suite (with coverage floor)
97         run: python run_all_tests.py --cov=backend --cov=training --cov-report=term
--cov-fail-under=70
98
99         # Cross-OS determinism guard for the committed golden regression fixtures
100        # (docs/GOLDEN_REGRESSION_FIXTURES.md). The CONTROL replay path is pure-stdlib
101        # (only numpy, no cv2/torch), so this is a FAST, focused job ? it does NOT run
102        # the full suite x3 OSes. It replays the Windows/Linux/macOS runner against the
103        # committed (parse-compared) snapshots, catching any cross-OS float drift beyond
104        # the 1e-6 tolerance before it can flake a contributor's PR on another machine.
105        golden-crossos:
106            name: Golden fixtures cross-OS (${ matrix.os })
107            strategy:
108                fail-fast: false
109            matrix:
110                os: [ubuntu-latest, windows-latest, macos-latest]
111            runs-on: ${ matrix.os }
112            steps:
113                - uses: actions/checkout@v6
114                - uses: actions/setup-python@v6
115                  with:
116                      python-version: "3.13"
117                - name: Install minimal deps (no GPU stack ? the CONTROL replay needs only numpy)
118                  run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-
genai numpy
119                - name: Golden-fixtures characterization (cross-OS determinism guard)
120                  run: |
121                      python -m backend.eval.golden_fixtures --check
122                      python -m pytest tests/test_golden_fixtures.py -q
123

```

21. assistant (2026-06-16T06:27:35.486Z)

Now let me look at the committed fixtures themselves, the manifest, and the README.

22. user (2026-06-16T06:27:37.728Z)

```

total 28
drwxr-xr-x 1 avidu 197609    0 Jun 16 11:54 .
drwxr-xr-x 1 avidu 197609    0 Jun 16 11:54 ..
-rw-r--r-- 1 avidu 197609 4018 Jun 16 11:54 README.md
drwxr-xr-x 1 avidu 197609    0 Jun 16 11:54 clean_single_rally
-rw-r--r-- 1 avidu 197609 1557 Jun 16 11:54 manifest.json

```

```
drwxr-xr-x 1 avidu 197609    0 Jun 16 11:54 multi_rally_with_deadtime
drwxr-xr-x 1 avidu 197609    0 Jun 16 11:54 occlusion_break
---SUBDIRS---
eval_baselines/fixtures/clean_single_rally/:
expected.json
labels.csv
provenance.json
trajectory.csv
```

```
eval_baselines/fixtures/multi_rally_with_deadtime/:
expected.json
labels.csv
provenance.json
trajectory.csv
```

```
eval_baselines/fixtures/occlusion_break/:
expected.json
labels.csv
provenance.json
trajectory.csv
```

23. user (2026-06-16T06:27:40.480Z)

```
1      # `eval_baselines/fixtures/` ? Golden Regression Fixtures (Tier-0, synthetic)
2
3      Video-free, **CONTROL-only** characterization fixtures for the rally-segmentation
pipeline.
4      A contributor on **any machine ? no video, GPU, WSL, or DB** ? replays these committed
5      structured files through the already-shipped eval stack and gets a red/green regression
6      signal. See `docs/GOLDEN_REGRESSION_FIXTURES.md` (deliverable **M1**; §4, §6, §7).
7
8      ## What's here
9
10     Each fixture is one directory:
11
12     | File | Purpose |
13     |-----|-----|
14     | `trajectory.csv` | Synthetic shuttle trajectory (`Frame,Visibility,X,Y`) ? the freeze
seam input. |
15     | `labels.csv` | Synthetic ground-truth rally intervals (`start,end`). |
16     | `expected.json` | Frozen replay snapshot (candidates + 5 shuttle features + CONTROL
segments + score). |
17     | `provenance.json` | Pinned windowing params + privacy/licensing tags + description. |
18
19     `manifest.json` lists every fixture; `backend/eval/golden_fixtures.py` is the shared
replay
20     module + freeze/refresh/check CLI.
21
22     ## Guardrails (non-negotiable)
23
24     - **Synthetic / owned only.** Every fixture is generated by deterministic formulae (NO
random)
25     ? zero third-party footage, zero provenance/licensing risk. Tagged `kind:
"synthetic"`,
26     `license: "synthetic"`, `minors_free: true`.
27     - **No biometric / identity streams.** Tier-0 carries the **5 shuttle feature columns
only**
28     (`duration, peak_velocity, mean_velocity, active_ratio, net_crossings`). There is, by
design,
29     **no person / pose / gait / face / audio / player field anywhere in the schema** ? a
test
30     asserts this positively (deletion-at-source, §4c). Minors and biometrics are excluded.
31     - **CONTROL / pure-stdlib only.** The replay uses `experiment.CONTROL` (`is_learned() ==
False`)
32     ? no numpy / logistic-regression. Learned variants' near-threshold decisions flip
```



```

across CPUs,
33     so they can **never** anchor a cross-machine snapshot ($4d / D5).
34     - **Parse-compare, never byte-compare.** The gate `json.load`'s both sides: exact
equality on
35     int/structural fields, abs-tolerance `1e-6` on floats (CRLF/float-format safe).
36     - **Windowing is pinned per fixture.** The `(vt, mg, mwd, max_win)` params live in each
37     `provenance.json` and are passed explicitly to `build_candidates` ? the replay does
**not**
38     depend on the named `windowing.SERVED` constant, so a future SERVED retune can't
silently
39     churn fixtures.
40
41     ## Fixtures are GENERATED, never hand-edited
42
43     Do **not** hand-edit `expected.json` / `provenance.json` / the CSVs. They are produced
44     correct-by-construction by the CLI.
45
46     ```bash
47     # Generate the synthetic trajectory.csv + labels.csv for all built-in scenarios, then
48     # freeze expected.json + provenance.json + (re)write manifest.json:
49     python -m backend.eval.golden_fixtures --freeze-synthetic
50
51     # Replay all fixtures and parse-compare vs the committed expected.json (exit 0/1):
52     python -m backend.eval.golden_fixtures --check
53
54     # Re-freeze expected.json from the COMMITTED trajectory+labels (NO regeneration). ? This
can
55     # rubber-stamp a real regression ? review the visible diff and record a reason ($6):
56     python -m backend.eval.golden_fixtures --refresh-snapshot --all --reason "<why>"
57     python -m backend.eval.golden_fixtures --refresh-snapshot --slug clean_single_rally
--reason "<why>"
58     ```
59
60     ## Honest limits ($6 ? rides in the test pass/fail messages, not only here)
61
62     - A green suite proves only that the **deterministic post-trajectory transforms** are
unchanged
63     for the frozen cases. It does **not** cover the WASB/TrackNet detector, frame
decoding, the
64     AI/provider handoff, chunking, the ffmpeg stitch, DB persistence, or the API filter.
65     - Characterization is **behaviour-locking, not correctness** ? it freezes current
behaviour
66     *including current bugs*. "Passing" = "behaviour unchanged", never "behaviour
correct".
67     - **Synthetic-tier F1 measures plumbing correctness, NOT field quality.** Never cite a
synthetic
68     fixture's F1 in `QUALITY_ITERATIONS.md` or the paper as rally-detection performance.
69

```

24. user (2026-06-16T06:27:41.419Z)

```

1      [
2      {
3          "fps": 30.0,
4          "frame_width": 1280.0,
5          "kind": "synthetic",
6          "paths": {
7              "dir": "clean_single_rally",
8              "expected": "clean_single_rally/expected.json",
9              "labels": "clean_single_rally/labels.csv",
10             "provenance": "clean_single_rally/provenance.json",
11             "trajectory": "clean_single_rally/trajectory.csv"
12         },
13         "schema_version": 1,
14         "slug": "clean_single_rally",

```

```

15     "windowing": {
16         "max_win": 0.0,
17         "mg": 2.0,
18         "mwd": 2.0,
19         "vt": 0.02
20     }
21 },
22 {
23     "fps": 30.0,
24     "frame_width": 1280.0,
25     "kind": "synthetic",
26     "paths": {
27         "dir": "multi_rally_with_deadtime",
28         "expected": "multi_rally_with_deadtime/expected.json",
29         "labels": "multi_rally_with_deadtime/labels.csv",
30         "provenance": "multi_rally_with_deadtime/provenance.json",
31         "trajectory": "multi_rally_with_deadtime/trajectory.csv"
32     },
33     "schema_version": 1,
34     "slug": "multi_rally_with_deadtime",
35     "windowing": {
36         "max_win": 0.0,
37         "mg": 2.0,
38         "mwd": 2.0,
39         "vt": 0.02
40     }
41 },
42 {
43     "fps": 30.0,
44     "frame_width": 1280.0,
45     "kind": "synthetic",
46     "paths": {
47         "dir": "occlusion_break",
48         "expected": "occlusion_break/expected.json",
49         "labels": "occlusion_break/labels.csv",
50         "provenance": "occlusion_break/provenance.json",
51         "trajectory": "occlusion_break/trajectory.csv"
52     },
53     "schema_version": 1,
54     "slug": "occlusion_break",
55     "windowing": {
56         "max_win": 0.0,
57         "mg": 2.0,
58         "mwd": 2.0,
59         "vt": 0.02
60     }
61 }
62 ]
63

```

25. user (2026-06-16T06:27:41.986Z)

```

1  {
2      "candidates": [
3          {
4              "end": 3.966667,
5              "shuttle": {
6                  "active_ratio": 1.008475,
7                  "duration": 3.933333,
8                  "mean_velocity": 1.27678,
9                  "net_crossings": 7,
10                 "peak_velocity": 1.970041
11             },
12             "start": 0.033333
13         }
14     ]
15 }

```

```

14     ],
15     "control_segments": [
16         [
17             0.033333,
18             3.966667
19         ]
20     ],
21     "gt_hash": "6c93801fd607",
22     "iou": 0.5,
23     "schema_version": 1,
24     "score": {
25         "f1": 1.0,
26         "false_negatives": 0,
27         "false_positives": 0,
28         "iou_threshold": 0.5,
29         "mean_end_error": 0.033333,
30         "mean_iou": 0.983333,
31         "mean_start_error": 0.033333,
32         "num_gt": 1,
33         "num_pred": 1,
34         "precision": 1.0,
35         "recall": 1.0,
36         "true_positives": 1
37     },
38     "slug": "clean_single_rally"
39 }
40

```

26. assistant (2026-06-16T06:27:46.185Z)

Interesting ? I notice `active\_ratio: 1.008475` is greater than 1.0, which the docstring describes as an "active-shuttle ratio." Let me look at the other expected.json files and check a few specific claims. Let me also examine the `gt\_loader` overlap behavior, the `\_scan\_forbidden` logic edge cases, and whether `test\_golden\_real\_fixtures.py` runs in the cross-OS job.

27. user (2026-06-16T06:27:48.097Z)

```

1  {
2      "candidates": [
3          {
4              "end": 4.0,
5              "shuttle": {
6                  "active_ratio": 1.008403,
7                  "duration": 3.966667,
8                  "mean_velocity": 1.282558,
9                  "net_crossings": 7,
10                 "peak_velocity": 1.970041
11             },
12             "start": 0.033333
13         },
14         {
15             "end": 10.966667,
16             "shuttle": {
17                 "active_ratio": 1.008475,
18                 "duration": 3.933333,
19                 "mean_velocity": 1.27678,
20                 "net_crossings": 7,
21                 "peak_velocity": 1.970041
22             },
23             "start": 7.033333
24         }
25     ],
26     "control_segments": [
27         [
28             0.033333,

```

```

29         4.0
30     ],
31     [
32         7.033333,
33         10.966667
34     ]
35 ],
36 "gt_hash": "bbec003aed74",
37 "iou": 0.5,
38 "schema_version": 1,
39 "score": {
40     "f1": 1.0,
41     "false_negatives": 0,
42     "false_positives": 0,
43     "iou_threshold": 0.5,
44     "mean_end_error": 0.016667,
45     "mean_iou": 0.9875,
46     "mean_start_error": 0.033333,
47     "num_gt": 2,
48     "num_pred": 2,
49     "precision": 1.0,
50     "recall": 1.0,
51     "true_positives": 2
52 },
53 "slug": "multi_rally_with_deadtime"
54 }
55

```

28. user (2026-06-16T06:27:48.727Z)

```

1  {
2      "candidates": [
3          {
4              "end": 3.966667,
5              "shuttle": {
6                  "active_ratio": 1.008475,
7                  "duration": 3.933333,
8                  "mean_velocity": 1.27678,
9                  "net_crossings": 7,
10                 "peak_velocity": 1.970041
11             },
12             "start": 0.033333
13         },
14         {
15             "end": 10.966667,
16             "shuttle": {
17                 "active_ratio": 1.008475,
18                 "duration": 3.933333,
19                 "mean_velocity": 1.27678,
20                 "net_crossings": 7,
21                 "peak_velocity": 1.970041
22             },
23             "start": 7.033333
24         }
25     ],
26     "control_segments": [
27         [
28             0.033333,
29             3.966667
30         ],
31         [
32             7.033333,
33             10.966667
34         ]
35     ],

```

```

36     "gt_hash": "bbec003aed74",
37     "iou": 0.5,
38     "schema_version": 1,
39     "score": {
40         "f1": 1.0,
41         "false_negatives": 0,
42         "false_positives": 0,
43         "iou_threshold": 0.5,
44         "mean_end_error": 0.033333,
45         "mean_iou": 0.983333,
46         "mean_start_error": 0.033333,
47         "num_gt": 2,
48         "num_pred": 2,
49         "precision": 1.0,
50         "recall": 1.0,
51         "true_positives": 2
52     },
53     "slug": "occlusion_break"
54 }
55

```

29. user (2026-06-16T06:27:49.327Z)

```

1      {
2      "description": "One continuous ~4s rally; shuttle oscillates across the net ? 1
window.",
3      "fps": 30.0,
4      "frame_width": 1280.0,
5      "iou": 0.5,
6      "kind": "synthetic",
7      "license": "synthetic",
8      "minors_free": true,
9      "note": "Synthetic, deterministic, owned (no real footage). Tier-0 / CONTROL-only.
Plumbing-correctness fixture ? NOT a measure of rally-detection quality (§6).",
10     "paths": {
11         "expected": "expected.json",
12         "labels": "labels.csv",
13         "trajectory": "trajectory.csv"
14     },
15     "schema_version": 1,
16     "slug": "clean_single_rally",
17     "windowing": {
18         "max_win": 0.0,
19         "mg": 2.0,
20         "mwd": 2.0,
21         "vt": 0.02
22     }
23 }
24

```

30. assistant (2026-06-16T06:27:56.164Z)

Now I have a strong understanding. Let me verify some specific claims by examining a few key things:

1. Whether ``test_golden_real_fixtures.py`` runs in the cross-OS job (the lens explicitly asks).
2. The ``_scan_forbidden`` substring logic for false negatives/positives.
3. The ``gt_hash`` collision claim and whether the two-rally fixtures share a hash.

Let me check the CI job ? it runs only ``tests/test_golden_fixtures.py``, NOT ``test_golden_real_fixtures.py``. That confirms the lens's hypothesis. Let me verify the determinism claim more carefully and check the ``_scan_forbidden`` against the actual scan order (FORBIDDEN_KEYS before the schema-literal exclusion).

Let me check one subtle thing: the order of checks in ``_scan_forbidden`` and whether ``player`` is

a substring issue, and whether the forbidden-key scan could false-positive on legitimate content.

31. user (2026-06-16T06:27:58.821Z)

```
Frame,Visibility,X,Y
0,1,640.000000,360.000000
1,1,723.164676,372.202099
...
start,end
0.000000,4.000000
===occlusion traj head===
Frame,Visibility,X,Y
0,1,640.000000,360.000000
1,1,723.164676,372.202099
2,1,802.694657,382.294345
3,1,875.114101,388.531695
```

32. assistant (2026-06-16T06:28:06.339Z)

Now let me verify the `\_scan\_forbidden` source-path token logic for a subtle false-negative case. The scan lowercases tokens and checks `t` in `low`. But a critical question: what if a single-character or very-short source-path token matches? More importantly, let me check whether the source-path scan handles the `paths` block ? the writers emit only the schema literals in `paths`, but the surrogate slug `fx\_<hex>` IS written into the provenance. Let me trace whether a source filename could survive.

Let me also examine whether the determinism claim holds: the synthetic trajectory uses `math.sin`. Let me check if `math.sin` is guaranteed identical cross-platform ? and whether the fixtures were frozen on Windows but the docstring claims cross-OS determinism is "enforced."

Let me verify the key over-claim concern: does the CI `golden-crossos` job actually exercise the privacy/refusal-gate path (P1), or only M1?

33. user (2026-06-16T06:28:07.832Z)

```
122:          python -m pytest tests/test_golden_fixtures.py -q
```

34. user (2026-06-16T06:28:08.972Z)

```
11: active_ratio, net_crossings] -> LogisticRegression(prob >= threshold) -> merge
41:FEATURE_NAMES = ["duration", "peak_velocity", "mean_velocity", "active_ratio",
"net_crossings"]
70:          float(w["active_frame_count"]) / win_frames,          # active-shuttle ratio ? fps-
invariant
```

35. assistant (2026-06-16T06:28:16.742Z)

Confirmed: the cross-OS job runs ONLY `tests/test\_golden\_fixtures.py`, not `test\_golden\_real\_fixtures.py`. Now let me verify the determinism question more rigorously. The replay path uses `math.sin` only for *generating* synthetic fixtures (freeze time), but the committed CSVs are static. At replay/check time, the math is: parse CSV ? velocity computation (sqrt, division) ? score. Let me trace whether the CI actually validates cross-OS for the `freeze` operation or just the `replay`. The committed fixtures were frozen once (on Windows, per the doc's #189 note "CI Linux replayed Windows-frozen fixtures"). So cross-OS only validates *replay*, not *freeze*. That's correctly scoped.

Let me now examine the subtle determinism concern around the `\_split\_overlong\_groups` and the velocity sqrt computation, and verify the active_ratio > 1.0 issue isn't a hidden bug being locked in. Let me check the run to confirm the actual numbers and whether tests would catch a sign issue.

Let me verify the key honesty claims by running the actual check and a couple of targeted experiments.

36. user (2026-06-16T06:28:21.583Z)

```
3.13.13 (tags/v3.13.13:01104ce, Apr 7 2026, 19:25:48) [MSC v.1944 64 bit (AMD64)]
[ok] clean_single_rally: 1 candidates, 1 segments, F1=1.000000
[ok] multi_rally_with_deadtime: 2 candidates, 2 segments, F1=1.000000
[ok] occlusion_break: 2 candidates, 2 segments, F1=1.000000
[golden-fixtures] 3/3 fixtures match (synthetic-tier = plumbing correctness, not field quality ?
$6)
```

37. assistant (2026-06-16T06:28:26.903Z)

The CLI works and carries the \$6 caveat in the message. Now let me verify a critical determinism concern: the `\_scan\_forbidden` source-path token false-negative. Let me trace a realistic leak scenario ? what if the source path's stem is a common substring that DOESN'T appear in the output but the absolute path uses backslashes on Windows. Let me test the actual scan behavior with edge cases, and check the `gt\_hash` rounding (3dp) vs the snapshot's 6dp floats.

Let me examine one real concern: the positive privacy scan runs only on `expected.json` and `provenance.json` ? NOT on the committed `trajectory.csv` or `labels.csv`. Let me verify.

38. user (2026-06-16T06:28:28.298Z)

```
13: parse_trajectory_csv ? distill_local.build_candidates(proposal=pinned)
137: "labels.csv". For a synthetic fixture they line up with the rallies by construction,
194: def write_trajectory_csv(rows: Sequence[Tuple[int, int, float, float]], path: Path) -> None:
202: def write_labels_csv(gts: Sequence[Interval], path: Path) -> None:
300: ``trajectory.csv`` + ``labels.csv`` + ``provenance.json``, then:
302: parse_trajectory_csv ? build_candidates(proposal=pinned) ? VideoCandidates(5 shuttle
311: traj_csv = fdir / "trajectory.csv"
312: labels_csv = fdir / "labels.csv"
328: gts = load_gt_intervals(str(labels_csv), strict=True)
393: "trajectory": "trajectory.csv",
394: "labels": "labels.csv",
407: ``trajectory.csv`` + ``labels.csv`` via the replay (so expected is correct-by-
construction).
413: traj_csv, labels_csv = fdir / "trajectory.csv", fdir / "labels.csv"
414: if not traj_csv.is_file() or not labels_csv.is_file():
416: f"{slug}: trajectory.csv/labels.csv missing ? run --freeze-synthetic to generate
them")
430: """Generate trajectory.csv + labels.csv for every built-in scenario, freeze
expected.json +
439: write_trajectory_csv(rows, fdir / "trajectory.csv")
440: write_labels_csv(spec.gts, fdir / "labels.csv")
449: """Re-freeze expected.json (+ provenance.json) from the COMMITTED trajectory.csv +
labels.csv ?
454: if d.is_dir() and (d / "trajectory.csv").is_file()
494: #: false-positive against the literal ``trajectory.csv``/``labels.csv``/``expected.json``.
496: "trajectory", "trajectory.csv", "labels", "labels.csv", "expected", "expected.json",
501: def _scan_forbidden(text: str, *, where: str, source_paths: Sequence[str]) -> None:
529: # (e.g. a source stem ``traj`` ? ``trajectory.csv``) ? that collision is not a
leak.
559: trajectory_csv_path: Any,
560: labels_csv_path: Any,
607: traj_path = Path(str(trajectory_csv_path))
608: labels_path = Path(str(labels_csv_path))
614: points = TrackNetRunner.parse_trajectory_csv(str(traj_path))
635: write_trajectory_csv(rows, fdir / "trajectory.csv")
636: write_labels_csv(reset_gts, fdir / "labels.csv")
658: "trajectory": "trajectory.csv",
659: "labels": "labels.csv",
678: _scan_forbidden((fdir / "expected.json").read_text(encoding="utf-8"),
680: _scan_forbidden((fdir / "provenance.json").read_text(encoding="utf-8"),
700: "trajectory": f"{slug}/trajectory.csv",
701: "labels": f"{slug}/labels.csv",
735: "trajectory": f"{spec.slug}/trajectory.csv",
```

736: "labels": f"{spec.slug}/labels.csv",

39. assistant (2026-06-16T06:28:35.946Z)

Confirmed a finding: `\_scan\_forbidden` (lines 678?681) re-reads and scans only `expected.json` and `provenance.json` ? NOT the written `trajectory.csv` or `labels.csv`. The doc §4c lists "Raw-trajectory exclusion ? never commit the per-frame CSV" but the P1 freeze-real tool DOES write `trajectory.csv` (line 635) and commits it. Let me verify this is intentional vs a doc contradiction, by checking the doc's Tier-0 contents claim.

Looking at the doc §4b table: Tier-0 contents = "5 shuttle scalars... No person/audio/pose." And §4c says "Raw-trajectory exclusion ? never commit the per-frame (Frame,Visibility,X,Y) CSV." But the shipped `freeze\_real\_fixture` writes the full per-frame trajectory.csv to the fixture dir (line 635), the same as synthetic. This is a real doc-vs-code contradiction worth flagging, though it's privacy-adjacent rather than the HONESTY lens specifically.

Let me focus on the lens. Let me now test the actual cross-platform determinism concern empirically and check the `gt\_hash` rounding interplay. Let me also confirm the test assertion messages actually carry the §6 caveat (the lens's core question).

40. user (2026-06-16T06:28:40.234Z)

first-step velocity: 1.9700406698719501
sin(2pi/30): 0.20791169081775931

41. assistant (2026-06-16T06:28:52.969Z)

Now let me verify whether the `--check` CLI exit-code path and the per-fixture FAIL message actually carry the §6 caveat (the lens question: "is the caveat in the test PASS/FAIL messages, not only docs?"). I already saw:

- Line 892-893: FAIL message carries "characterization = behaviour-locking, NOT correctness ? §6"
- Line 900-901: summary carries "synthetic-tier = plumbing correctness, not field quality ? §6"
- test_golden_fixtures.py line 38: assertion message carries "characterization, not correctness ? §6"

But let me check the OTHER assertions. The lens asks specifically: "Is the §6 'behaviour-locking NOT correctness' caveat actually carried in the test PASS/FAIL MESSAGES (not only docs)?" Let me check which assertions carry it and which don't ? particularly the privacy/score-invariance assertions in test_golden_real_fixtures.py.

Looking back at the test files:

- `test\_golden\_fixtures.py::test\_replay\_matches\_committed\_expected` ? carries §6 ?
- `test\_no\_person\_or\_audio\_keys`, `test\_synthetic\_provenance\_tags`, `test\_schema\_version\_le\_current`, `test\_perturbing...`, `test\_freeze\_is\_idempotent`, `test\_check\_cli\_passes` ? do NOT carry §6 (but these aren't quality claims; they're privacy/structural, so that's fine)
- `test\_golden\_real\_fixtures.py` ? only the module docstring mentions §6; the `test\_time\_origin\_reset\_is\_metric\_invariant` assertion messages make metric-invariance claims.

Now, the key honesty question on F1=1.0. Let me check whether the `--check` "[ok]" line prints `F1={sc['f1']:.6f}` (line 899) ? yes it prints `F1=1.000000`. The summary line does carry the "not field quality" caveat. But the per-fixture `[ok]` line prints F1=1.000000 WITHOUT a per-line caveat. That's a minor over-claim surface but the summary line right after carries it. Let me assess severity.

Let me check the DONE criteria claim in §9 vs what shipped, and the tracker's "[verified]" vs "[design]" honesty. Let me look at one more thing ? whether MX (cross-OS) tracker row over-claims given test_golden_real_fixtures isn't in the job.

42. user (2026-06-16T06:28:55.245Z)

8d3e185 feat(eval): golden fixtures P1 ? de-attribution/minimization tool + M2 refusal gate (#192)


```

adb88f4 ci(eval): cross-OS determinism guard for golden fixtures + reconcile project tracker
(#191)
950106b fix(fusion): W2 net-crossings gate is do-no-harm ? never empty a non-empty video (#188
follow-up) (#190)
64494ef feat(eval): golden regression fixtures M1 ? video-free CONTROL-only characterization
harness (#189)
a471745 feat(eval): Tier 1 (W2) ? net-crossings rally-state gate on W1 (?F1 +0.019, 6/0) (#188)
1de2817 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc template (#186)
a3b209a feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-merge (?F1 +0.139, 6/0)
(#187)
90ab087 feat(eval): Tier 0 (E1) ? over-segmentation-aware rally-seg eval + frozen baseline
(#185)
===PRs touching golden_fixtures===
8d3e185 feat(eval): golden fixtures P1 ? de-attribution/minimization tool + M2 refusal gate
(#192)
955e902 feat(eval): golden fixtures P1 ? de-attribution/minimization tool + M2 refusal gate
64494ef feat(eval): golden regression fixtures M1 ? video-free CONTROL-only characterization
harness (#189)
84f8428 feat(eval): golden regression fixtures M1 ? video-free CONTROL-only characterization
harness

```

43. assistant (2026-06-16T06:29:06.602Z)

Now let me examine a determinism subtlety the lens raises: "cross-OS determinism claim overreach given test_golden_real_fixtures.py is NOT in the cross-OS job." Let me verify what test_golden_real_fixtures.py actually exercises that could be OS/platform-sensitive but is NOT in the cross-OS guard. The freeze_real path uses `os.urandom` (non-deterministic by design, so it can't be in a snapshot gate) but `test\_supplied\_slug\_is\_honored` and `test\_time\_origin\_reset\_is\_metric\_invariant` use fixed slugs and DO produce deterministic snapshots ? yet none of them run cross-OS.

More importantly: the `freeze\_real\_fixture` code path itself (the `\_reset\_trajectory\_rows`, the `\_scan\_forbidden`, the M2 gate) is NEVER exercised cross-OS. The doc M1/MX rows and §4d claim "CONTROL path only across machines" ? but the P1 freeze path produces committable fixtures via the SAME float math (`\_round`, `\_fmt`, velocity sqrt) on the producer's machine, and a real fixture frozen on macOS then replayed on Linux is only protected by the 1e-6 parse tolerance, which IS tested for the synthetic fixtures. So the determinism guarantee transfers. But let me confirm there's no real fixture committed (there isn't ? P3 is ?).

Let me now check the doc's §9 DONE checkbox honesty and one final thing: whether the `--refresh-snapshot` reason is actually *required* (doc §6 says "Require a recorded reason" and tracker says "the CLI requires one"). Looking at the code (lines 952-960): if no `--reason`, it only prints a WARNING and proceeds. The reason is NOT required ? it's a soft warning.

44. user (2026-06-16T06:29:07.752Z)

```

446
447     def refresh_snapshot(slug: Optional[str] = None, all_slugs: bool = False,
448                           fixtures_dir: Path = FIXTURES_DIR) -> List[str]:
449         """Re-freeze expected.json (+ provenance.json) from the COMMITTED trajectory.csv +
labels.csv ?
450         NO regeneration of the trajectory/labels. ? §6: this can rubber-stamp a real
regression; the
451         mitigation is reviewing the visible diff + recording a reason (the CLI requires
one)."""
452         if all_slugs:
453             targets = [d.name for d in sorted(fixtures_dir.iterdir())
454                        if d.is_dir() and (d / "trajectory.csv").is_file()]
455         elif slug:
456             targets = [slug]
457         else:
458             raise ValueError("refresh_snapshot needs --slug S or --all")
459         for s in targets:
460             freeze_fixture(s, fixtures_dir=fixtures_dir)
461         return targets

```

45. assistant (2026-06-16T06:29:20.352Z)

Found a clear over-claim. The `refresh\_snapshot` docstring (line 451) states "***recording a reason (the CLI requires one)***", and the tracker §6 / refresh_snapshot docstring imply the reason is mandatory. But the actual CLI (lines 955?957) only prints a WARNING when `--reason` is absent and proceeds to refresh anyway ? exit 0. The reason is NOT required. This is a docstring over-claim: it asserts a guardrail (mandatory reason) that the code does not enforce.

Let me double-check by tracing `main()` once more for the refresh path ? confirmed at lines 952-961: `if not args.reason:` ? prints WARNING ? still calls `refresh\_snapshot` ? returns 0. So the docstring's "(the CLI requires one)" is false.

Now let me verify the `compare\_expected` honesty about what it covers vs the snapshot. One subtle gap: `compare\_expected` does NOT compare the `provenance.json` at all (kind/license/minors_free/windowing) during `--check`. The `--check` only re-reads `expected.json` and replays. So a fixture whose `provenance.json` got its `minors\_free` flipped to false, or `windowing` params changed, would NOT be caught by `--check` unless it changed the replayed numbers. The windowing IS read from provenance during replay (line 292-293), so a windowing change would change the snapshot and be caught. But `kind`/`license`/`minors\_free`/`gt\_hash`-in-provenance drift is not checked by `--check`. The privacy tags are only asserted in `test\_synthetic\_provenance\_tags` (synthetic only), and that test is NOT in the cross-OS job either (only `test\_golden\_fixtures.py` is ? wait, that test IS in test_golden_fixtures.py). Let me confirm.

`test\_synthetic\_provenance\_tags` is in `test\_golden\_fixtures.py` (line 69-77), which IS run in the cross-OS job. Good ? so synthetic provenance tags are checked. But for `cleared\_real` fixtures (P3), the provenance-tag assertions live in `test\_golden\_real\_fixtures.py`, which is NOT in the cross-OS job, and `test\_synthetic\_provenance\_tags` asserts `kind=="synthetic"` so it would actually FAIL on a committed `cleared\_real` fixture. Let me verify that interaction.

46. user (2026-06-16T06:29:21.303Z)

```
69     @pytest.mark.parametrize("slug", SLUGS)
70     def test_synthetic_provenance_tags(slug):
71         """Tier-0 hard tags: kind=synthetic, license=synthetic, minors_free=true (constraint
5)."""
72         prov = gf.load_fixture(slug)["provenance"]
73         assert prov["kind"] == "synthetic"
74         assert prov["license"] == "synthetic"
75         assert prov["minors_free"] is True
76         # Windowing pinned in the fixture (not the named SERVED constant) ? constraint 3.
77         assert set(prov["windowing"]) == {"vt", "mg", "mwd", "max_win"}
```

47. assistant (2026-06-16T06:29:32.430Z)

Confirmed a latent issue: `test\_synthetic\_provenance\_tags` is parametrized over ALL slugs in the manifest (including any future `cleared\_real` fixture from P1's `freeze\_real\_fixture`, which appends to the same manifest). It hard-asserts `kind == "synthetic"` and `license == "synthetic"`. So the moment a P3 real fixture is committed (the explicit purpose of the shipped P1 tool), this test will FAIL for that fixture. This is a forward-incompatibility / mislabeled-test gap: the test name says "synthetic" but it gates EVERY manifest entry. It's not a today-bug (no real fixture committed yet), but it's a gap the per-PR verification missed since P1 shipped the tool that produces exactly the fixtures this test rejects. Worth flagging as low/medium.

Now let me confirm the `active\_ratio > 1.0` is a locked-in oddity but NOT being misrepresented (the docstring calls it "active-shuttle ratio" implying ?1, but it's 1.0085). This is a characterization-locks-current-behavior situation, which §6 explicitly covers ("freezes current behaviour including current bugs"). The ratio exceeds 1.0 because `active\_frame\_count` (count of active inter-frame steps) can exceed `win\_frames = duration\*fps` due to the `max(1.0,...)` and off-by-one in window-frame derivation. This is honestly disclosed by the §6 framing, so it's not an over-claim ? it's exactly what characterization is for. I'll note it as clean under the lens but mention it.

Let me now verify the final honesty claim: "owner-accepted (researched risk), not retained-

counsel" framing consistency. Let me grep the doc for any place that might contradict this.

48. user (2026-06-16T06:29:34.306Z)

```
3:> **Status:** `IN PROGRESS` ? owner-gated . **Owner:** Avi . **Created:** 2026-06-16 . **Last
updated:** 2026-06-16 . **Legal:** owner **accepted the researched risk** 2026-06-16 (conditions
§3 stand; this is the owner's informed acceptance, **not** a retained-counsel opinion)
7:> **Honesty note:** §3 (Legal) is **researched general information, NOT legal advice** ? it
gates on retained counsel (§8). §2/§4/§7 mark which claims are **[verified]** against code vs
**[design]** proposals.
52:## 3. Legal foundation (researched ? gates on counsel, §8)
133:- De-attribution does **not** cure provenance/licensing ? it sits *behind* the per-record
provenance gate + counsel sign-off.
151:| **P2** | **Owner-accepted (researched-risk), 2026-06-16.** Owner accepted the researched
IP/privacy analysis (§3) to proceed **under its conditions** (owned/Fork-A source · minors
excluded · biometric/person streams excluded · de-attributed). Owner's informed risk acceptance
? **NOT a retained-counsel opinion** ; optional future counsel confirmation per §8. | P1 | Owner-
accepted | ? | ? |
157:[Omitted long matching line]
163:| [#191](https://github.com/avidullu/badminton-highlight-indexer/pull/191) | MX +
tracker/legal reconcile | full suite green | 80.41% | `golden-crossos` CI job green on
ubuntu+windows+macos ? cross-OS **enforced** ; legal ? owner-accepted |
168:## 8. Open owner / counsel questions
170:**Counsel questions ? documented basis (the owner accepted the researched risk on
2026-06-16; see P2). These remain the record, and a retained-counsel confirmation is
optional/recommended before scaling beyond a small owned, minors-free, de-attributed subset:**
179:7. **Scope of first public commit:** synthetic-only (zero provenance risk, recommended) vs a
counsel-cleared real subset.
192:- [ ] **Either** the public commit is synthetic-only, **or** (**P2 owner-accepted ? + P3**)
for a real-data commit, under the §3 conditions enforced **in code** by P1's refusal gate.
203:**Legal authorities:** see §3 (full URLs in the research workflow artifact). *This section
is researched general information, not legal advice.*
```

49. assistant (2026-06-16T06:29:45.697Z)

The "owner-accepted (researched risk), not retained-counsel" framing is honest and consistently stated across the header, §3, §8, P2 row, and §10. However, I notice one inconsistency worth flagging: §9 DONE criterion (line 192) says a real-data commit is allowed via "P2 owner-accepted ? + P3" under §3 conditions ? but §8 question 4 ("Sign-off on the final committed Tier-0 subset") and §8 q7 ("counsel-cleared real subset") still frame the real subset as counsel-gated/recommended. And the P3 row's "Counsel-gated?" column says "conditions §3". This is mostly consistent (owner accepted risk, counsel optional), so not a contradiction ? just slight tension. I'll note it as clean-ish.

Now the cross-OS "enforced" claim (line 163): "cross-OS **enforced**". Let me assess. The job runs on all 3 Oses and IS in CI on every PR. But it's a *separate* job ? is it a *required* status check? The doc §9 explicitly lists "CI leak-scan is a required status check" as a TODO (unchecked), and there's no mention that `golden-crossos` is a required check. If the job is not a required/branch-protection check, "enforced" slightly overstates ? a red `golden-crossos` wouldn't block merge. But this is a GitHub branch-protection setting outside the repo, unverifiable here. The doc's own §9 leaves "required status check" as future work. I'll flag the "enforced" wording as a mild over-claim given the doc itself doesn't claim branch protection.

Let me now verify the determinism claim around `json.dumps` float formatting cross-platform and the parse-compare's handling of `gt\_hash` (which uses 3dp rounding internally vs the snapshot's 6dp). Let me confirm gt_hash is stable.

50. user (2026-06-16T06:29:50.401Z)

```
multi/occ gts: bbec003aed74
single gts: 6c93801fd607
```

51. assistant (2026-06-16T06:30:06.532Z)

Confirmed: `multi\_rally\_with\_deadtime` and `occlusion\_break` share `gt\_hash = bbec003aed74` because they have identical labels `((0,4),(7,11))`. This is fine ? `gt\_hash` only fingerprints labels by design. The trajectory differences ARE captured in the `candidates`/`control\_segments` snapshot, so cross-fixture confusion is caught. Not a finding.

Now let me do one final check on the strongest privacy/determinism angle: the `\_scan\_forbidden` false-negative. The scan checks `key in low` for FORBIDDEN_KEYS = ("person", "pose", "gait", "face", "audio", "player"). But a real leak vector ? the `label` and `source\_note` free-text fields ? are written into provenance and scanned. Good. But the scan is substring-based and case-insensitive, which could *false-POSITIVE*: e.g., if a legitimate `source\_note` or `label` contained "face" as part of "surface" or "interface", it would be refused. That's an over-strict edge but not a privacy hole. More concerning for honesty: the scan does NOT detect a leaked *player name* unless it happens to be a source-path token or contains a forbidden substring. A real human name (e.g., "Lin Dan") in `source\_note` or `label` would pass the scan silently. The tool relies on the operator not putting names in `--label`/`--source-note`. The docstrings DO say "non-identifying scenario label" / "NEVER a path/name/id", so it's disclosed as operator responsibility ? but the positive scan does NOT actually catch arbitrary names. This is a gap between the §4c "positive privacy assertion" framing and what it mechanically verifies.

Let me confirm the test that "proves" the positive scan (`test\_positive\_privacy\_assertion\_catches\_leak\_in\_source\_note`) only tests the *path* leak case, not a name leak.

Looking at test line 155-164: it only tests `source\_note=str(traj)` (a path). There's no test that a human name in `label`/`source\_note` is caught ? because it can't be. The assertion's name "positive privacy assertion" overstates: it catches MD5/forbidden-stream-keys/source-path-tokens, but NOT arbitrary identifiers (names, match titles, dates). The doc §4c "Metadata strip ? no filename, video_id, match/event/player identity, capture date" implies these are stripped/blocked, but mechanically only path-tokens, MD5, and the 6 forbidden substrings are scanned. A player name in free text is not detected.

I have enough for a thorough audit. Let me do a final verification that the full real-fixtures test suite passes locally and check the `active\_ratio` framing isn't asserted as ?1 anywhere.

52. user (2026-06-16T06:30:12.326Z)

..... [100%]
29 passed in 2.77s

53. assistant (2026-06-16T06:30:15.762Z)

Both test files pass (29 tests). Now let me verify the one forward-incompatibility concern empirically: would `test\_synthetic\_provenance\_tags` fail on a committed `cleared\_real` fixture? Let me simulate by freezing a real fixture into the actual fixtures dir temporarily and running the test, then clean up.

54. user (2026-06-16T06:30:25.759Z)

cleared_real kind= cleared_real license= owned
would FAIL test_synthetic_provenance_tags -> kind assertion (kind is 'cleared_real')

55. assistant (2026-06-16T06:30:42.230Z)

Confirmed empirically: a `cleared\_real` fixture (the exact output of the shipped P1 `freeze\_real\_fixture` tool) would FAIL `test\_synthetic\_provenance\_tags` because that test is parametrized over ALL manifest slugs and hard-asserts `kind == "synthetic"`. The moment P3 commits a real fixture, the suite breaks. This is a real forward-gap the per-PR verification missed (P1 shipped the producer; the consumer test rejects its output).

I now have a complete picture. Let me compile findings. Summary of what I verified:

****Over-claims / honesty gaps found:****

1. `refresh\_snapshot` docstring + tracker say the CLI "requires" a reason ? it does NOT (only

warns, exit 0).

2. `\_scan\_forbidden` is called the "positive privacy assertion" but only catches MD5/6-forbidden-substrings/source-path-tokens ? NOT arbitrary names/match titles/dates in `--label`/`--source-note`. The only test of it covers the path case only.
3. `test\_synthetic\_provenance\_tags` (in the cross-OS job) hard-rejects any `cleared\_real` fixture ? the exact output of the shipped P1 tool ? breaking the suite the moment P3 lands.
4. Doc P1 §4c "Raw-trajectory exclusion ? never commit the per-frame CSV" contradicts the shipped `freeze\_real\_fixture`, which writes the full per-frame `trajectory.csv` (line 635) into the committable fixture dir.
5. The per-fixture `[ok] ... F1=1.000000` CLI line (line 899) prints synthetic F1 without the "not field quality" caveat on that line (the caveat is on the summary line that follows).
6. "cross-OS enforced" (line 163) slightly overstates given the doc itself lists "required status check" as unchecked TODO in §9.

****Verified clean:****

- §6 caveat IS carried in the primary characterization test assertion (test line 38) and both `--check` FAIL + summary lines.
- "owner-accepted (researched risk), not retained-counsel" is honest and consistent across header/§3/§8/P2/§10.
- Time-origin reset metric-invariance is real and tested.
- CONTROL avoids numpy; gt_hash label-only fingerprint is fine (shared hash is by design, trajectory diffs captured in snapshot).
- Determinism: replay math is pure-stdlib, 1e-6 parse tolerance, cross-OS guard covers the committed synthetic fixtures correctly.

Let me check one more: does the lens's specific concern "could F1=1.0 be mistaken for field quality anywhere" ? is the F1=1.0 ever surfaced WITHOUT a caveat in a place a reader would see it as quality? The `[ok]` line and the snapshot files themselves show `f1: 1.0`. The snapshot JSON has no inline caveat, but the README + docstring + provenance "note" all carry it. Acceptable. The most exposed surface is the per-`[ok]` CLI line.

56. user (2026-06-16T06:31:37.479Z)

Structured output provided successfully

57. assistant (2026-06-16T06:31:47.763Z)

I've completed the honesty/over-claim audit of the shipped Golden Regression Fixtures feature and returned my findings via StructuredOutput.

Key results: the feature's core honesty discipline is genuinely strong (the §6 "behaviour-locking, not correctness" and "synthetic ? field quality" caveats really do ride in test assertion messages and the `--check` FAIL/summary lines, and the "owner-accepted researched-risk, not retained-counsel" framing is consistent throughout). But the per-PR verification missed six items, four of them medium severity:

1. `refresh\_snapshot` docstring claims "the CLI requires" a reason ? it only warns and exits 0 (golden_fixtures.py:451 vs :955-960).
2. The "positive privacy assertion" (`\_scan\_forbidden`) cannot catch arbitrary names/match-titles/dates in `--label`/`--source-note` ? only MD5/forbidden-stream-substrings/source-path tokens; its sole test covers only the path case.
3. `test\_synthetic\_provenance\_tags` (in the cross-OS CI job) hard-asserts `kind=="synthetic"` over ALL manifest slugs ? I confirmed empirically it will FAIL on a `cleared\_real` fixture, which is exactly what the shipped P1 tool produces, so committing P3 will red the suite.
4. Doc §4c "never commit the per-frame CSV" contradicts `freeze\_real\_fixture`, which writes the full per-frame `trajectory.csv` into the committable fixture dir (line 635).

Plus two low-severity wording over-reaches (the per-fixture `[ok] F1=1.000000` line lacks the inline caveat that the summary line carries; "cross-OS enforced" overstates given §9 still lists "required status check" as a TODO).

Verdict: NOT CLEAN ? no critical privacy/determinism hole, but two forward-facing traps should be fixed before any P3 real-fixture commit.